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Development of a Suspended Aerobic Membrane Bioreactor (SAMBR) for Sustainable Wastewater Treatment on Early Planetary Bases

by

Talon James Bullard

A dissertation submitted in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy
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Dedication

To my wonderful family, Olivia, Lottie, and Eve, who were with me every step of the way.

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Abstract

Environmental Control and Life Support Systems (ECLSS) ideally should treat all "waste" streams for recovery in a sustainable and near closed-loop manner. This will be critical for Partial Gravity Habitats (PGH) and Early Planetary Bases (EPB) that will be established on the Moon and Mars due to the lack of readily available resources. Currently, the Water Recovery System (WRS), onboard the International Space Station (ISS), recovers and purifies water from urine and graywater. While efficient, this system is not optimized for future missions and utilizes physical-chemical technologies that operate with single-use and hazardous consumables. In response to the lack of alternative flight ready technologies capable of sustainably treating urine and gray water, a Suspended Aerobic Membrane Bioreactor (SAMBR) was developed between the University of South Florida and Kennedy Space Center. SAMBR was optimized for PGH/EPB scenarios and served as a hybrid alternative, coupling biological treatment with physical membrane filtration.

With its modular design, SAMBR's operation was customized to suit varying treatment objectives, with a primary focus on carbon and nitrogen removal from urine. The scope of work presented here covered the design, development, preliminary evaluation of SAMBR's treatment of urine and gray water. SAMBR advancement from a concept to a prototype tested with real urine brought it from a Technology Readiness Level (TRL) of one to six. SAMBR was challenged to remove carbon and nitrogen under high-strength conditions and achieved respective removals of 76%, and 93% during optimal operation. Deeper evaluation of its performance under high strength conditions is needed to understand the effect of potential

inhibiting factors such as nitrogen, pH, and salinity. Recommended future research includes further optimization of system safety and reliability, expanding treatment capabilities, measurement of additional parameters, specifically phosphorous and salinity.

Chapter 1: Introduction

1.1 Motivation

The creation of the Artemis program from Space Policy Directive 1, which aims to return humans to the Moon, marks another progressive step within the Space Age (*Space Policy Directive-1, Reinvigorating America's Human Space Exploration Program*, 2017). Per this Directive, Artemis III should deliver the next man and first woman to the Moon and will be the first of many long-duration, deep-space, human exploration missions. These missions aim to maintain a continuous presence on the Moon and pioneer the way to Mars(*NASA's Plan for Sustained Lunar Exploration and Development*, 2020). Such an ambitious undertaking presents NASA and its collaborators with unique and trying technological challenges not encountered since the Space Race. Many of these challenges stem from the harsh and dynamic environments encountered in space, on the Moon, and on Mars. With little to no resources readily available to support human life, planetary bases in these environments will need to be robust, independent, and provide all the basic requirements to support life that are naturally granted on Earth.

In the case of a thirty-month mission, a single crew member (CM) will require 2250 kg of drinking water, 6525 kg of hygiene water (oral, handwash, shower, shave, etc.), and produce over 2000 kg of metabolic waste (Muirhead et al., 2022b; Pickett et al., 2019). With a current payload cost of approximately $10,000/lb. (~$4,535/kg), the cost of just water for a single astronaut exceeds $39 million (Boen, 2008). Additionally, Environmental Control and Life Support Systems (ECLSS) technologies used in space are not optimized for Partial Gravity Habitats and Early Planetary Base (PGH/EPB) scenarios. Urine is notoriously difficult to treat,

primarily due to its high nutrient and ion content, accounting for up to 80% of nitrogen found in municipal wastewater while constituting only 1% of the total volume (Jalalieh et al.; Randall & Naidoo, 2018; Tian et al., 2019). While the formulations and compositions of PGH/EPB waste streams (humidity, hygiene, urine/flush, and laundry, etc.) are still under investigation, it is estimated that urine will account for approximately 97% of the nitrogen(Muirhead & Carter, 2021; Muirhead et al., 2022b). The Water Recovery System (WRS) aboard the International Space Station (ISS) utilizes multiple consumable inputs that are not optimized for such applications beyond Low Earth Orbit (LEO) (Pickett et al., 2019). Consequently, the WRS is highly dependent on a harsh chemical pretreatment method that must be constantly resupplied in order to recover water from the urine(Pickett et al., 2019). These current challenges highlight the need for the development of alternative treatment technologies which can operate with minimum external inputs. For these reasons alternative treatment technologies need to be considered for PGH/EPB scenarios. In space, treatment technologies have been strictly physical-chemical as they provide rapid and consistent treatment (Pickett et al., 2019). However, biological treatment systems are well established terrestrial technologies and require comparatively less input. To date, no flight-ready biological system has been developed for space. Biological treatment is a promising alternative for PGH/EPB as it requires less energy and inputs, leading to reduced cost (Pickett et al., 2019).

1.2 Currently Existing Treatment Approaches for Space

The ISS is the primary testbed for current and developing ECLSS technologies. For wastewater, the Urine Processor Assembly (UPA), Brine Processor Assembly (BPA), and Water Processor Assembly (WPA) are the primary subsystems that make up the WRS. Prior to entering the UPA, urine is pretreated with hexavalent chromium and phosphoric acid to prevent the

precipitation of uric acid and inhibit biological growth(Muirhead, 2018). Upon entering the UPA, the pretreated urine is subjected to Vapor Compression Distillation (VCD) to separate the water from the brine solution which recovers approximately 85% of the water found in the urine(Carter et al., 2018; Pickett et al., 2019). The WPA achieves approximately 85% water recovery but does not recover any nutrients (i.e., C and N) and utilizes consumables including disposable filters and stored oxygen(Pickett et al., 2019).

A relatively recent technology, the BPA aims to treat the concentrated brine produced by the UPA and has been on the ISS for testing since March 2021. Utilizing forced convection from ambient air coupled with membrane distillation, water is separated from the brine as vapor. This water vapor is recovered using the existing condensate system while the concentrated brine is stored for disposal. Preliminary results indicate that the BPA increases the water recovered from urine from 85% to >90%(Williamson et al., 2022). In June 2023, the WRS achieved 98% water recovery of the currently treated waste streams (Gaskill, 2024) but still requires further optimization. This achievement currently comes at the expense of consumable inputs (i.e., chemical pretreatment and filters) and loss of nutrients - .

Tied into the WRS is the Oxygen Generation System (OGS). A portion of the treated water produced by the WPA is diverted to the Oxygen Generation Assembly (OGA). The OGA electrolyzes the water to produce gaseous hydrogen and oxygen (Rxns. 1.1-1.3)(Takada et al., 2021; Wang et al., 2021). The oxygen is used to revitalize the cabin air, while the hydrogen is vented out to space or sent alongside carbon dioxide from the crew to the Carbon dioxide Reduction System (CRS)(El Arwadi, 2023; Takada et al., 2021). Contained within the CRS, a Sabatier reactor converts hydrogen and carbon dioxide to methane and water (Rxn. 1.4) (Yu et al., 2022). The methane is vented overboard while the water is sent to the WPA for treatment.

Water Oxidation: $2H_2O_{(l)} \Leftrightarrow O_{2\,(g)} + 4H^+_{(aq)} + 4e^-$ (1.1)

Water Reduction: $4H^+_{(aq)} + 4e^- \Leftrightarrow 2H_{2\,(g)}$ (1.2)

Overall Water Electrolysis Reaction: $2H_2O_{(l)} \Leftrightarrow O_{2\,(g)} + 2H_{2\,(g)}$ (1.3)

Sabatier Reaction: $CO_2 + 4H_2 \Leftrightarrow CH_4 + 2H_2O$ (1.4)

While efficient, consistent, reliable, and well documented, current space ECLSS technologies are still being investigated for improved recovery, broader treatment capabilities, reduced odor, and reduced consumable inputs to achieve a near closed-loop architecture(Boyce et al., 2022; Williamson et al., 2022). The hazardous urine pretreatment chemical, hexavalent chromium, creates safety concerns and prevents any possibility for additional resources recovery. Recovered elements such as carbon and nitrogen will be vital for sustainable in-situ food production. Constant use of consumable inputs (i.e., chemical pretreatment, disposable filters, etc.) takes up valuable cargo space, requires expensive resupply missions, and creates waste that requires disposal. Most current ISS ECLSS technologies are not suitable for EPB, as loading requirements and waste streams are vastly different such as laundry wastewater stream. EPB/PGH scenario waste stream types and volume estimates are compared to the ISS in Table 1.1 to highlight these differences per crew member. The requirements of ECLSS technologies in EPB/PGH missions include a 98% water recovery and the capability of treating all forms of waste (Pickett et al., 2019). To achieve these mandated goals, which are primarily outlined in NASA's Technology Taxonomy (*2020 NASA Technology Taxonomy*, 2020), alternative treatment technologies need to be explored to reduce consumable inputs, expand resource recovery beyond water, and maintain a similar level of efficiency and reliability.

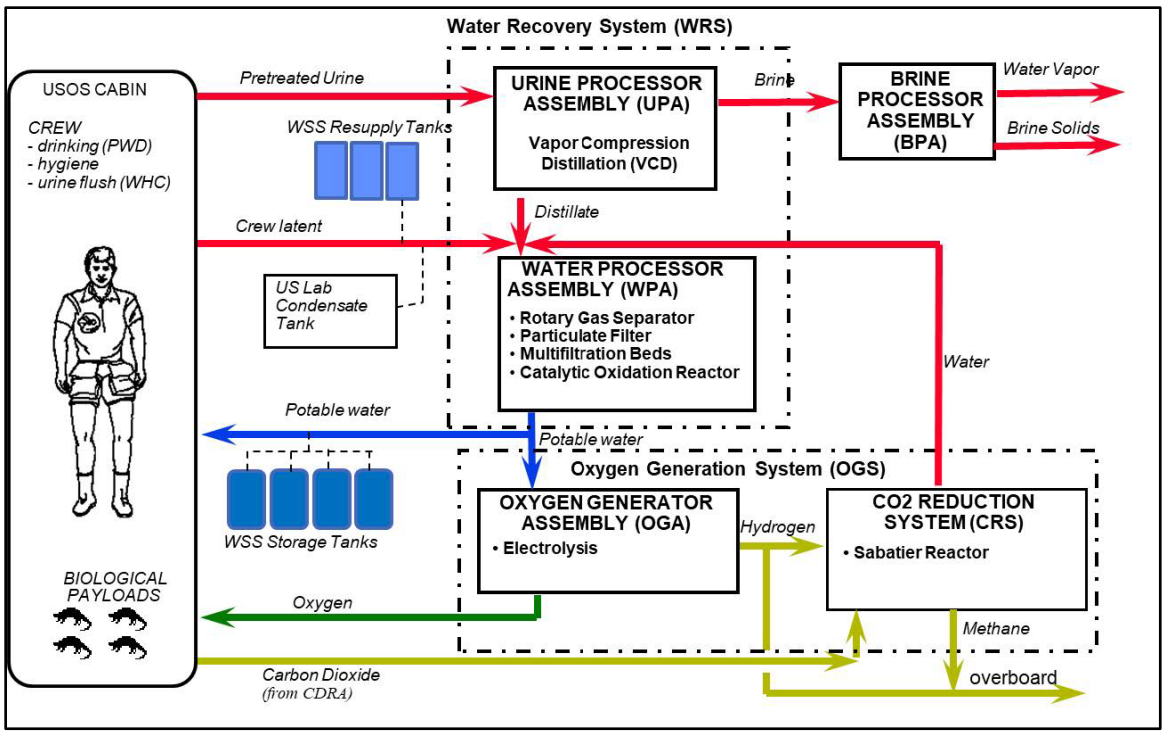


Figure 1.1 Schematic of water recovery and management for the ISS US segment. (Williamson et al., 2022)

Table 1.1 ISS and EPB/PGH crew generated waste streams. (Anderson et al., 2015; Muirhead et al., 2023; Muirhead et al., 2022a)

Crew Generated Waste Streams (L/CM-d)	ISS	EPB/PGH Scenario		
	Nominal	Nominal	Min.	Max.
Feces	0.1	0.23	0.16	0.3
Urine	1.2	2	1.5	2.5
Flush	0.3	0.3	N/A	N/A
Humidity Condensate	2.3	2	1.3	2.4
Handwash	N/A	1	N/A	N/A
Shower	N/A	6	N/A	N/A
Oral	N/A	0.2	N/A	N/A
Shave	N/A	0.0125	N/A	N/A
Laundry	N/A	1.07	N/A	N/A
Total Blackwater	1.6	2.5	2.0	3.1
Total Graywater	2.3	10.3	9.6	10.7
Total	3.9	12.8	11.5	13.8

Chapter 2: Background

2.1 Challenges Treating Urine and Graywater

The treatment of urine presents a unique set of challenges and rewards due to its concentration and variability. While the recovery of water and nutrients as a potential fertilizer are considerable, treatment systems must be able to handle scaling, corrosion, fouling, eutrophication, and inhibition caused by urine(Jurga et al., 2023; Ozair, 2012; Pourbavarsad et al., 2022). Table 2.1 compares the characteristics of urine (fresh and hydrolyzed) and municipal wastewater, demonstrating the extreme concentration of source separated urine compared to what municipal technologies on Earth are designed to treat. The primary challenges stem from the hydrolysis of urea, the main form of nitrogen in urine. Urea hydrolyses with water (Rxn. 2.1) in the presence of the enzyme, urease, to create ammonium, bicarbonate, and hydroxide(Ray et al., 2018). The creation of hydroxide results in a pH rise, while the ammonium and bicarbonate increase ion concentration (electrical conductivity) and alkalinity(Ray et al., 2018).

Urea Hydrolysis: $(NH_2)_2CO + 3H_2O \overset{urease}{\Longleftrightarrow} 2NH_4^+ + HCO_3^- + OH^-$ (2.1)

Complications with urine are ubiquitous in many treatment systems and environments, making it a major subject of research (Faust et al., 2023; Larsen et al., 2021; Randall et al., 2022; Udert et al., 2006; Udert & Wächter, 2012). Precipitation and scaling can cause blockages and fouling in physical and chemical systems(Jurga et al., 2023; Ozair, 2012; Pourbavarsad et al., 2022). In biological systems, the high nitrogen content often results in inhibition of treatment(Pourbavarsad et al., 2022). The high level of nutrients (N, P, K) can cause eutrophication in biological systems and fouling in membrane filtration systems(Akinnawo,

2023). The highly basic pH (>9) of hydrolyzed urine presents challenges for biological treatment systems, which typically prefer a neutral pH (~6.5-8) (Zhou et al., 2019). To counteract the high pH caused by hydrolyzed urine, urine hydrolysis can be prevented, or a storage system can implement a pH control system for the hydrolyzed urine. A plethora of smaller components including ions and nutrients found in urine contribute to its dynamic profile such as urea, phosphate, potassium, sulfate, calcium, magnesium, as well as biological constituents, with a comprehensive list found in Table 4.27 (Anderson et al., 2015).

Graywater includes a broad variety of wastewaters, typically shower, handwash, shave, oral rinse, and laundry streams. Compared to blackwater, graywater generally contains less contaminants (i.e., pathogens, organic matter, and nutrients) and, therefore, is comparatively less hazardous, easier to treat, reuse, and recycle(Al-Jasser; Khajvand et al., 2022; Noutsopoulos et al., 2018). However, graywater is still challenging to treat, as its characteristic profile is highly variable. While lower in contaminants overall, graywater can contain a higher concentration of household chemicals including surfactants which may cause foaming or inhibition of biological treatment systems. Municipal wastewater treatment plants commonly employ biological treatment technologies for domestic wastewater (including graywater) but often require the addition of an external nutrient source (namely carbon/COD), as the wastewater is limited in the nutrients required for the biological treatment consortia to achieve the desired performance(Li et al., 2021). Generation rates of graywater are generally much higher than blackwater and can strain the loading and treatment capability of graywater treatment systems and washout the active organisms responsible for treatment (Filali et al., 2022).

Table 2.1 Characteristics of urine and municipal wastewater. (Hassan et al., 2023; Levlin, 2007; O'Neal & Boyer, 2013; Pronk et al., 2006; Rose et al., 2015; Santoro et al., 2020; Udert et al., 2006; Von Sperling, 2007)

Waste Stream	TN (mg-N/L)	COD (mg/L)	Alkalinity (mg/L)	Electrical Conductivity (us/cm)	pH
Urine (fresh)	9,350	11,450	2,200	13,500	6.2
Urine (hydrolyzed)	9,200	10,000	49,000	35,200	9.1
Municipal Wastewater	45	600	200	522	7

2.2 Treatment Approaches

2.2.1 Biological Nitrogen Removal

A considerable number of options exist to use biological technologies for waste treatment, resource recovery, and biological nitrogen removal (BNR). Though numerous in approaches, BNR technologies typically revolve around conversion of nitrogen within the nitrogen cycle (Fig. 2.1). Most nitrogen in waste treatment originates from urine, accounting for ~80% found in municipal wastewater(Wuang et al., 2013). With select microorganisms and environmental conditions, nitrogen can be converted to various forms to achieve the desired form, typically as dinitrogen which can be off-gassed into the atmosphere. Development and use of microorganisms for biological treatment of wastewater are typically categorized as attached or suspended growth. Attached growth processes have the organisms growing on fixed surfaces to accumulate a biofilm that treats surrounding wastewater and are typically used for biofilters and biological contactors. Suspended growth cultivates organisms floating freely in solution and treating in the mixture; they are commonly used in activated sludge systems. Some applications

utilize both attached and suspended growth to enhance treatment and stability but may increase complexity and maintenance.

In Fig. 2.1, depicting the nitrogen cycle, (1) dinitrogen gas is fixed as ammonium, commonly referred to as nitrogen fixation; (2) ammonium is oxidized by microorganisms; (3) nitrite is oxidized to nitrate; (4) nitrate (or nitrite if not fully nitrified) is denitrified to dinitrogen gas through a nitrous oxide intermediate; (5) anaerobic ammonium oxidation (anammox) converts ammonia to dinitrogen; and (6) nitrate/nitrite is reduced to ammonium. (Jetten, 2008)

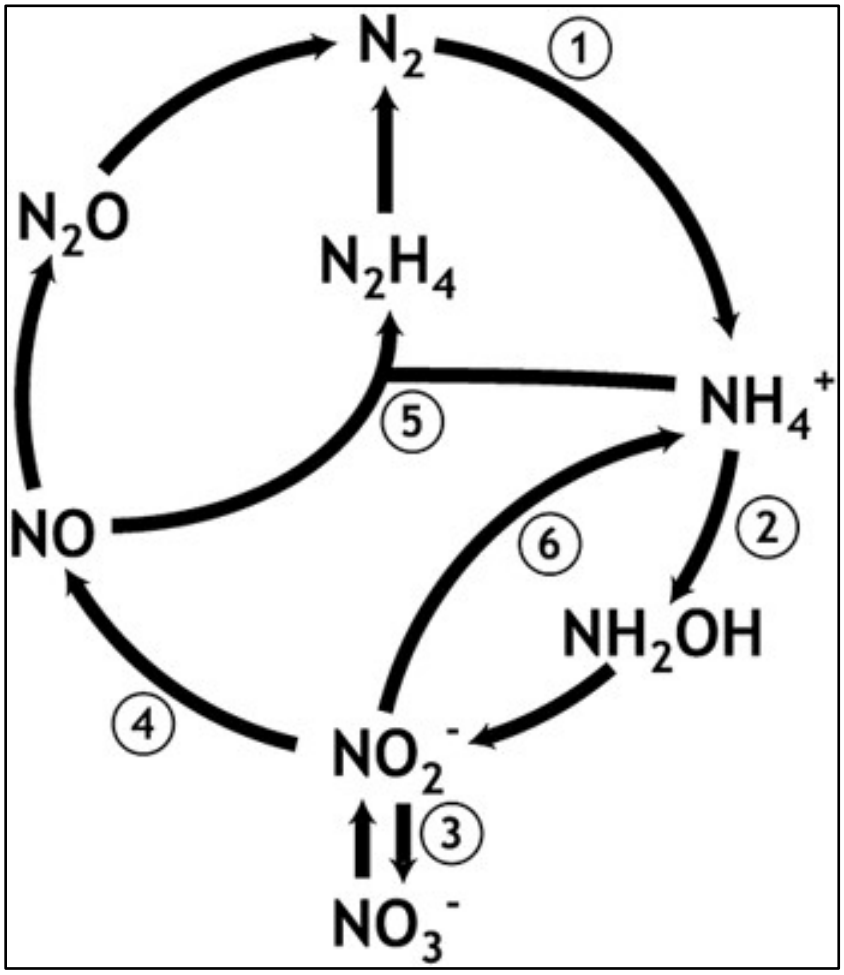


Figure 2.1 Biological nitrogen cycle. (Jetten, 2008)

One of the most common configurations of BNR for municipal use is the Modified-Ludzack Ettinger (MLE) process, which is a coupling of nitrification and denitrification reactions to convert ammonia to nitrogen gas (Fig. 2.2a)(Metcalf & Eddy, 2003). The influent enters an anoxic zone, where nitrogen is predominantly in the form of organic nitrogen and ammonia. Additionally, an internal recycle introduces nitrate from downstream for nitrogen removal via denitrification and is discussed more in the following paragraph. A relatively small fraction of nitrogen is used in biomass uptake. Once the ammonia nitrogen enters the aerobic stage, nitrification converts ammonia to nitrate. This is a coupled process utilizing Ammonia Oxidizing Bacteria (AOB) that oxidizes ammonia to nitrite (Rxn. 2.2) and Nitrite Oxidizing Bacteria (NOB) that oxidizes the nitrite into nitrate (Rxn. 2.3). The overall process is often combined into a general nitrification process (Rxn. 2.4). These autotrophic bacteria use ammonia and nitrite as their inorganic electron donor and oxygen as the acceptor(Baskaran et al., 2020). Consequently, heterotrophic microorganisms oxidize organic constituents, resulting in COD reduction and is commonly in wastewater treatment(Verma et al., 2020).

Nitritation: $NH_4^+ + 1.5O_2 \Leftrightarrow NO_2^- + 2H^+ + H_2O$ (2.2)

Nitrification: $NO_2^- + 0.5O_2 \Leftrightarrow NO_3^-$ (2.3)

Overall Nitrification Reaction: $NH_4^+ + 2O_2 \Leftrightarrow NO_3^- + 2H^+ + H_2O$ (2.4)

Near the end of the aerobic zone, a pump recycles a portion of the nitrified contents back to the beginning of the nitrification zone. This internal recirculation (IR) distinguishes MLE from the LE process and is typical pumped at twice the flowrate of the influent but can vary anywhere from 50–500%(Ergas & Aponte-Morales, 2014; Rossinskyi, 2018). Under anoxic conditions, denitrifying microorganisms convert the recycled nitrate into diatomic nitrogen that off-gasses into the atmosphere through a series of reduction reactions (Rxns. 2.5-2.8) and the overall

denitrification reaction (Rxn. 2.9)(Rittmann & McCarty, 2001). Organic carbon is utilized as the electron donor in these reactions. As it is often the rate limiting step, additional carbon sources are typically needed(Ergas & Aponte-Morales, 2014; Rittmann & McCarty, 2001). The recently converted nitrogen off-gasses into the atmosphere, while the reactor contents flow through the aerobic zone again and eventually to a clarifier where the treated effluent exits the system. A portion of the settled sludge is returned to the influent as returned activated sludge (RAS), and the remaining is removed as wasted activated sludge (WAS).

Nitrate Reduction: $NO_3^- + 2H^+ + 2e^- \overset{Nitrate\ reductase}{\Longleftrightarrow} NO_2^- + H_2O$ (2.5)

Nitrite Reduction: $NO_2^- + 2H^+ + e^- \overset{Nitrite\ reductase}{\Longleftrightarrow} NO + H_2O$ (2.6)

Nitric Oxide Reduction: $2NO + 2H^+ + 2e^- \overset{Nitric\ oxide\ reductase}{\Longleftrightarrow} N_2O + H_2O$ (2.7)

Nitrous Oxide Reduction: $N_2O + 2H^+ + 2e^- \overset{Nitrous\ oxide\ reductase}{\Longleftrightarrow} N_2 + H_2O$ (2.8)

Overall Denitrification Reaction:

$2NO_3^- + 12H^+ + 10e^- \Leftrightarrow N_2 + 6H_2O$ (2.9)

Considered an extension of the MLE process, a 4-stage Bardenpho provides additional denitrification with another anoxic zone after the oxic zone (Fig. 2.2b). A shorter oxic zone is the last step in traditional activated sludge system to improve settling in the clarifier. Typically, the additional carbon source that is needed for complete denitrification can be sourced from excess biomass or an external source such as methanol.

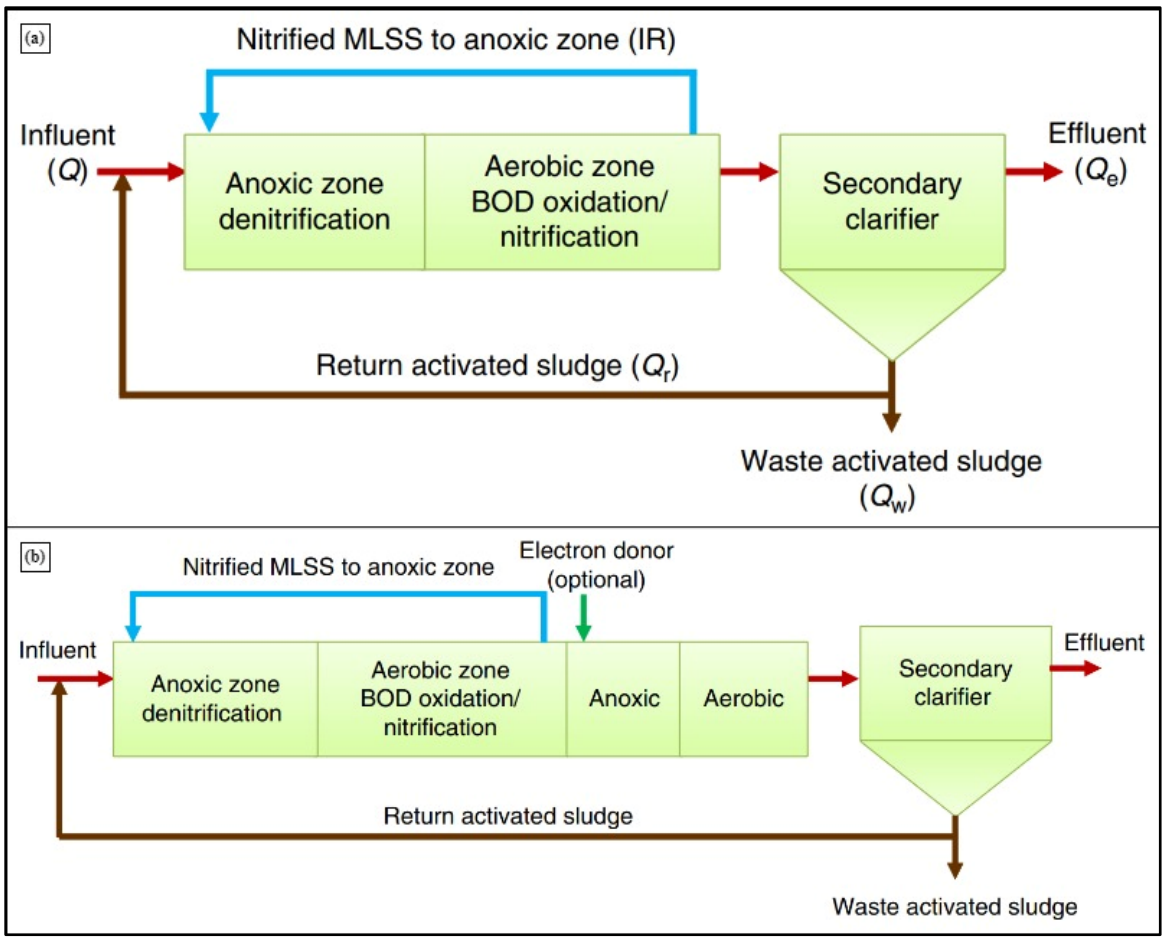


Figure 2.2 Common biological treatment approaches; (a) MLE process; (b) 4-stageBardenpho process. (Ergas & Aponte-Morales, 2014)

2.2.2 Texas Tech University's Bioreactor Systems

Although treatment technologies in space have predominantly been physical and chemical in principle, increasing efforts have been made to develop bioregenerative technologies as sustainable approaches. Texas Tech University (TTU) has developed a biological treatment systems termed: Counter Diffusion Membrane Aerated Nitrifying Denitrifying Reactor or the CoMANDR(Sevanthi et al., 2018). Based on Membrane Aerated Bioreactor (MABR) technology, the CoMANDR has successfully treated various combinations of urine, flush, hygiene, humidity, condensate, and laundry waste streams in pseudo ISS, transit, and EPB/PGH

scenarios continuously for nearly three years (Christenson et al., 2018; Sevanthi et al., 2018). As a biological system, it can immediately treat crew generated wastes (removing the need for intermediate storage tanks) and requires only aeration, acids, and salts as inputs (Barta et al., 2015). While analysis and optimization of mass, volume, and energy are ongoing, challenges for this system include biofouling mitigation, overcoming extended startup following inoculation, improving salt rejection, mitigating water permeance, implementing automated control, and increasing nitrogen removal as it currently achieves a maximum removal of 50%(Sevanthi et al., 2018).

More recently, TTU and Pancopia have developed a single stage biological treatment system (NDX) combining nitrifications, denitrification, and anammox to remove C and N present in PGH/EPB waste streams (Fig. 2.3a) (Jalalieh et al., 2021). This system uses minimal energy and consumable inputs to treat urine, flush, humidity condensate, hygiene, and laundry waste streams (Jalalieh et al., 2021). The nitrogen removal ranged from 49%-66% and averaged 93% for DOC removal (Jalalieh et al., 2021). However, the system is volumetrically inefficient and initial testing was with waste streams equivalent to less than one crew member (Jalalieh et al., 2021).

A newer MABR based system was designed to treat crew urine and flush water specifically. Focused on nitrification, the UMABR was designed for N conversion and C removal (Fig. 2.3b) (Hooshyari et al., 2023; Jackson et al., 2022). Using real urine and flush, the system was capable of achieving approximately 50% conversion of nitrogen and 90% removal of DOC (Hooshyari et al., 2023; Jackson et al., 2022). With no immediate storage tanks needed and efficient aeration from the membrane delivery, energy and volume requirements are reduced (Hooshyari et al., 2023; Jackson et al., 2022). However, the system was coupled with a

distillation system with acid trap to prevent N carryover and HNO_2 decomposition in the distillation system (Hooshyari et al., 2023; Jackson et al., 2022). Brine solids from the distillation process are stored/accumulated in the distillation unit (Hooshyari et al., 2023; Jackson et al., 2022).

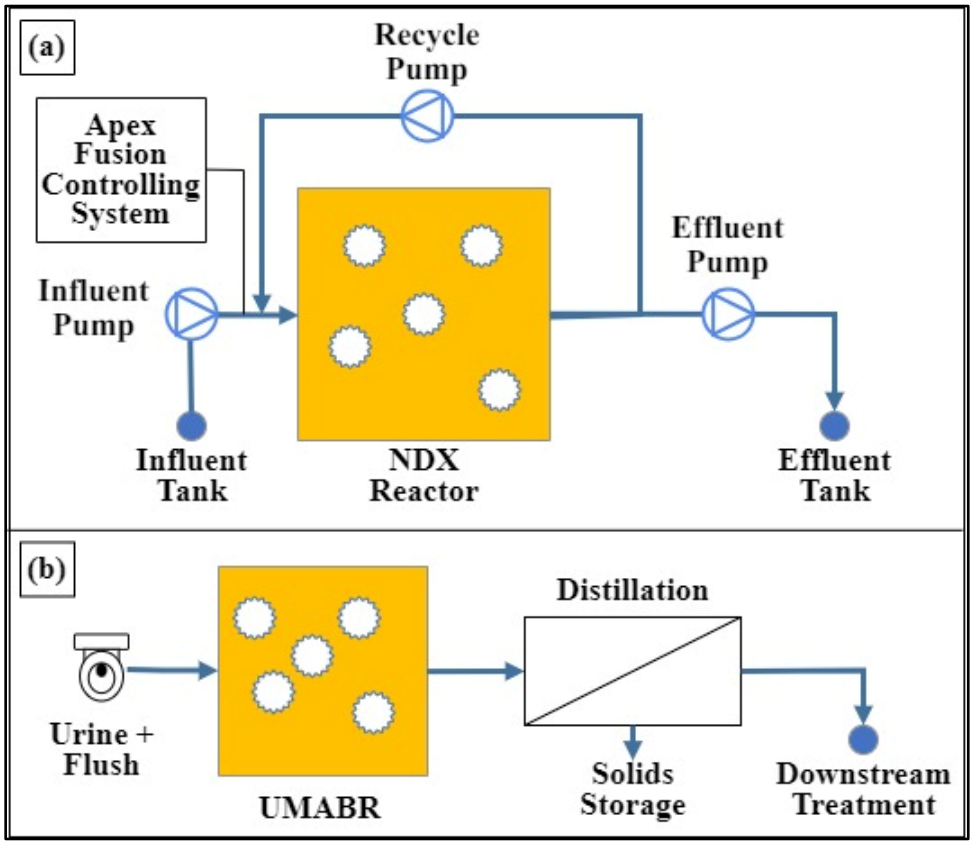


Figure 2.3 Schematic of TTU's NDX and UMABR systems. (Hooshyari et al., 2023; Jackson et al., 2022; Jalalieh et al., 2021)

2.2.3 European Space Agency's Micro-Ecological Life Support System Alternative

One of the most prominent partnerships within the European Space Agency (ESA) led to the development of the Micro-Ecological Life Support System Alternative (MELiSSA) (Fig. 2.4)(Walker & Granjou, 2017). Utilizing four main treatment compartments: thermophilic anaerobic treatment, photoheterotrophic bacteria, nitrifying bacteria, and photoautotrophic

bacteria with higher order plants, the comprehensive system aims “to construct autonomous habitats in deep-space, supplying astronauts with fresh air, water, and food through continuous microbial recycling of human wastes”(Hendrickx et al., 2006; Lasseur et al., 2010; Walker & Granjou, 2017). Waste, including urine, enters compartment one to begin treatment, but urine is not the treatment focus until compartment three. Utilizing nitrifying bacteria in a fixed-bed bioreactor, ammonia is nitrified into nitrate, which is the desired form to be utilized by the higher order plants and cyanobacteria downstream in compartment four. Nitrogen conversion efficiencies as high as 95–100% have been reported(Verbeelen et al., 2021). However, it is stated that “[Compartment III] cannot be used to directly process urine at present”(Verbeelen et al., 2021). Additional challenges are related to the energy efficiency primarily related to lighting requirements for phototrophic bacteria, culturing appropriate consortia tolerant of a dynamic ammonia influent, and lack of architecture integration(Hendrickx et al., 2006).

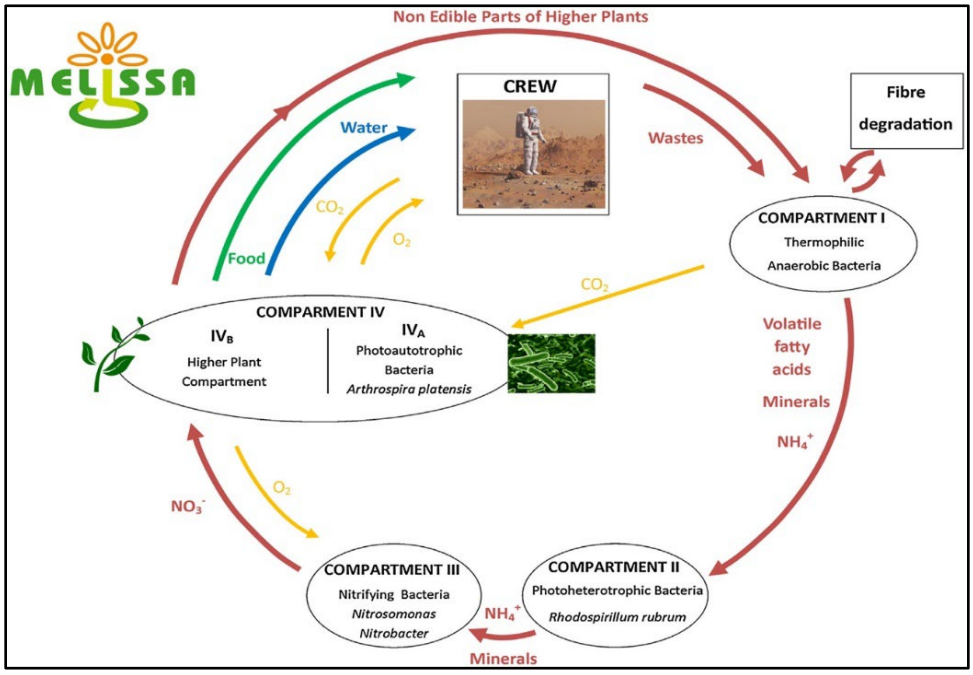


Figure 2.4 Schematic of MELiSSA process. (Walker & Granjou, 2017)

2.2.4 CO_2 Carbonation

A primary problem in urine treatment, aside from inhibition caused by high nitrogen content, is the increase in pH when it hydrolyzes. Since the desired pH range for AOB and NOB is 6–7.5, inhibition from pH is also likely(Sharif Shourjeh et al., 2021). Traditionally acid addition is used when pH reduction is necessary; however, if a solution has a high alkalinity, (like hydrolyzed urine) it can require substantial amounts of acid. In the case of space treatment, the need for large volumes of acid would reduce the sustainability of a bioregenerative treatment technology(Udert & Wächter, 2012). The addition of dissolved CO_2 is a novel method that has been demonstrated to reduce pH in urine(Courtney et al., 2021; Randall et al., 2022). As shown in Rxn. 2.10, gaseous CO_2 dissolves and becomes aqueous, then reacts with water to form carbonic acid (Rxn. 2.11). The carbonic acid then dissociates to form bicarbonate and releases a hydrogen ion (Rxn. 2.12. The bicarbonate also dissociates to produce carbonate and release another hydrogen ion (Rxn. 2.13). The release of these hydrogen ions contributes to the reduction of pH. The production of these carbonate species, especially calcium carbonate (Rxn. 2.14), are some of the primary constituents that contribute to the alkalinity of a solution

Carbon Dioxide Dissolution: $CO_{2(g)} \Leftrightarrow CO_{2(aq)}$ (2.10)

Carbonic Acid Formation: $CO_2 + H_2O \Leftrightarrow H_2CO_3$ (2.11)

Bicarbonate Formation: $H_2CO_3 \Leftrightarrow H^+ + HCO_3^-$ (2.12)

Carbonate Formation: $HCO_3^- \Leftrightarrow H^+ + CO_3^{2-}$ (2.13)

Calcium Carbonate Formation: $Ca^{2+} + CO_3^{2-} \Leftrightarrow CaCO_3$ (2.14)

Ammonium Bicarbonate Formation: $CO_2 + NH_3 + H_2O \Leftrightarrow (NH_4)HCO_3$ (2.15)

While CO_2 alone is not sufficient for urine treatment beyond pH reduction, it has the potential to be coupled with complementary treatment technologies. (Bavarella et al., 2020) investigated the use of CO_2 with a gas-liquid hollow-fiber membrane to produce crystalized ammonium-bicarbonate (Rxn. 2.15), which in urine treatment could be precipitated as a product. However, this method is difficult to investigate due to the high vapor pressure of CO_2 and ammonia(Sutter & Mazzotti, 2017). Most existing precipitation technologies are designed for larger molecules, do not consider evaporation of the precipitating substance, and are not adequate for such an application(Sutter & Mazzotti, 2017).

2.2.5 Membrane Bioreactors

While MABR technology struggles with limitations such as biofilm thickness sloughing and slow startup, other technologies should be explored to find a complementary alternative. One such candidate is a Membrane Bioreactor (MBR) which is capable of on demand treatment and smaller footprint due to its membrane filtration.

A MBR is created by coupling a biological process and a semi-permeable membrane. MBRs are primarily favored in wastewater applications under tight space constraints, providing a distinct advantage not offered by conventional wastewater treatment solutions (Bair, 2016; Xiao et al., 2019). The smaller footprint capability is primarily due to complete retention of biomass by the membrane, concentration of microbial activity, and allowing for comparable treatment in a smaller volume, such as compared to a MABR. The ability of MBRs to recover water in a compact footprint makes them valuable in water-scarce environments and have become common in decentralized applications(Bair, 2016; Xiao et al., 2019). Much of the research to improve MBRs focuses on decreasing their cost, use of chemicals, fouling mitigation, operational complexity, and increasing energy efficiency(Bair, 2016; Xiao et al., 2019).

MBR membrane separation relies on a pressure gradient, typically created via pumps, between the two sides of the membrane, known as Transmembrane Pressure (TMP). TMP is used as a measure of the pressure gradient between the two sides of the membrane. Increased TMP indicates an increased pressure gradient and can suggest an issue with the membrane such as extensive fouling. However, TMP is influenced by many characteristics including the membrane material, pore size, configuration, operation, and waste characteristics. The flowrate of the permeate divided by the surface area of the membrane is known as the flux of the membrane. The flux is used to obtain the rate of permeate production per unit area of the membrane and is an indicator of the health of the membrane as well as its efficiency.

Two main operational configurations of membrane operation exist: submerged and side stream (Fig. 2.5). As implied, submerged configurations operate the membrane directly in the bioreactor. This configuration requires less volume for the membrane, allowing the system to be more compact and have a lower energy demand as the pumping requirements are lowered(Judd, 2006). However, this configuration faces difficulties, as it cannot be cleaned in place and is not easily accessible for maintenance or replacement. Side stream configurations pump the bioreactor's contents to an externally placed membrane, with the reject water recycled back into the bioreactor. These configurations have an inherently higher disposition for fouling and energy expenditure, as an increase in flux consequently increases fouling; it is believed that the characteristic turbulent and shear force caused by pumping reduces particle size and contributes to the release of foulants in the membrane(Judd, 2006). While increased fouling is observed relative to immersed configurations, the external configuration permits the membranes to be cleaned in place (CIP), is much easier maintain, and mitigates fouling.

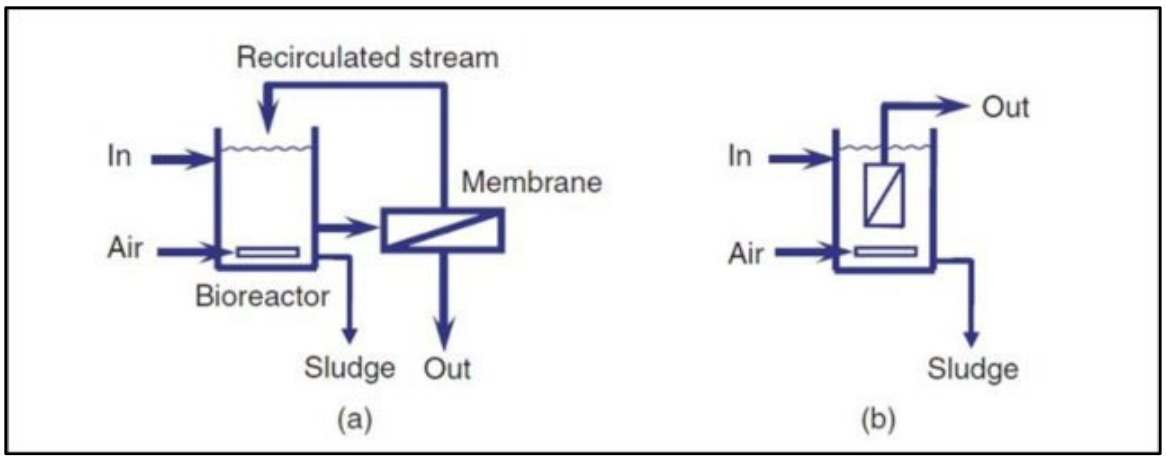


Figure 2.5 MBR configurations; (a) Side stream (sMBR); (b) Immersed (iMBR). (Judd, 2006)

MBRs are ideal for various applications including remote and space restricted environments due to their retention of biomass and can provide high quality treatment with an absolute barrier provided by the membrane. This has been demonstrated in multiple cases, such as the development of a Non-Sewered Sanitation (NSS) system termed the Nutrients, Energy, and Water generator (NEWgenerator or NEWgen). Centered around an AnMBR, the NEWgen couples it with ion exchange and electro-chlorination to form a decentralized treatment system capable of meeting ISO 30500 requirements and is summarized by Fig. 2.6(Bair et al., 2015; Shyu et al., 2021; Watabe et al., 2023). Long-term field tests in India and South Africa demonstrated NEWgen's capability to treat a wide variety of wastes, achieve high resource recovery, consistently produce a high-quality product water, all in a self-sustaining manner in remote environments(Bair et al., 2015; Shyu et al., 2021; Watabe et al., 2023).

NEWgen's success as a NSS system in remote applications influenced the development of a MBR system as a bioregenerative life support technology for space applications. Also centered around an AnMBR, the Organic Processor Assembly (OPA), was designed to sustainably treat the fecal waste generated by a crew of four during long-duration, deep-space

missions, where mass and volume restrictions are necessary. The prototype was tested with simulated and real fecal waste and reduced the organic and solids content by over 90%. The solid-free and nutrient rich effluent from the OPA is ideal for downstream treatment systems that would otherwise be unable to treat such concentrated waste without dilution(Bullard, 2020; Bullard et al., 2021; Smith et al., 2022). Fig. 2.7a depicts the OPA in the ECLSS architecture and Fig. 2.7b shows the prototype.

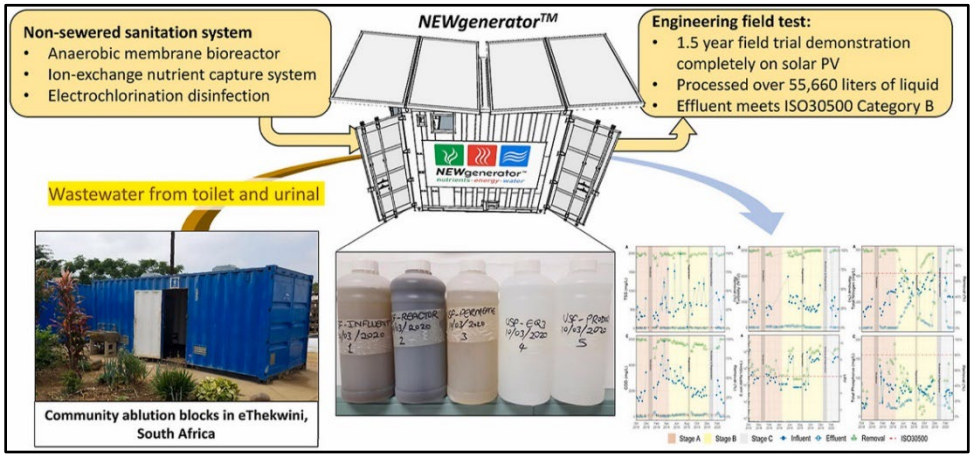


Figure 2.6 Visual abstract of NEWgen system. (Shyu et al., 2021)

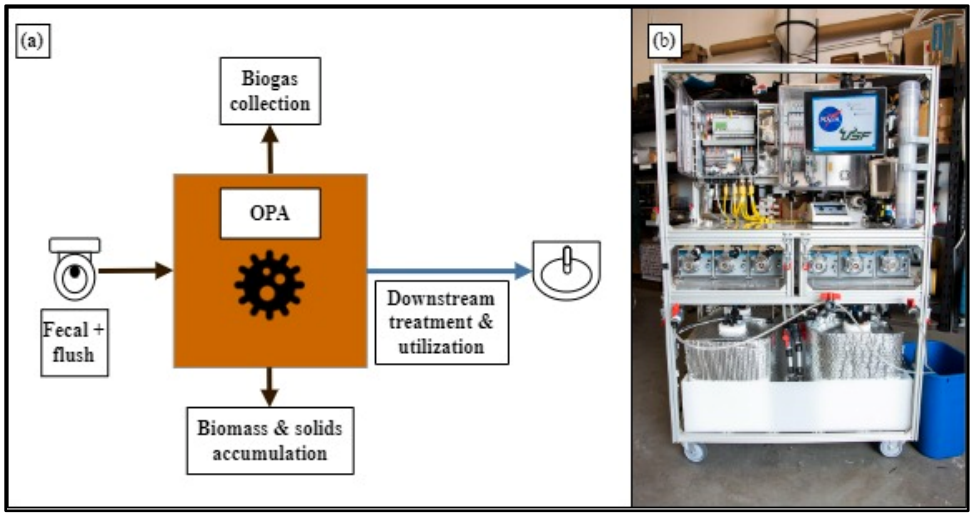


Figure 2.7 OPA system; (a)in ECLSS architecture; (b) prototype. (Bullard et al., 2021; Smith et al., 2022)

2.3 Statement of Research

To advance the limited field of sustainable ECLSS technologies for future exploration, an MBR-based system is proposed as a bioregenerative alternative for the treatment and recovery of key resources in urine and graywater. Fig. 2.8 depicts how SAMBR is envisioned to integrate into future ECLSS architecture. Using conditioning, BNR, and membrane filtration technologies, this hybrid system is termed the SAMBR. As mass and volume constraints are paramount for space exploration, a small footprint will be required for this technology. Reduced consumable inputs will be needed to make this system competitive against similar technologies. As MLE and 4-stage Bardenpho configurations are typically industrial scale biological treatment processes, scaling down will be necessary and is not known to have been done for space applications. The process of developing and converting a series of industrial scale terrestrial technologies to small, intensive, and concentrated treatment in space will expand the known boundary of space waste treatment. The research presented here aims to investigate SAMBR's capability to address many of these challenges and to be a progressive step towards a sustainable ECLSS architecture.

To assess the feasibility of developing SAMBR for space applications, the following questions were investigated during the course of this research:

1. What will be required to achieve optimal influent conditioning and what will the effects be throughout the system?
2. Are there additional inhibiting factors that the influent conditioning subsystem is unable to address and how might these be addressed in the future?
3. What design parameters for the system process and architecture are needed to meet the restraints for space while maintaining a sufficient hydraulic retention time (HRT) to adequately treat the urine and hygiene waste streams generated by a crew of four?

4. How will the biological consortia respond to increasing concentrations of crew waste, namely nitrogen (as ammonia)?
5. What will be the maximum amount of nutrient conversion and removal the system can achieve? And how will this compare against similar technologies?
6. What will be fates of influent COD and nitrogen throughout the system?

The associated research tasks are as follows:

1. Investigate the development of subsystems to condition the urine and graywater influent to create ideal conditions for the biological consortia.
2. Using NASA designated design constraints, design the system process and architecture for a maximum HRT while maintaining a footprint within said constraints.
3. Fabricate a model housing and bioreactor, then integrate the bioreactor and associated hardware into the housing.
4. Hydraulicly test the system to verify design, assembly, and viability of the biological consortia.
5. Operate the system with increasing strengths of urine for consortia activity until the full-strength of urine and flush is achieved.
6. Utilizing a modeling approach, assess the fate of COD and nitrogen throughout the system.

The overall objective is to design, develop, and evaluate the SAMBR concept for its potential to provide a bioregenerative alternative for treatment of urine and graywater in space. As a first-generation, proof of concept technology, preliminary research will revolve around four

primary research thrusts to investigate the resulting research questions and objectives, summarized in Table 2.2.

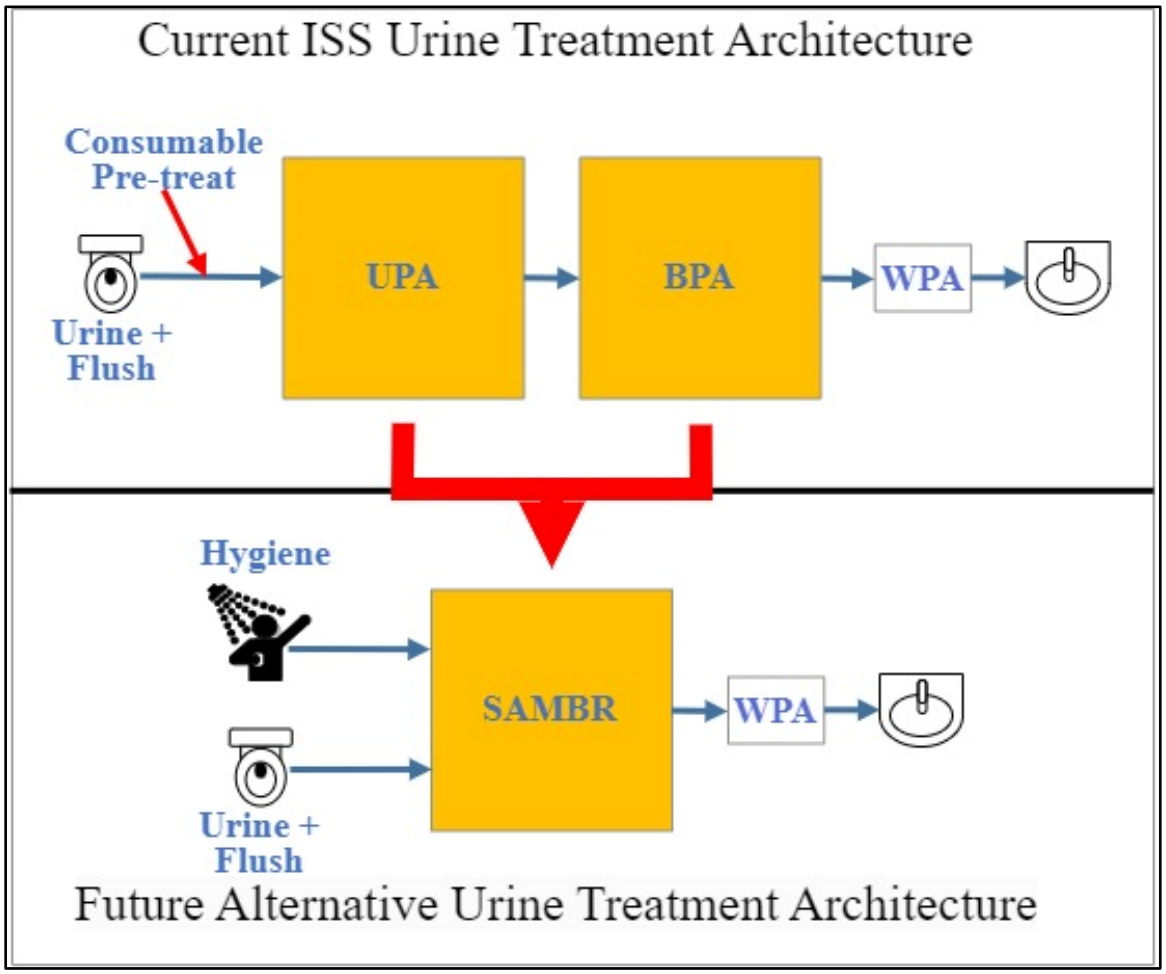


Figure 2.8 SAMBR placement in future ECLSS architecture.

Table 2.2 Summary of research questions, thrusts, and tasks.

Research Thrusts and Associated Tasks		Corresponding Research Question(s)
Thrust A: SAMBR Design, Development, and Testing	SAMBR process design	Q3
	SAMBR hardware design with CAD	
	SAMBR fabrication	
	Hydraulic verification	
	Oxic zone verification	
	Anoxic/oxic coupling verification	
	Operation using increasing strengths of urine and flush	Q4-5
	Operation using full strength urine and flush coupled with urine carbonation	
Thrust B: Urine Conditioning	Benchtop feasibility study of urine carbonation	Q1
	Urine carbonation column design, fabrication, and integration	Q1-2
	Urine carbonation column testing	
Thrust C: Hygiene Graywater Conditioning	Assessment and characterization of ISS No-Rinse	Q1
	Benchtop feasibility study of foaming mitigation technologies	Q1-2
Thrust D: Fate of Organics and Nitrogen in SAMBR	COD balance	Q6
	Nitrogen balance	
	Incorporation of handwash and shower	

Chapter 3: SAMBR Design and Operation

The SAMBR system was designed and operated as a hybrid bioregenerative alternative treatment system for urine, flush, and graywater waste streams generated by crew members during long term habitation missions. Starting from initial concept design, SAMBR was scaled to a pilot technology for preliminary testing. The design and fabrication investigated the ability for the treatment system to be created while fitting into the constraints (primarily volumetric) required for ECLSS technology development. Assessment of SAMBR's performance was primarily monitored through water quality testing of samples taken throughout the system. The execution of this major research thrust provided insight into SAMBR's ability to tolerate high strength waste, optimizing treatment, and conversion of contaminants, nutrients, and gain an understanding of the fate of constituents of interest throughout SAMBR.

3.1 Materials and Methods

3.1.1 Sample Collection

Samples were drawn using syringes directly from the testing apparatus and were stored in 50- and 15-mL conical polypropylene tubes at 4°C. All samples were analyzed approximately within one week after being taken. Sample containers were soaked in a soap and disinfecting bath for at least one hour, rinsed with tap water, rinsed with distilled water, and dried prior to sample collection. Most analyses were performed in duplicate with triplicate readings, yielding six readings per sample unless indicated otherwise.

3.1.2 Sample Preparation and Storage

Samples for solute analysis were subjected to centrifugation at approximately 3660 RPM for at least 15 minutes. After centrifugation, the supernatant was pipetted into a new container. Samples for solute analysis were subsequently brought to ambient temperature for analysis if previously stored at 4°C. Samples were diluted with deionized (DI) water to within range of the corresponding parameter as needed.

3.1.3 Bioreactor Inoculation

Waste Activated Sludge (WAS) was collected from the South Cross Bayou Advanced Water Reclamation Facility in St. Petersburg, FL. The sludge was sieved to remove any debris and macro-particulates. Based on initial solids analysis the sludge was diluted with water to achieve a solids content of approximately 1%. The SAMBR reactor was filled with ~110 L of sludge including added packing media. At this stage the reactors are inoculated and prepared for operation.

3.1.4 Urine Collection

Human urine was collected from consenting donors in specimen containers without any identifying markings or information. These specimen containers were anonymously placed in a designated collection site. The donations were combined, at random, into 5 L carboys to create a urine blend that was impossible parse out any identifying information regarding the donors. As the analyses conducted on any urine samples were solely related to water quality coupled with the anonymous donation and preparation methods, all data was completely de-identified with regards to donors. Therefore, Institutional Review Board (IRB) approval of this research is not necessary, nor does it violate any of the following: U.S. Department of Health and Human

Services (DHHS), Food and Drug Administration (FDA), Health Insurance Portability and Accountability Act (HIPAA), and USF's Human Research Protection Program (HRPP).

3.1.5 Chemical Oxygen Demand

Chemical Oxygen Demand (COD) is defined as the amount of oxygen necessary for the complete oxidation of carbon in organic matter and is commonly used to characterize wastewater strength. The COD measured for these samples examined the dissolved or soluble fraction of COD. COD analyses was conducted in accordance with Hach Method 8000 utilizing Hach COD HR dichromate digestion kits. A 2 mL sample was pipetted into the digestion vial with 2 mL of DI water pipetted into a separate digestion vial to serve as a blank calibration. The digested vial was inverted several times to mix the contents and heated in a COD reactor for two hours at 150°C. After the reaction time, the samples were cooled to room temperature, wiped clean, and concentration measured using a Hach DR/4000U spectrophotometer.

3.1.6 Total Nitrogen

Total Nitrogen (TN) analyses measured the total concentration of nitrogen species (organic N, ammonia, nitrate, etc.) present in the sample. Soluble TN analyses were conducted in accordance with Hach Method 10072 and utilized Hach HR Total Nitrogen kits. A reactive agent powder pillow of persulfate was added to the vials. A 0.5 mL sample was pipetted into the vial with 0.5 mL of DI water pipetted into a separate digestion vial to serve as a blank calibration. The vials were inverted several times to mix the contents and heated in a COD reactor for 30 minutes at 150°C. After the reaction time, the samples were cooled to room temperature. The contents of a reagent A powder pillow were added to each vial, inverted to mix, and let sit to reactor for three minutes. This process was repeated with a reagent B powder pillow and let react for two minutes. 2 mL of this digested vial was pipetted into a reagent C vial, inverted to mix,

and let sit to react for five minutes. Lastly, the vials were wiped clean and concentration measured using a Hach DR/4000U spectrophotometer. Only soluble analyses were conducted as total fractions are a source of interference and would have resulted in unreliable results.

3.1.7 Ammonia Nitrogen

Soluble ammonia nitrogen (AN) analyses were conducted in accordance with Hach Method 10031 and utilized Hach HR Ammonia Nitrogen kits. A 0.1 mL sample was pipetted into the vial with 0.1 mL of DI water pipetted into a separate digestion vial to serve as a blank calibration. Two reactive agent powder pillows of ammonia salicylate and cyanurate were added to each vial and inverted several times to mix the contents and set out to react for 20 minutes. After the reaction time, the sample was wiped clean and measured transmittance and correlated to a concentration using a Hach DR/4000U spectrophotometer. Only soluble analyses were conducted as total fractions are a source of interference and would have resulted in unreliable results.

3.1.8 Nitrate Nitrogen

Nitrate analyses measure the total nitrate concentration present in samples. This was conducted in accordance with Hach Method 10020 using Hach Nitraver X High Range test kits. A 1 mL sample was pipetted into the vial and then inverted 10 times to mix. The vial was wiped clean and used to serve as a blank calibration. The contents of a reagent B powder pillow were added to each vial, inverted to mix, and let sit to react for five minutes. Lastly, the vials were wiped clean, and concentration measured using a Hach DR/4000U spectrophotometer.

3.1.9 pH

pH analyses were conducted using ion selective probes. Prior to and after analysis, the probes were rinsed thoroughly with DI water and dried. The pH reading was taken during a 5–

10-minute window to allow for stabilization and when little drift was observed. Calibration of the pH probe was checked on a regular basis and recalibrated if necessary. When not in use, the probe was stored in a pH probe storage solution.

3.1.10 Oxidation Reduction Potential

Oxidation Reduction Potential (ORP) analyses were conducted using an electron sensitive probe. Prior to and after analysis, the probes were rinsed thoroughly with DI water and dried. The ORP reading was taken during a 15–20-minute window to allow for stabilization and when little drift was observed. Only single samples conducted for analysis and singular readings taken due to the inherent nature of method for this analysis. Calibration of the probe was checked on a regular basis and recalibrated if necessary. When not in use the probe was stored in an ORP probe storage solution.

3.1.11 Dissolved Oxygen

Dissolved oxygen analyses were conducted using an oxygen sensitive probe. Prior to and after analysis, the probes were rinsed thoroughly with DI water and dried. The reading was taken during a 5–10-minute window to allow for stabilization and when little drift was observed. Only single samples conducted for analysis, and singular readings taken due to the inherent nature of this method. The probes used came with a factory calibration and was checked on a regular basis and recalibrated if necessary. When not in use, the probe was stored in a DO probe storage solution.

3.1.12 Electrical Conductivity

Electrical conductivity analyses were conducted using an ion sensitive probe. Prior to and after analysis, the probes were rinsed thoroughly with DI water and dried. The electrical conductivity reading was taken after a few minutes to allow for stabilization and when little drift

was observed. Only single samples conducted for analysis and singular readings taken due to the inherent nature of method for this analysis. The probes used came with a factory calibration and was checked on a regular basis and recalibrated if necessary. When not in use, the probe was stored dry as instructed by the manufacturer.

3.1.13 Turbidity

Turbidity analyses were conducted in accordance with Hach Method 8237. A compatible cuvette cell was filled with the sample. The cell was wiped clean and measured turbidity using a Hach 2100Q spectrophotometer. Calibration of the spectrophotometer was checked on a regular basis and recalibrated if necessary. Effluent turbidity was the only sample measured as the rest of the samples were beyond the range of the spectrophotometer.

3.1.14 Transmembrane Pressure

To measure pressure differentials within the membranes used for this investigation, in line pressure transducers measured the pressure at the influent into the membrane, the concentrate that was recycled back, and the effluent that filtered through the membrane. Each transducer was individually calibrated, using a PVC tube connected to pressure gauge. A syringe was used to change the pressure and the corresponding voltage reading of the transducer at five different pressures. These five different readings were used to create a linear calibration used to calculate pressure. The pressure readings were recorded using a HOBO U30 automated data logging system. The TMP or the pressure difference between the two sides of the membrane that drives the filtration process is calculated by Eqn. 3.1.

$$\text{Equation 3.1 } TMP = \frac{P_c + P_c}{2} - P_p$$

where:

TMP = transmembrane pressure (bar)

P_f = feed pressure (bar)

P_c = concentrate (retentate) pressure (bar)

P_p = permeate (effluent) pressure (bar)

The flux of a membrane is defined as the volumetric flowrate of liquid that passes through the membrane per the surface area of the membrane (Eqn. 3.2),

$$\text{Equation 3.2 } J = \frac{Q_p}{A_m}$$

where:

J = membrane flux ($L/m^2/hr.$, LMH)

Q_p = volumetric flowrate of permeate membrane effluent (L/hr.)

A_m = total membrane surface area (m^2)

The specific flux (permeability) which is used to determine the integrity and effectiveness of the constructed membrane module is define in Eqn. 3.3

$$\text{Equation 3.3 } J_s = \frac{J}{TMP}$$

where:

J_s = specific membrane flux ($L/m^2/hr./bar$, LMH/bar)

3.2 Design and Fabrication

3.2.1 Process Design

One of the main treatment objectives for SAMBR was to convert and remove the high levels of nitrogen inherently present in urine. A suspended activated sludge system was

developed that was informed primarily from MLE and 4-stage Bardenpho processes. Fig. 3.1 shows the process diagram of SAMBR. The influent (In) was pumped into an initial anoxic zone, termed pre-anoxic (Pr-Ax) where the influent was mixed with a nitrate rich stream from the downstream recirculation. Aeration delivered via fine diffusion stones provided oxygen to create an oxic zone (Ox) and encouraged nitrification of ammonia. The oxic zone was kept to about half the size of the anoxic zones which allowed for the oxygen to expire sufficiently and created a post anoxic zone (Po-Ax) where an IR pump recirculated a portion of the stream at a rate approximately four times the influent flowrate. The IR stream returned nitrate back to the pre-anoxic zone for further denitrification. A high flow pump fed the treated effluent into the UF membrane with the rejected concentrate returned back to SAMBR. A pump connected to the permeate side of the membrane created a pressure differential to drive the filtration process. The permeate was collected into a product column for sampling before exiting as the final effluent. The gases produced were mainly nitrogen gas produced by denitrification, carbon dioxide, and excess oxygen which off-gassed into the atmosphere. Traditional activated sludge systems have a clarifier for solids removal and retention. The settled solids are removed as Waste Activated Solids (WAS) with a portion recycled back into the system. In a similar fashion, a solids wasting pump was installed near the end of the treatment process for removing accumulated solids, though it was never needed throughout the course of operation.

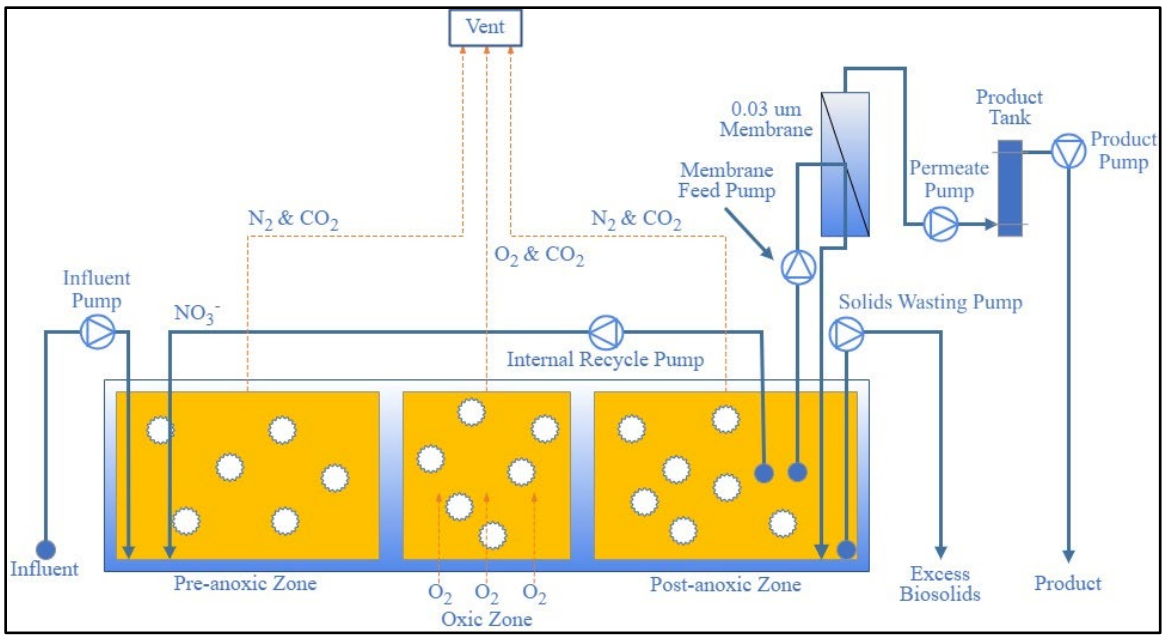


Figure 3.1 Diagram of designed SAMBR treatment process.

3.2.2 Hardware and Software Design

Design challenges primarily stemmed from optimizing reactor geometry for optimal performance and incorporating an automated control and monitoring scheme. SAMBR's design was developed over several iterations to optimize its application and operation in varying space environments, primarily EPB/PGH scenarios. The system's housing was based on NASA's EXPRESS rack and designed to be compact, easily accessible, and modular for easy integration for possible flight demonstration (*EXpedite the PRocessing of Experiments to Space Station (EXPRESS) Rack Payloads Interface Definition Document*, 2013). The EXPRESS rack system is the most current and appropriate as no future form factors have been officially announced, and as a NASA standard, any future form factor will likely be derivative of the EXPRESS rack to ease hardware transition. While intricately comprised of many fragments, the active volume of the EXPRESS rack is composed of two shelves and powered drawers each that can be configured with the powered drawers at the bottom like in Fig 5.2 or in between each of the system shelves.

For the allotted volume in an EXPRESS rack, a single rectangular baffled tank was chosen as it was deemed to be the most optimal for utilization of the geometry and volume (Fig. 3.3). Multiple slots allowed for easy reconfiguration of baffle spacing desired and further contributed to the system's modularity. The total active reactor volume ranged from 100-110L and was primarily sized on the desired retention time and the urine, flush, and graywater waste streams generated by a crew of four, an estimated 38 L/d (Muirhead et al., 2022a).

Liquid level sensors maintained a proper volume and overflow sensors in each quadrant paused the system when the liquid level became too high. Additional pumps were installed in a short loop to provide sampling and monitoring points of each zone in the system. An aeration pump was operated at approximately 10 LPM and ran to diffusion stones in the oxic zone. The membrane module consisted of multiple tubular (5.2 mm diameter), ultrafiltration, polyvinylidene fluoride (PVDF) membranes from Pentair with an average pore size of 0.03 μm and an approximate active surface area of 0.31 m^2. Clear PVC piping and fittings were used to construct the housing for the membranes. This clear PVC provided direct observation of the membrane and allowed the membrane to be monitored for any biofilm growth or anomalous changes.

To mitigate fouling of the membrane, it was subjected to physical relaxation and backwashing at regular intervals during operation. For relaxation the permeate pump was temporarily paused. During this time, the membrane feed continued to cycle through the membrane. The high-flow membrane feed pump created a strong cross flow velocity that extricated foulants caked onto the membrane. For backwashing, the permeate pump was reversed, which consequently reversed the flow through the membrane. The flow reversal ejects many of the particulates clogging the pores and restores active membrane surface area. These

methods were run for three minutes for relaxation and for 15 seconds for backwashing after 15 minutes of operation. Figure 3.4 shows how the membrane was operated during these events.

The ending result was a dynamic system that is easily modified to become compatible with the future ECLSS requirements (Fig. 3.5). As technology advances, the system can be easily modified and expanded upon to evolve its capabilities. Adaptation and safety were cornerstones that guided the approach when SAMBR was fabricated as the projected criteria currently used will inevitably vary as future missions evolve. A robust control and monitoring system ensured that nominal operational parameters were maintained and enacted several safeguards whenever anomalies arose to prioritize crew health and safety in long term habitation missions.

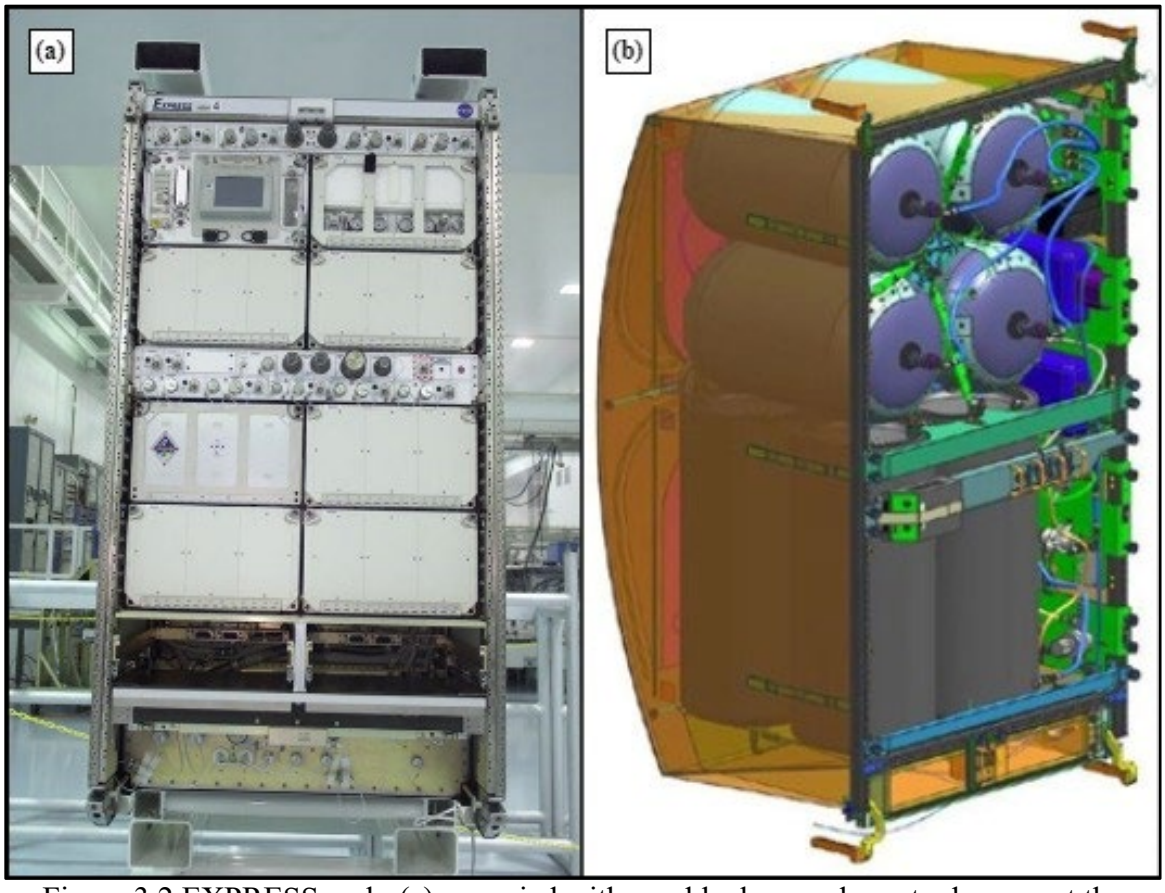


Figure 3.2 EXPRESS rack; (a) occupied with quad lockers and empty drawers at the bottom: (b) occupied with WRS components and empty drawers at the bottom. (Carter et al., 2018; Roy, 2001)

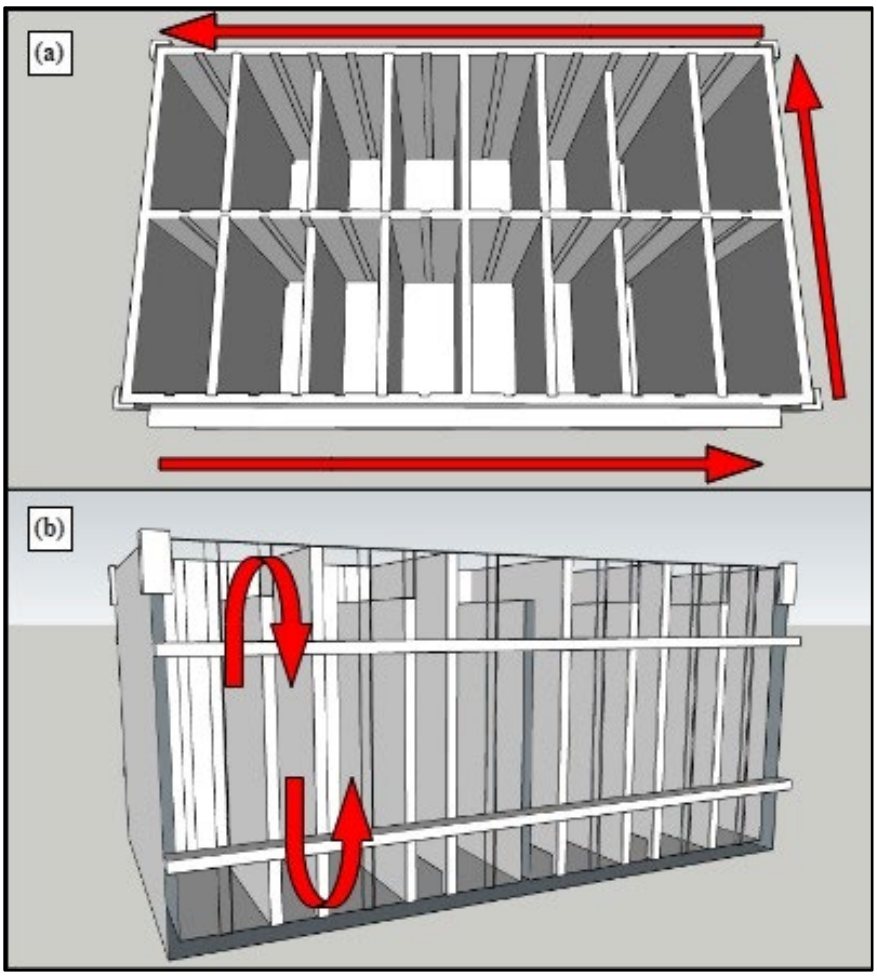


Figure 3.3 Baffled bioreactor design

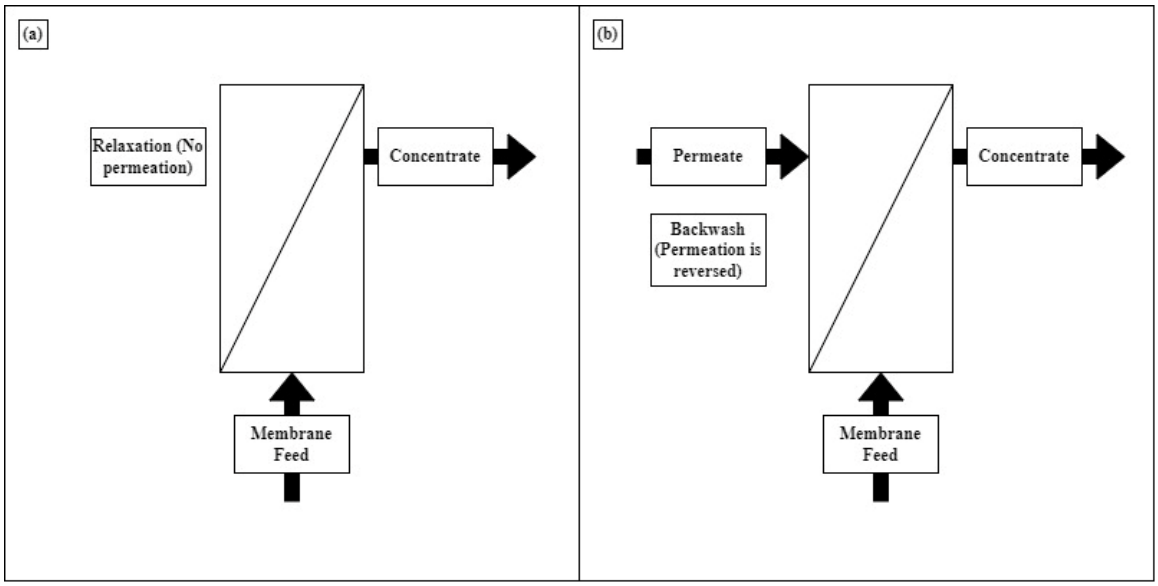


Figure 3.4 Membrane maintenance operations (a) relaxation (b) backwash.

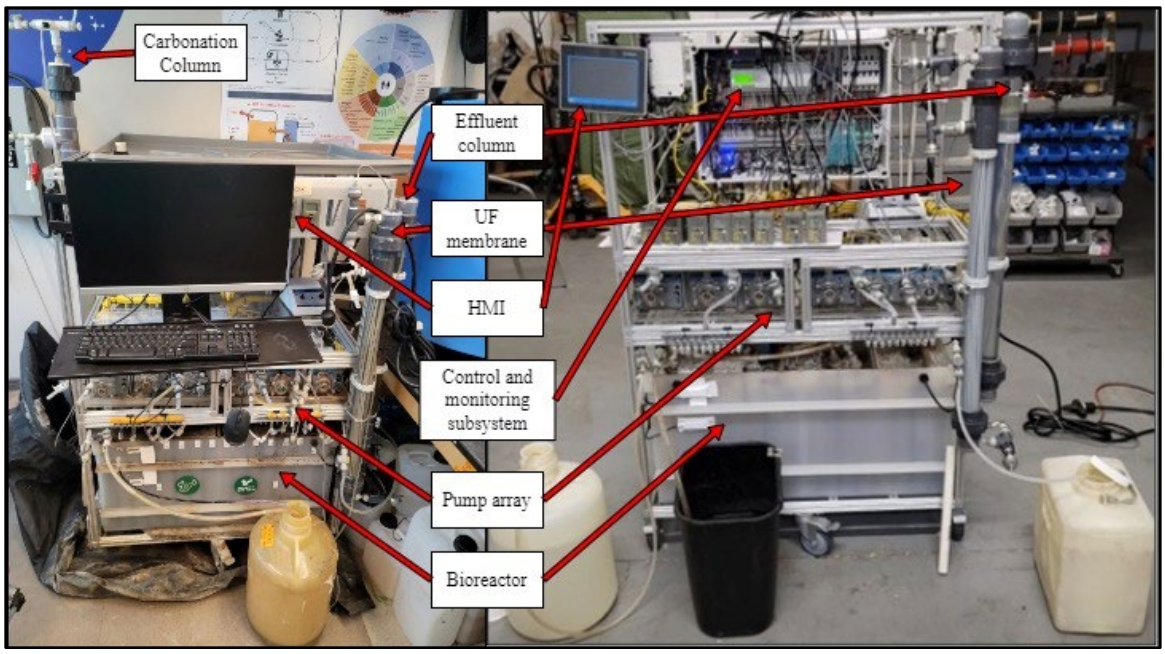


Figure 3.5 Assembled SAMBR system.

3.3 Experimental Plan

As previously stated, the core objective of SAMBR was to convert and remove the high levels of nitrogen found in urine. With an estimated urine and flush generation of 2 and 0.3 kg/CM-day respectively, the estimated nominal hydraulic loading for four crewmembers is 9.2 L/d (assuming ~1 kg/L density) (Muirhead et al., 2022b). However, to have immediately begun operating SAMBR with the high-nitrogen waste would have possibly resulted in complications stemming from Free Ammonia (FA) and Free Nitrous Acid (FNA) inhibition. If such inhibition had occurred early on, it might have resulted in perceived failure before SAMBR was fully evaluated. AOB and NOB, some of the primary treatment consortia of SAMBR, can be inhibited by FA at concentrations as low as 8 and 0.08 mg-N/L respectively(Pourbavarsad et al., 2022). For FNA, AOB and NOB can suffer inhibition at concentrations as low as 0.2 and 0.06 mg-N/L respectively(Pourbavarsad et al., 2022). However, there is a lack of study into the maximum tolerance of FA and FNA and at nitrogen levels exceeding 1000 mg-N/L and for scenarios such

as source separated urine (Pourbavarsad et al., 2022). Due to this knowledge gap, a phased approach, summarized in Table 3.1, was favored to allow for the consortia to acclimate increasing levels of nitrogen. The beginning of this approach verified the fundamental principles that were used to develop SAMBR, including design, operation, and treatment. As operation progressed through each stage, the biological consortia in SAMBR was challenged to tolerate increasing strengths of urine until full-strength urine and flush was achieved. Initially, a synthetic waste of ammonium bicarbonate (stage A-1) and acetate (stage A-2) was introduced at levels similar to that of domestic wastewater. These stages validated the performance and operation of the oxic zone and nitrification process. During the initial stages, the IR was not active, and samples of the anoxic zones were collected. Implementing the IR and engaging the anoxic zones followed in stage B and similarly validated the anoxic zones and denitrification process. Stage B was followed by replacing the synthetic influent with real urine and flush at increasing strengths of real urine until full strength was reached (stages C to H-2).

Inhibition from high levels of nitrogen and resulting high pH were known to be a possibility but not the severity. Evidence of inhibition was observed, discussed below, and led to a pivot in the research. Testing at full strength urine was integrated with two conditioning strategies: dosing with acetic acid (AA) dosing (stage H-1) and carbonation (stage H-2). The investigation, development, and evaluation of these strategies is discussed in Ch. 4.

Table 3.1 SAMBR operation schedule.

Phase	Preliminary	Evaluation		Experimental						
Stage	A-1	A-2	B	C	D	E	F	G	H-1	H-2
Base feed type	Synthetic: ammonium bicarbonate	Synthetic: ammonium bicarbonate + carbon source (acetate)		Urine (actual) and flush					Urine (actual) dosed with AA and flush	Urine (actual) and flush subjected to carbonation
Influent N (mg-N/L)	46	46	46	46	450	1700	3500	5000	7061	7061
Influent COD (mg/L)	0	70	70	70	693	2617	5388	7697	11685	10870
COD/N	0	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.60	0.65
N loading (g/d)	0.4	0.4	0.4	0.4	4	16	32	46	65	65
COD loading (g/d)	0	1	1	1	6	24	50	71	107	100
Hydraulic fraction of urine and flush as % of total influent	1%	1%	1%	1%	6%	24%	50%	71%	100%	100%
Q (L/d)	9.2	9.2	9.2	9.2	9.2	9.2	9.2	9.2	9.2	9.2
HRT (d)	12	12	12	12	12	12	12	12	12	12
Internal recirculation (IR)	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON
Duration of stage(d)	14	14	14	28	28	28	42	101	62	74
Cumulative duration (d)	14	28	42	70	98	126	168	269	331	405

3.4 Results, Discussion, and Critical Analysis

Throughout the course of the long-term operation of SAMBR, several unanticipated challenges occurred that are worth noting.

TMP was automatically recorded continuously, and the data extracted from the logging system. However, upon startup an IT error occurred that resulted in a lack of recorded data for TMP until late into stage C.

A few days into operation of stage G and the corresponding increase in urine strength, it was discovered that much of plumbing that delivered air to the diffusion stones in the oxic zone had disconnected from the air stones. Without the connection, air was delivered in much larger bubbles with less surface area. The plumbing was addressed partway into stage G and proper aeration was reestablished.

A 152-day stand-by operation was enacted between stages H-1 and H-2. During this period, the system was maintained but not fed influent or produced permeate through the UF membrane. Aeration was continuously provided to the oxic zone during this time in the same manner during active operation. This event provided a unique opportunity to assess SAMBR's ability to maintain an active consortia without a constant influx of substrate as might be necessary for space travel.

3.4.1 Chemical Oxygen Demand

The COD analysis (Fig. 3.6) across the operational stages displays SAMBR's ability to effectively remove organics present despite variations in the feed waste. A progressive reduction of COD was observed through each zone and across all stages of testing. The initial stage of A-1 did not include any added COD resulting in only already present organics in the initial inoculum to be captured in the analysis. The lack of initial added COD allowed for a baseline value of

present organics to be established and consequently allow for wash out of any remnant artifacts from the inoculum. Additionally, as the consortia was initially only provided nitrogen (as ammonium bicarbonate), this encouraged nitrification as nitrogen would need to be used as the electron donor in place of organic carbon. In each of the subsequent stages there is an increase in COD resulting from the increased strength of the feed, with the exception of stage H-2 which on average is lower than the H-2. The donated urine was stored in containers and allowed to hydrolyze so that the nitrogen would already be present as ammonia when fed into the system for nitrification (Liu et al., 2008). As the storage containers were reused, a variety of hydrolyzed urine blends was created and in turn created some biological activity as biofilms were observed in the containers during the later stages of testing. The inadvertent development of this biological activity is attributed to the large variation of COD in the influent and decreased presence of COD in H-2 relative to the prior stages. Despite the large variations in the feed, including in the same stages, SAMBR consistently reduced COD with removal ranging from 29% to 93% with an average removal of 75%. Even when challenged by the stand-by operation and high-strength stages, COD removal occurs across all treatment zones, indicating an active and robust consortium is present.

3.4.2 Nitrogen

TN measurements captured the presence of ammonia, nitrate, nitrite, and organic nitrogen components. As the urine in the influent was allowed to hydrolyze for considerable durations, much of the urea had been converted to ammonia. Though not explicitly discussed, nitrite was infrequently measured to check for accumulation but was always found to be at non-detectable levels. The remaining species, ammonia and nitrate, are believed to be the two predominantly present throughout the system and is largely supported by comparing TN levels (Fig. 3.7) to the

combined levels of AN (Fig. 3.8) and NN (Fig. 3.9). In most instances the AN and NN comprised a majority of the TN, leaving a relatively small fraction remaining to be unconverted organic nitrogen and even lesser nitrite. However, in the later stages of H-1 and H-2 the AN concentration is greater than the TN by several thousand mg-N/L on multiple occasions, an obvious impossibility. The reason for the observed anomaly is believed to be attributed to method interference. The method states that a large presence of nitrogen-free organic constituents can interfere with the accuracy of the method but does not specify a quantitative concentration at which interference begins or to what extent. The nature of this interference with the method would explain why this phenomenon was not observed in the earlier stages when nitrogen-free organic compounds were less concentrated. The AN of the influent reached the maximum observed concentration of 22,610 mg-N/L. The highest nitrogen concentration in urine was reported to be almost 14,000 mg-N/L during a review of literature. As the urine-based influent for SAMBR is slightly diluted to represent flush water, a value ~60% higher than the known literature is unexpected. Excess CO_2 and stripped gasses from the carbonation column were routed to the feed tank to provide additional, upstream conditioning and maximize utilization of the CO_2. A possible inadvertent effect was increased liquid evaporation which would result in concentration of the constituents in urine.

Similarly to the COD profile, a progressive reduction of TN was observed through each zone and across all stages of testing. The initial stages show an initial washout of legacy nitrogen present in the inoculum followed by utilization of the incoming nitrogen. The reduction of TN and AN combined with increasing NN concentrations verified the occurrence of nitrification and denitrification in the oxic and anoxic zones as designed. The switch to urine did not appear to have a significant impact as the occurrence of nitrification and denitrification continued to be

evident and continued with increasing urine strength up to 50% in stage F. Up to stage F, conversion of ammonia ranged from 85% to 99% with an average of 93%. Removal of TN ranged from 40% to 86% with an average of 64%.

The discovery and observed effect of the disconnected aeration plumbing coincides with the transition to stage G by a matter of days. The coincidence of these two events make it difficult to parse out the source of the observed effects in stage G. NN concentration rapidly decreased to single digit concentrations, indicating minimal nitrification activity. Once the aeration plumbing was restored, the system was operated nominally for the rest of stage G to see if nominal nitrification activity could be reestablished. During stage H-2, additional inoculum was added to the oxic zone on day 498 to aid in reinvigorating nitrification and is marked by the spike in nitrate. The recovery strategy only temporarily increased nitrification followed by a return to minimal activity.

Though nitrification activity did not return to optimal levels, the trend of nitrogen reduction continued into the later stages. Considering the low presence of nitrate, not all of the subsequent disappearance of the nitrogen could be attributed to biological treatment. The universal pH rise (Fig. 3.10) in SAMBR that started in stage G led to a shift in the ammonia-nitrogen species. With a pKa of ~9.25 for the ammonium/ammonia species and the system pH reaching as high as 9.6 (Fig. 3.10), a majority of the AN species was now present as ammonia, NH_4^+, instead of ammonium, NH_3. The shift in ammonium/ammonia gave the ammonia an increased propensity to volatilize out of solution. The volatilization is believed to have contributed to the removal of nitrogen without the corresponding increase in NN, associated with nitrification, observed in these late stages.

Despite the diminishment of nitrification, biological treatment is still evident by the continued reduction of COD. Regardless of the variations of biological treatment the membrane provides an absolute barrier and constant treatment, retaining biomass and rejecting particulate constituents.

3.4.3 pH

The pH remained relatively constant in the initial stages, varying from 7.5-8.3 for stages A-1 to C, a nominal range for activated sludge (Baldwin & Campbell, 2001). When nitrification was strongly active, the pH in the oxic zone and those downstream of it became more neutral and even mildly acidic in some instances, reaching a pH as low as 5.5. As hydrogen ions are released during the nitrification process, a decrease can be observed as was in these stages of SAMBR. The continued release of hydrogen ions likely led to the observed deviation where In and Pr-Ax remained neutral to mildly basic while Ox, Po-Ax, and Perm. trended down. The coinciding transition to stage G and aeration event resulted in rise in pH of the system to approximately 9.5. The pH management strategies implemented in stages H-1 and H-2 proved to be effective, gradually returning the pH to a more neutral level and creating a more ideal environment for the consortia.

3.4.4 Dissolved Oxygen

High levels of DO (Fig. 3.11) were present across the entire system at initiation, nearing saturation (assuming STP for water). Between collection and inoculation, the inoculum was stored and aerated in carboys, creating the initially high DO. The air pump continuously provided aeration at 10 LPM, maintaining the high DO. As nitrification can occur at DO levels less than 1 mg/L (Stenstrom & Poduska, 1980), the aeration was systemically adjusted until optimal aeration was achieved such that the moderate levels were only measured in the oxic

zone. Optimizing air delivery reduced excessive aeration and power consumption. The aeration was modified by adjusting the valves of air manifold, downstream of the air pump, which connected to each diffusion stone. The optimal aeration was achieved in stage D and delivered enough oxygen to maintain aerobic conditions but significantly expired by the time it reached the anoxic zones.

Despite the high levels of DO measured in the anoxic zones during the initial stages, nitrogen removal was consistently observed, indicating denitrification was still occurring. Though the measured DO was high, it is unlikely to be uniform with depth. Without the turbulent aeration provided in the oxic zone, the consortia stratified and was predominantly present near the bottom where DO level were likely to be lower. This stratification allowed them to carry out denitrification in the apparent presence of higher DO levels.

3.4.5 Modeling

BioWin, a comprehensive modeling software tailored to wastewater treatment design, is commonly used to design and optimize treatment processes ranging from industrial treatment plants to MBRs. The software possesses a relatively simple interface while maintaining a comprehensive modeling database capable of modeling numerous treatment applications and scenarios. Because of its simplicity and wide range of applications, BioWin was chosen to model SAMBR treating urine with flush. Modeling SAMBR with urine and flush gave insight to the model's accuracy of SAMBR's custom design and operation.

Fig. 3.12 shows how SAMBR was represented in BioWin and the carbon and nitrogen mass balance treating urine and flush. As BioWin has a built-in MBR system, it was used to represent both the Po-Ax zone and UF membrane. By default, the MBR has the input, permeate, and concentrate, without the ability to add more. As the solids wasting and IR streams also

originate in the Po-Ax zone it was decided to split the concentrate coming from the membrane to represent these streams as well. The IR stream is represented in red and solids wasting in brown. Table 3.2 summarizes the influent characteristics and was developed from literature and SAMBR's experimental data with the remaining settings and inputs left at default (Ray et al., 2018; Rose et al., 2015).

The resulting model and balance was found to be relatable to an extent. When treating urine and flush, the resulting COD and TN removal were 95% and 52% respectively. The model more closely resembled SAMBR during stage F in terms of COD measuring 95% removal of COD compared to SAMBR's actual average of 93% but was an overestimate compared to an average of 72% removal in stage H-2. For TN the model underestimated SAMBR's removal, citing 52% compared to 76% (stage F) and 25-96% (stage H-2).

While the model was able to accurately predict COD removal, further refinement would benefit its validity for nitrogen treatment. As SAMBR was acclimated to tolerate a high degree of nitrogen, inhibition calculations likely activated at lower levels than what SAMBR was actually able to tolerate. The experimental COD and nitrogen loading approximately ranged from 14 – 63 mg/L and 29 – 127 mg-N/L respectively while the model assumes a constant loading rate. As the generation rates of urine and hygiene are dynamic, setting the model to account for this non-steady state would more accurately replicate the variable loading. Additionally, the model places a significant portion of the treatment burden on the membrane which is not reflected in SAMBR to the same degree. The default assumptions regarding the particulate and soluble fractions could explain the deviation. Other default settings and assumptions could contribute and would likely need to be tailored to SAMBR. For to achieve that level of customization, detailed microbial studies to obtain specific growth rates, yields, and other

various kinetic parameters. This initial model developed for SAMBR provides a foundation to build off of for future refinement and increasing its validity.

3.4.6 Critical Analysis

SAMBR displayed considerable success when it was challenged to treat a urine dominated, high-strength influent, with Table 3.3 summarizing the treatment and performance through each stage. SAMBR was capable of removing COD fed into the system at all stages of testing, averaging 93% removal at full strength (Fig. 3.6). Prior to the events of stage G, a large fraction of the ammonia was nitrified with an average conversion of 91% (Fig. 3.8). Given that the pH (Fig. 3.10) was consistently below the 9.25 pKa of the ammonium/ammonia equilibrium, there is presumed to be minimal ammonia volatilization. Evidence of nitrification occurring was further supported by increasing levels of nitrate in the system, as high as ~250 mg-N/L were observed (Fig. 3.9). A majority of the nitrogen present in the effluent was predominantly nitrate, further supporting that minimal ammonia escapes untreated. TN measurements (Fig. 3.7) support evidence of N removal via denitrification with an average TN removal of 76%.

Following the integration of influent conditioning for full-strength urine, pH returned to acceptable levels. Decreases in total nitrogen and ammonia was still observed but due to the minimal nitrate levels, it is unclear to what degree of removal is attributed to treatment, volatilization, and other such losses. The coinciding events of transitioning to stage G and temporary aeration loss make it difficult to dissect the causality of the cascading effects.

The temporary lack of aeration could have shifted the oxic zone to anoxic conditions from which the nitrifying consortia was unable to re-dominate. The increase in ammonia likely contributed to the resulting increase in pH (>9), and jointly exceeded the tolerance threshold of the consortia, resulting in significant inhibition. If the diminishment of nitrification was solely

due to nitrogen inhibition, the highest N concentration the consortia was able to tolerate was ~3000mg-N/L, significantly above known tolerance studies of the past.

Supplemental discussion of ORP, EC, turbidity, and TMP data can be found in Appendix B.

3.4.6.1 Comparison of SAMBR to Compatible Work

The work of TTU was identified to have the most compatible research for comparison. TTU has conducted multiple studies regarding biological treatment systems, primarily based on MABR technology, for space applications. Some of the studies cover operation and evaluation of their systems over multiple years. Table 3.4 summarizes the comparisons of TTU's NDX (Jalalieh et al., 2021) and UMABR (Hooshyari et al., 2023; Jackson et al., 2022) systems against SAMBR during stage F and H-2.

The NDX system treated a wide variety of waste streams beyond just urine and flush, with no requirement for pH management. The lack of recycle flow(s), single stage treatment, and minimal aeration requirement make NDX's operation relatively simple and energy efficient. However, its small scale (sized for <1 CM), volumetric inefficiency, and comparatively lower N conversion to previous studies identify areas for future optimization.

Like SAMBR, the UMABR was primarily designed for urine treatment and also used real human urine in its evaluation. Taking the urine directly into the MABR eliminates intermediate storage tanks, reducing mass and volume considerations. Membrane delivery of air to the biofilm ensures efficient aeration and optimal energy consumption. The main goal of the UMABR was organic removal and nitrogen conversion, which then was sent to a distillation vessel in order to achieve any kind of nitrogen removal. The coupled distillation unit experienced nitrogen carry over and decomposition of HNO_2 leading to the addition of an acid trap between the two systems

to mitigate this. The lifespan of the acid trap has not been evaluated. Brine and solids produced during distillation are stored in the vessel as they accumulate.

The preliminary evaluation of SAMBR was focused to just urine + flush and has not yet incorporated other waste streams into its treatment scope. SAMBR's aeration and filtration operations comprise a majority of its energy consumption and have not yet been optimized. Though the projected life of the membrane is estimated to last multiple years (based on the experimental TMP data), it is one of the few consumables in the system. Due to the inevitable fouling spares and replacements would eventually be required. However, the membrane provides several advantages including retaining and concentrating biomass, decoupling SRT and HRT, and is typically a more compact system.

In terms of performance SAMBR, during stage F, had similar organic removal as both the NDX and UMABR systems but less so during stage H-2 due to aforementioned events. The DOC and COD measurements are strongly correlated and are both commonly used as measurements of available organic substrate, but a direct comparison would require testing both metrics in all three systems to develop a custom correlation. SAMBR, during stage F, had higher performance than both systems in terms of nitrogen treatment, achieving significantly higher conversion and removal but at a lower N concentration compared to the UMABR. Further evaluation would be needed for a confident comparison of nitrogen treatment between these systems and SAMBR during stage H-2 because, as stated before, it is unclear what degree the change in nitrogen is attributed to treatment due to minimal nitrate concentrations to support the presence nitrification occurring.

SAMBR's conversion and removal of nitrogen coupled with substantial organic removal when challenged with high-strength waste even in comparison to literature highlight its robust

ability to handle stress. With a compact footprint and minimal consumables, SAMBR can be easily configured to meet the mass and volume restrictions of space travel. Though further evaluation is necessary to primarily optimize energy consumption and nitrogen tolerance, SAMBR has proven to be a competitive technology for sustainable treatment of waste streams during long-habitation missions.

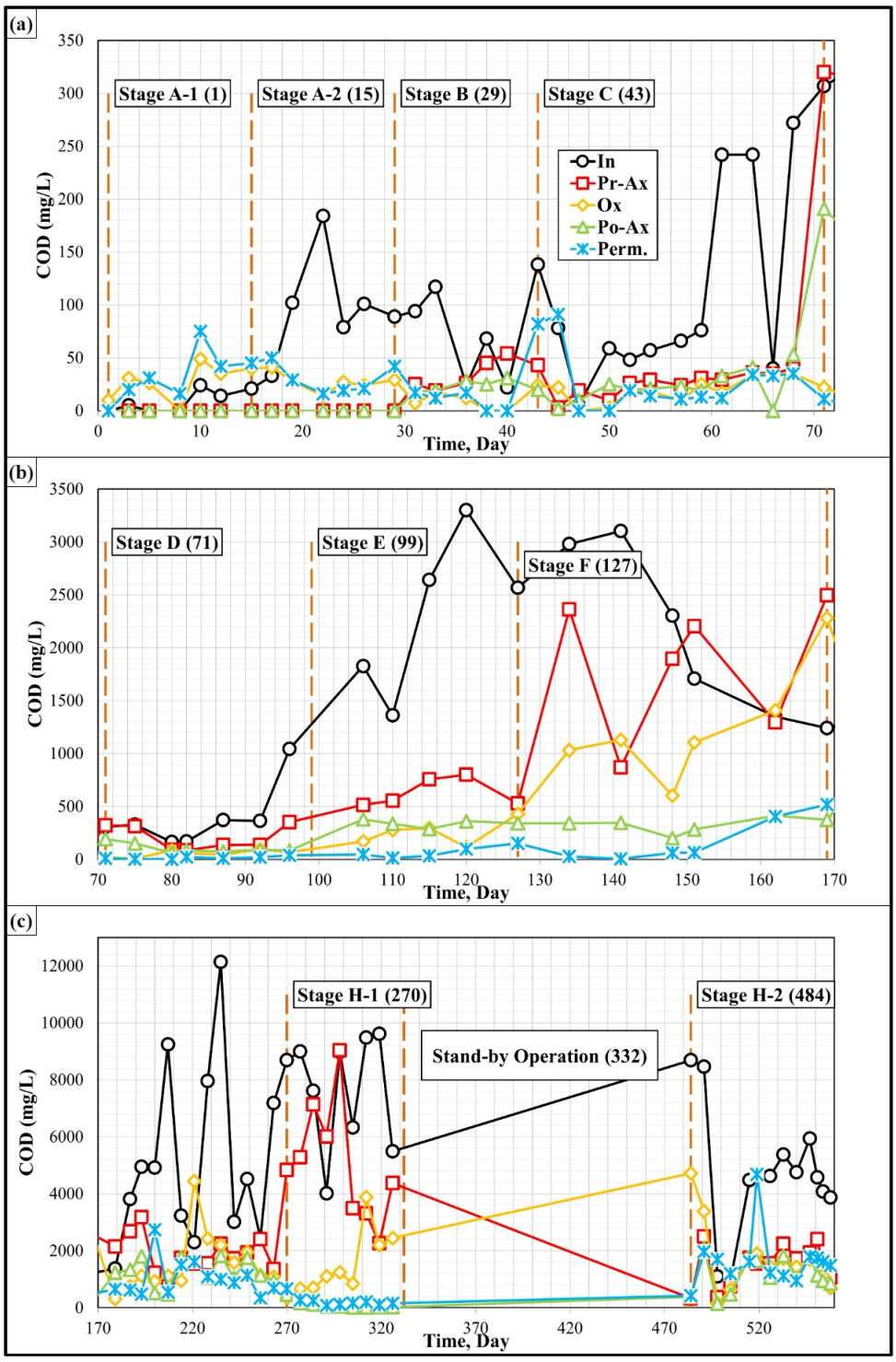


Figure 3.6 COD profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

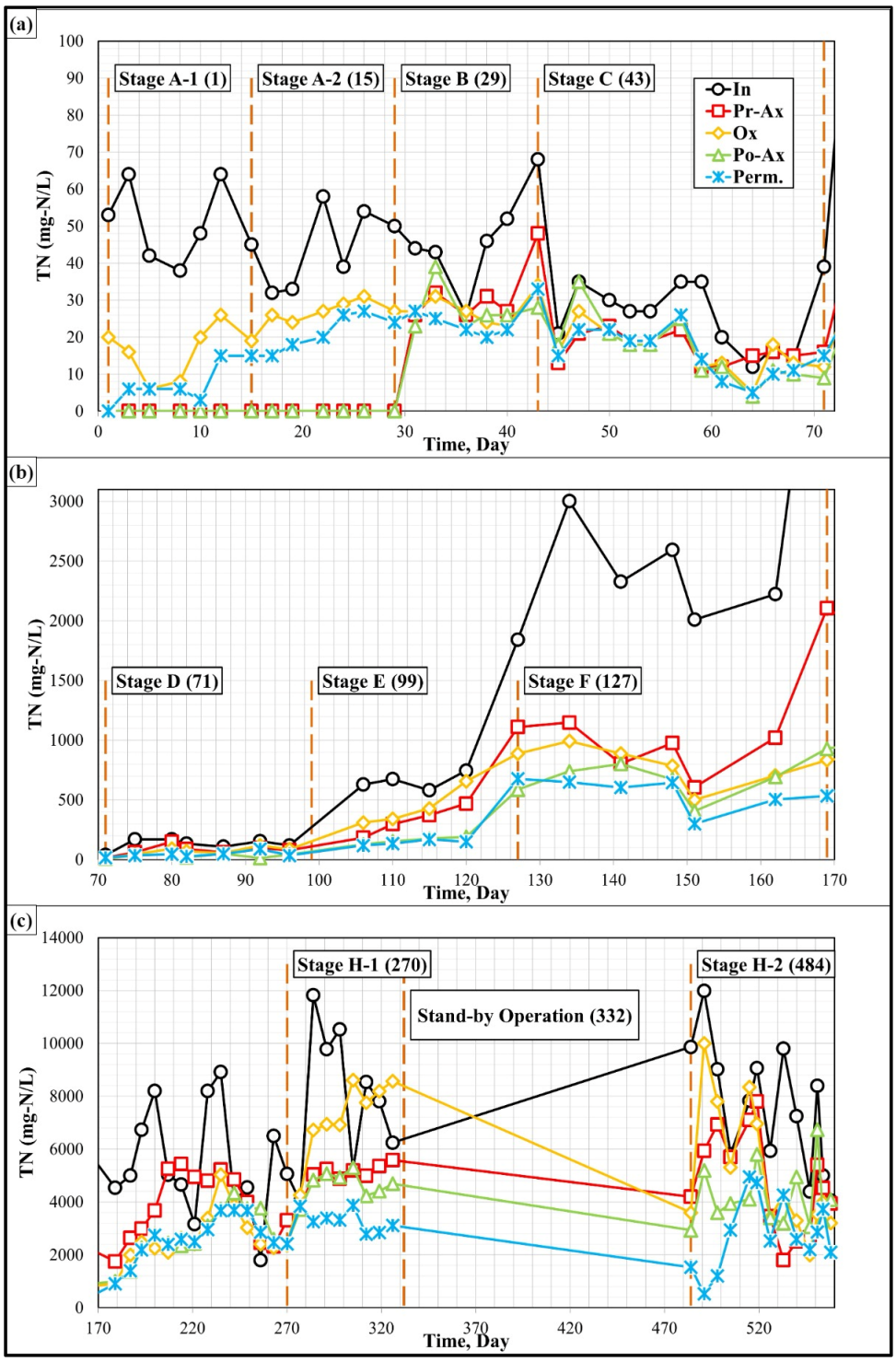


Figure 3.7 TN profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

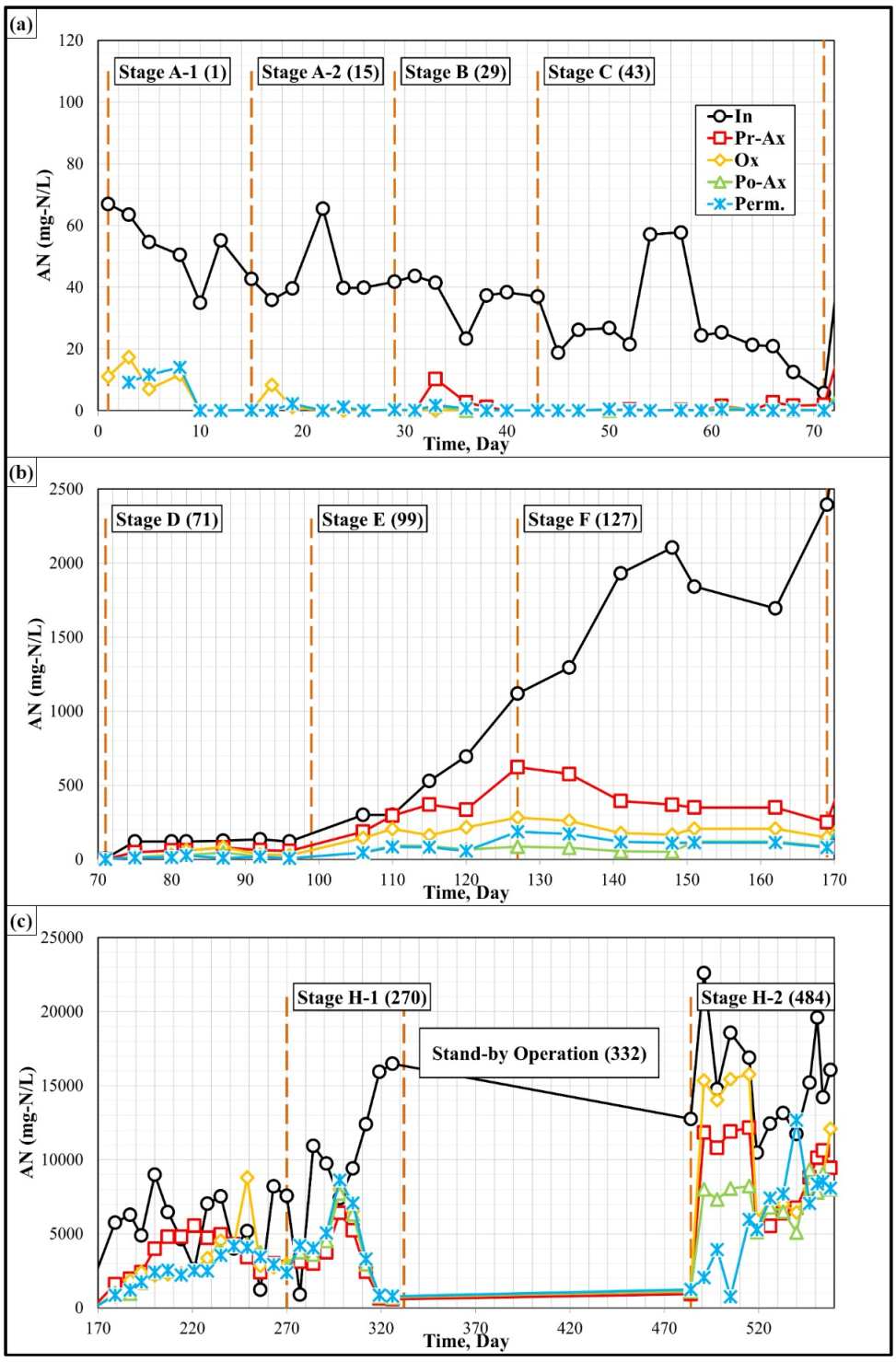


Figure 3.8 AN profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

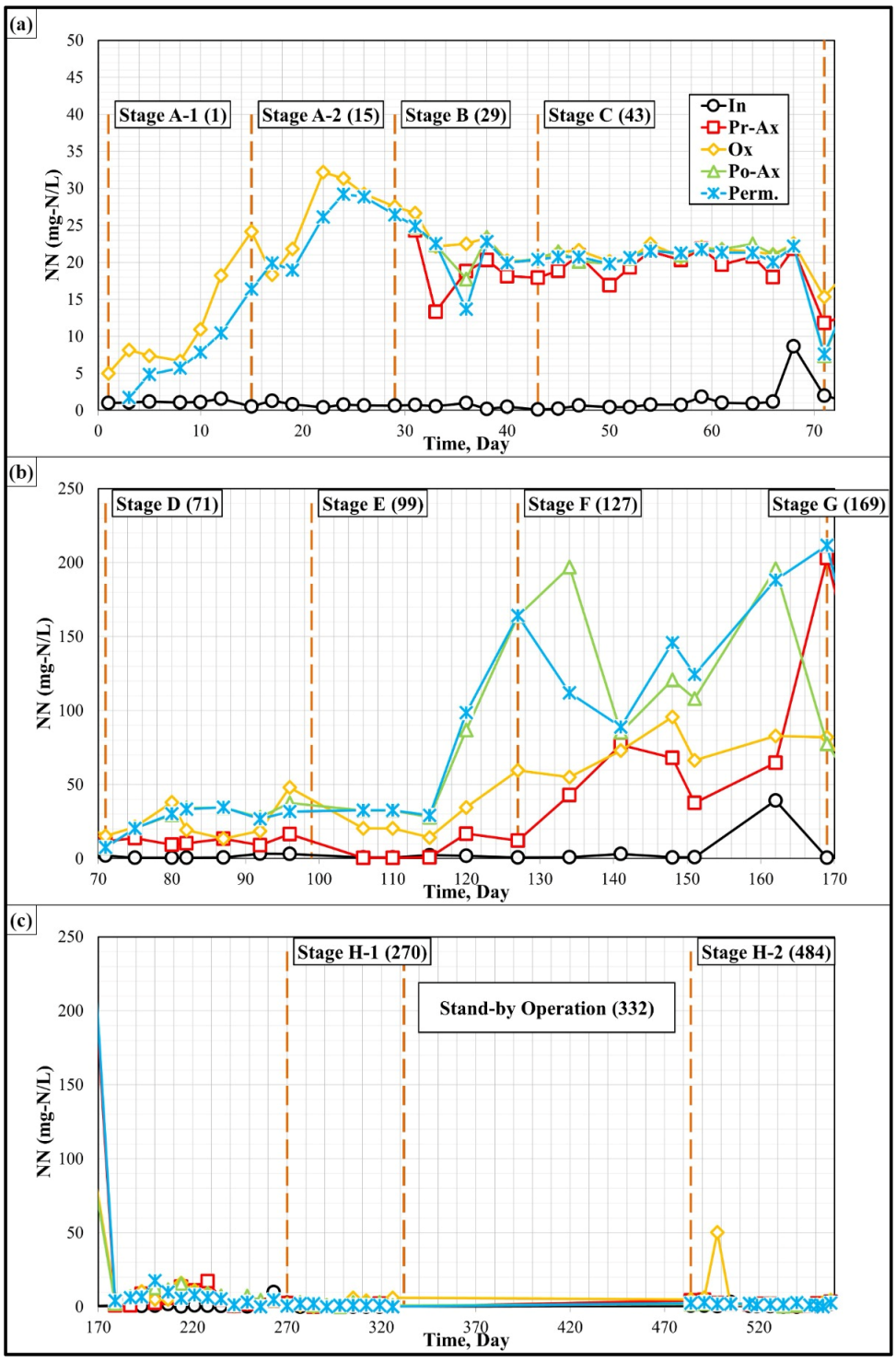


Figure 3.9 NN profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

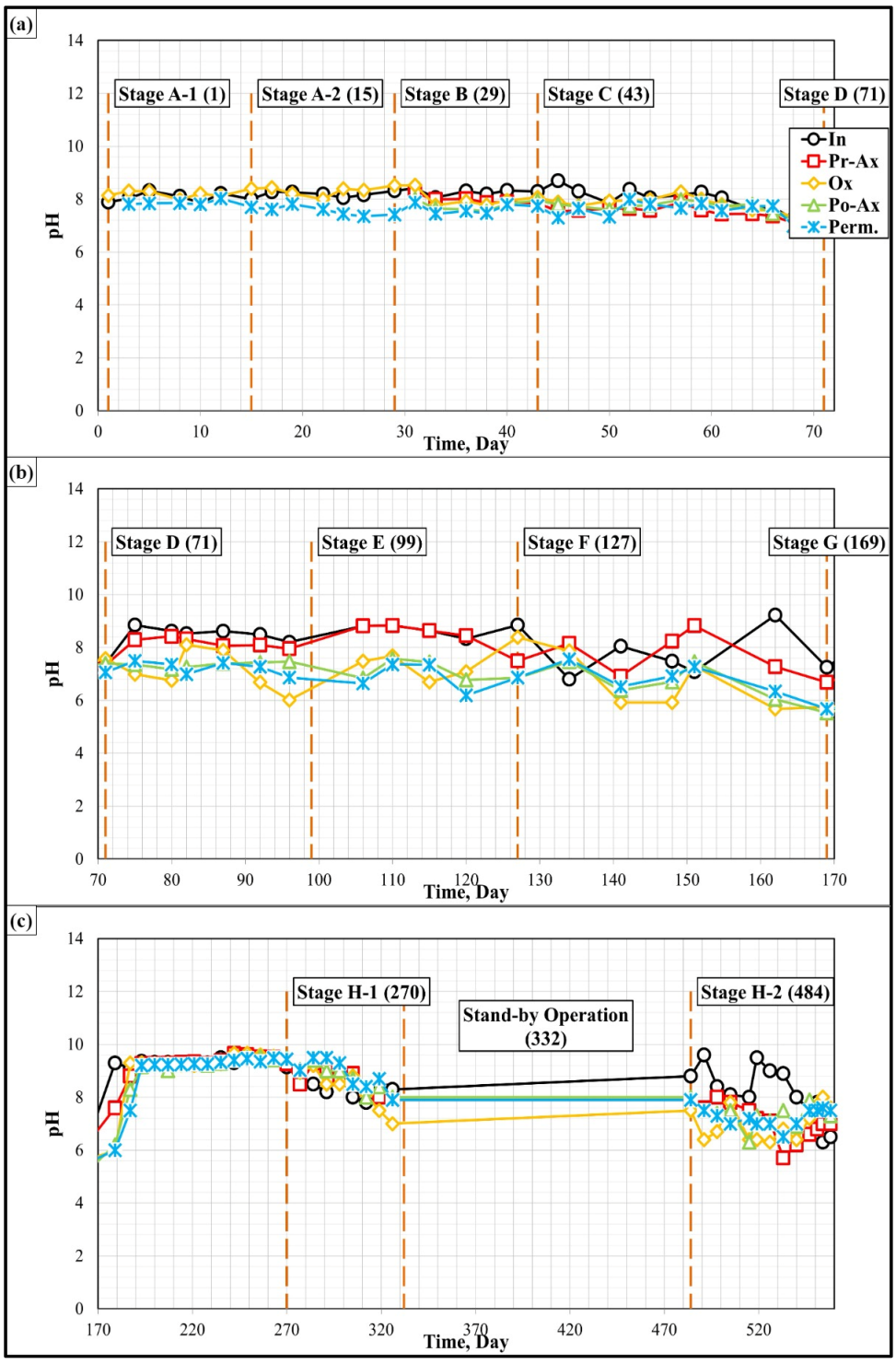


Figure 3.10 pH profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

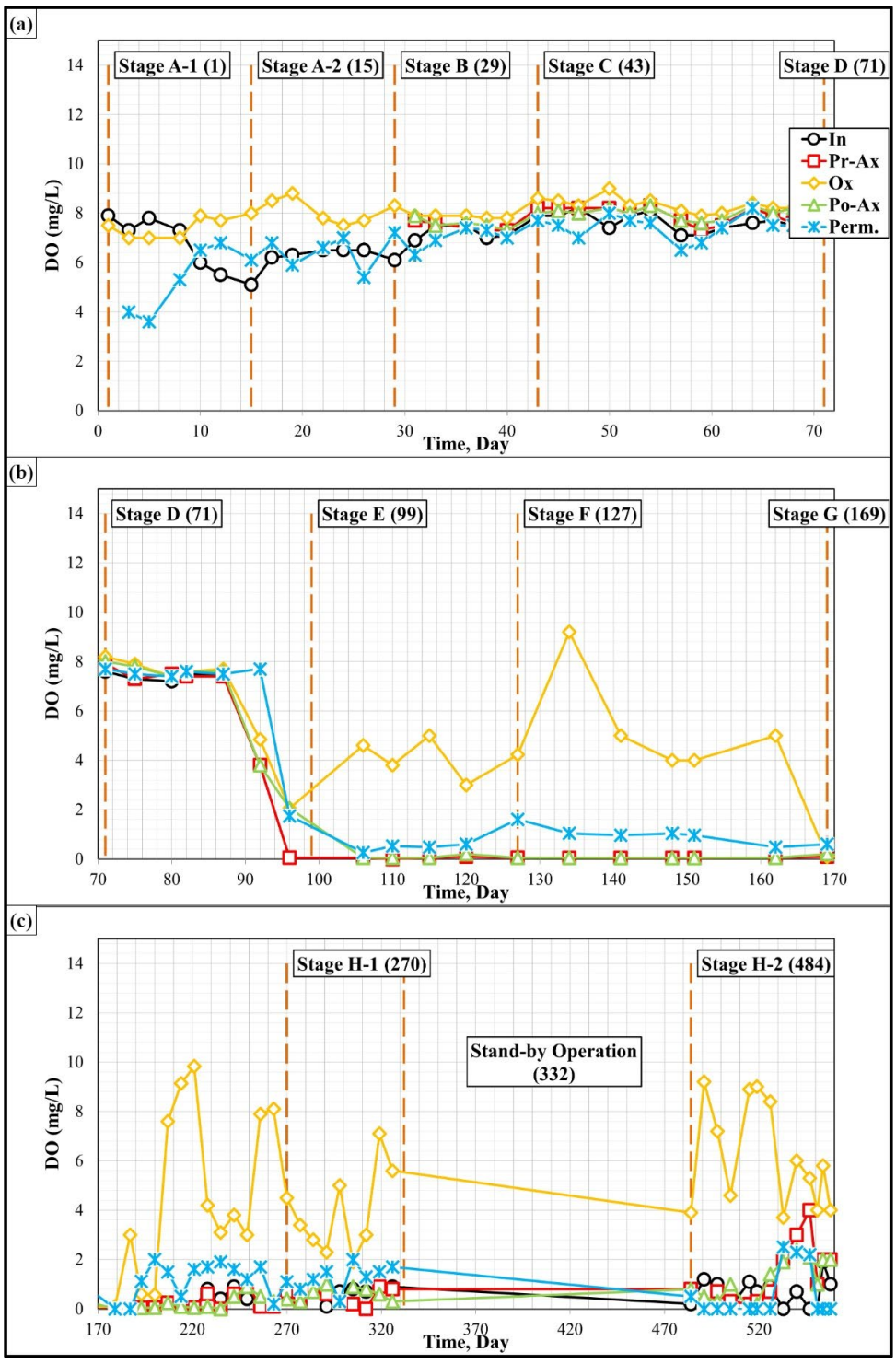


Figure 3.11 DO profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

Table 3.2 Summary of BioWin input data. (Ray et al., 2018; Rose et al., 2015)

Input Metric	U+F+H+S
Influent (L/d)	37.2
IR (L/d)	148.8
Solids wasting (L/d)	0.4
UF SA (m^2)	0.311
Aeration (LPM)	20
Membrane Feed (LPM)	2.4
COD (mg/L)	1692
TKN (mg-N/L)	3761
TP (mg-P/L)	258
pH	7.5
Alkalinity (mM CaCO3/L)	81

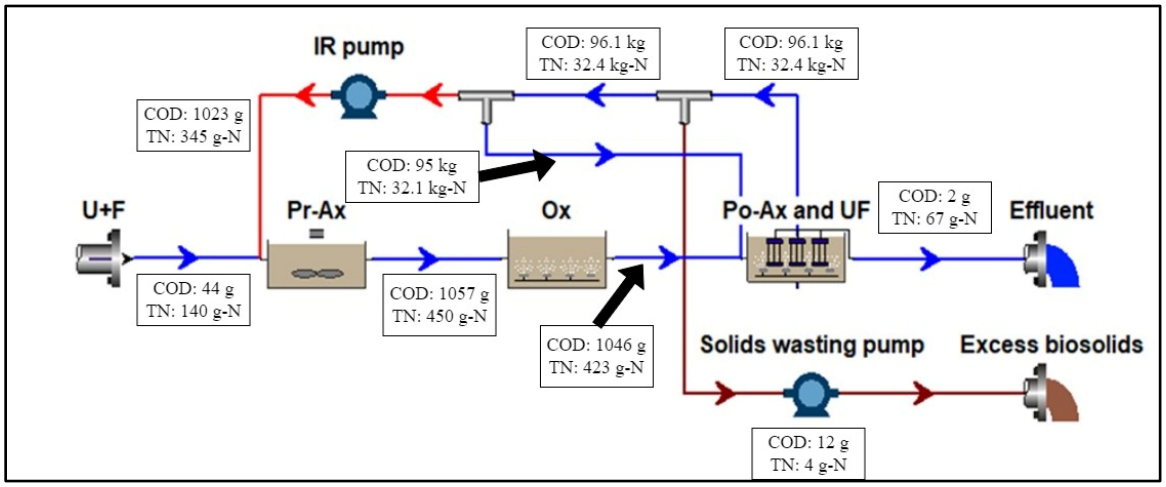


Figure 3.12 SAMBR mass balance treating U+F.

Table 3.3 SAMBR influent/effluent water quality averages.

Day, Operation	Stage	In.	Eff.	In.	Eff.	In.	Eff.	In.	Eff.	Eff.	In.	Eff.	TMP (bar)
		COD (mg/L)		TN (mg-N/L)		AN (mg-N/L)		NN (mg-N/L)		Turb. (NTU)	pH		
1	A-1	7	37	52	7	54	7	1	6	1.2	8.1	7.9	N/A
15	A-2	87	30	44	20	44	1	1	23	0.7	8.2	7.6	
29	B	69	15	44	23	38	ND	1	22	0.6	8.3	7.6	
43	C	110	29	28	17	29	ND	1	21	0.6	8.0	7.6	0.31
71	D	393	15	128	42	107	12	2	26	0.8	8.4	7.2	0.28
99	E	2283	47	658	143	456	67	1	48	2.0	8.7	6.9	0.27
127	F	2335	119	2334	563	1664	136	8	137	2.3	7.9	6.9	0.29
169	G	4881	986	5489	2470	5370	2445	1	21	146.8	9.2	8.7	0.31
270	H-1	7691	222	7667	3205	10084	4032	ND	1	12.4	8.5	8.9	0.38
332	H-2	4813	1668	7864	2841	15200	5916	1	2	4.3	8.3	7.3	0.48

*Influent value exceeded measurable range (>1000 NTU). Only effluent value is shown.

Table 3.4 Comparison of SAMBR to similar technologies.

Name	NDX (Jalalieh et al., 2021)	UMABR (Hooshyari et al., 2023; Jackson et al., 2022)	SAMBR Stage F	SAMBR Stage H-2
Base Technology	Biological system (unspecified)	MABR	MBR	MBR
Primary Function(s)	N conversion N removal Org. C removal	N conversion Org. C removal	N conversion N removal Org. C removal	N conversion N removal Org. C removal
Biological Mechanisms	Nitrification Denitrification Anammox	Nitrification	Nitrification Denitrification	Nitrification Denitrification
Design Scenario	EPB/PGH	EPB/PGH	EPB/PGH	EPB/PGH
Crew size	<1	2	4	4
Active Treatment Volume (L)	42	114	100-110	100-110
HRT (d)	21-53	25	11	11
Influent Medium	Urine + Flush + Humidity Condensate + Hygiene + Laundry	Urine + Flush	Urine (50%) + Flush	Urine + Flush
pH Management	No	Yes	No	Yes
Influent N (mg-N/L)	590-740	6000-7000*	1,120-3,000	10,480-22,610
N conversion	60-72%	50%	83-95% (Avg: 91%)	40-96%**
N removal	49-66%	N/A	63-85% (Avg: 76%)	25-96%**
Org. C removal	93% (DOC)	90% (DOC)	70-99% (COD) (Avg: 93%)	60-95% (COD) (Avg: 72%)

Table 3.4 (Continued)

Advantages	• High SRT • Efficient aeration • No recycle flows • Single stage system	• No intermediate storage • Efficient aeration • Supported by prior long-term studies	• Compact and volumetrically efficient • Absolute retention of suspended solids and pathogens • Decouple SRT and HRT • Process and control are easily modified • Resistant to shock loading
Limitations	• Volumetrically inefficient • Small scale • Lower N conversion compared to prior studies	• Coupled with static distillation to achieve N and ion removal • Phosphoric acid trap needed prevent N carryover and HNO_2 decomposition • N trapped in acid is wasted • Brine/solids accumulate/stored in distillation unit	• Aeration and filtration operations need to be optimized to minimize energy consumption • Membrane fouling • Max tolerance to N inhibition needs further investigation

*Range estimated from figure in publication as numerical data was not provided. **Includes possible volatilization

Chapter 4: Urine Conditioning

Due to the concentrated nature of the urine waste stream, particularly the high pH, nitrogen, and ion content, SAMBR's biological consortia may have experienced inhibition from optimal performance. In response to this potential effect, a subsystem capable of improving the condition of the urine and flush waste stream was investigated to be coupled with SAMBR.

Initially an automated acid addition process was considered. Dosing the waste stream prior to entering the bioreactor would reduce the increase in pH resulting from urea hydrolysis. AA was identified in literature as an effective acid dosing approach (Saetta et al., 2020). Preliminary tests indicated that a 5% AA solution was capable of reducing and maintaining the pH of urine from 8.8 to 7.0. However, the mass of pure AA required to continuously treat the urine generated by a crew of four for a 30-month mission was estimated to be almost 56 kg and therefore considered to be too great as it would diminish the sustainability of the system. Several other technologies were considered including HCl, zeolite, and electrodialysis. However, carbonation of the urine by bubbling carbon dioxide was selected for its generally non-hazardous properties, ubiquitous availability, and simplicity.

As hydrolyzed urine is characterized by a relatively higher pH (>9) than what typical biological treatment systems require (<8), the carbonation system should be able to reduce the pH of the urine to below eight. However, as carbon dioxide could off-gas in between the carbonation treatment and reaching the consortia present in SAMBR, this could result in a rebound of pH. As such the pH of the urine after carbonation should be considerably lower to

account for the potential pH rebound and could be achieved by maximizing mass transfer, contact time, and treatment time.

4.1 Materials and Methods

4.1.1 Sample Collection

Sample collection was conducted similar to the method described in section. 3.1.1.

4.1.2 Sample Preparation and Storage

Sample preparation was conducted similar to the method described in section. 3.1.2.

4.1.3 Chemical Oxygen Demand

COD measurements were conducted similar to the method described in section. 3.1.5.

4.1.4 Ammonia-Nitrogen

AN measurements were conducted similar to the method described in section. 3.1.7.

4.1.5 pH

pH measurements were conducted similar to the method described in section. 3.1.9.

4.1.6 Alkalinity

Alkalinity analyses were conducted in accordance with titration color change method 2320 from the Standard Methods for the Examination of Water and Wastewater. 50 mL of sample was measured out in small beaker with a stir bar on a magnetic stir plate set at a low speed. A burette was filled with 0.02N of sulfuric acid as the titrant. 0.2 mL of Bromocresol Green indicator was added to the sample and allowed to mix. The sulfuric acid was slowly titrated into the beaker of sample (while still being stirred continuously) until a color change is observed and the color did not change further. The alkalinity is then calculated by Eqn. 4.1.

$$\text{Equation 4.1 } Alk = (V_t \cdot N_t \cdot 50{,}000)/V_s$$

where:

Alk = Alkalinity as mg/L of $CaCO_3$

V_t = mL of acid titrated into sample

N_t = normality of acid titrated (0.02)

V_s = mL of sample (50)

4.1.7 Urine Collection

Urine collection was conducted similar to the method described in section. 3.1.4.

4.2 Benchtop Investigation

4.2.1 Experimental Design and Plan

A series of benchtop experiments were conducted to assess the feasibility of carbonation to condition the pH of the urine for SAMBR, with the experiments and setup summarized in Table 4.1 and Fig. 4.1, respectively. A cylindrical container was filled with 0.5 L of urine and a 15 mL sample was taken. Industrial-grade carbon dioxide was delivered to the urine through a fine diffusion air stone at the bottom of the container. The flowrate was controlled by a mechanical flowmeter. The urine was carbonated for one hour, after which the carbonation was stopped, and a second 15 mL sample was taken. The carbonated urine was left to rest for an additional 23 hours, for a total of 24 hours, and a third and final 15 mL sample taken. This experiment was conducted at three different volumetric flowrates of carbon dioxide: 1, 5, and 10 liters per minute (LPM). pH was the primary metric under consideration but COD, AN, and alkalinity were measured to analyze any other potential effects such as increasing alkalinity by carbonate species formation or soluble nitrogen reduction by ammonium bicarbonate precipitation.

As a control group, three containers of urine were setup: one that was left to sit without external inputs of any kind (stagnant), one that was stirred continuously with a magnetic stir bar and stir plate (agitated), and one that was subjected to aeration at a flowrate of 1 LPM using an identical fine diffusion air stone.

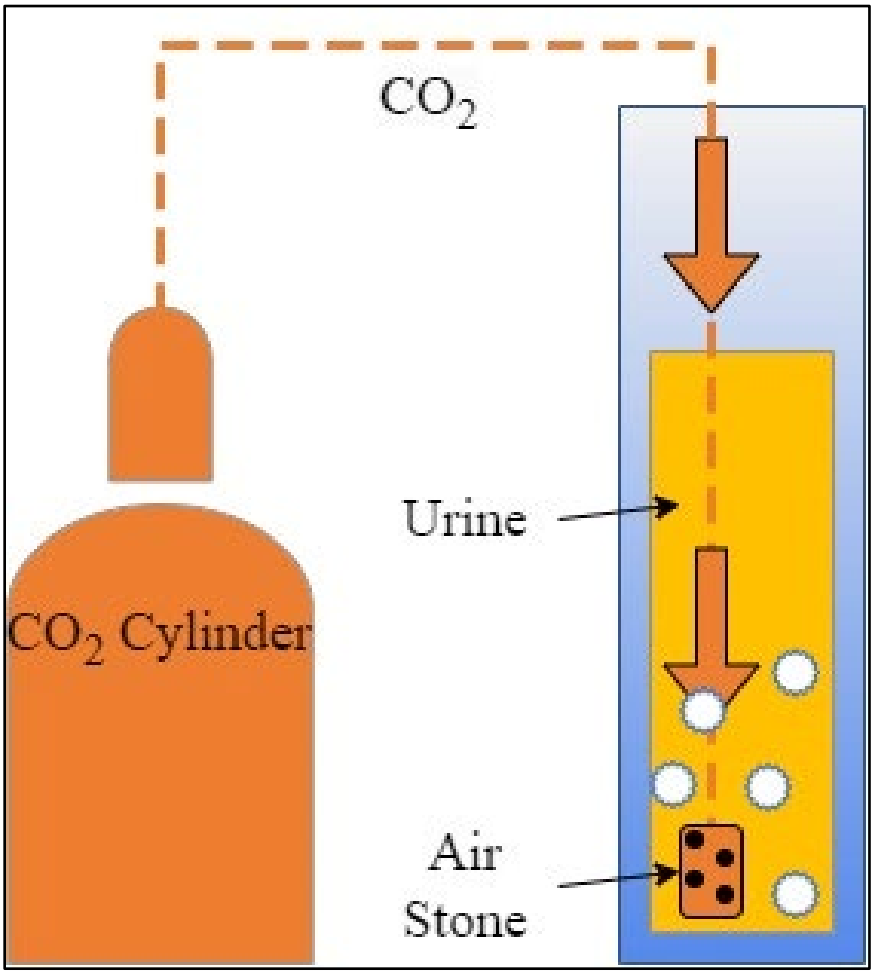


Figure 4.1 Benchtop urine carbonation setup.

Table 4.1 Urine carbonation experimental setup.

Time Sample Taken (hr.)
t = 0
t = 1
t = 24
CO_2 Delivery Rates Tested (LPM)
1
5
10
Controls
Stagnant
Agitated
Aerated

4.2.2 Results and Discussion

All three carbonation rates were capable of reducing the urine pH approximately from 9.5 to 7.5 which is in the optimal range for biological treatment (Fig. 4.2). The pH of all three of the controls were consistently around 9.5, supporting that the pH reduction is correlated to the introduction of carbon dioxide. However, at the 24-hour mark (23 hours after carbonation ceased) the pH of the carbonated samples had almost returned back to their original pH value. While the carbon dioxide appears to have dissolved, it seems to almost immediately begin to off-gas in a matter of hours. To better investigate the off-gassing behavior, the experiment was repeated at a 1 LPM carbonation rate with a continuous logging of pH and is displayed in Fig. 4.3. Immediately, after carbonation ceased, the pH began to rebound and appears to follow a logarithmic trend. At 1.9 and 13.1 hours after carbonation ceased the pH reached 8 and 9 respectively and eventually leveled off back to 9.5, indicating that all carbon dioxide had off gassed.

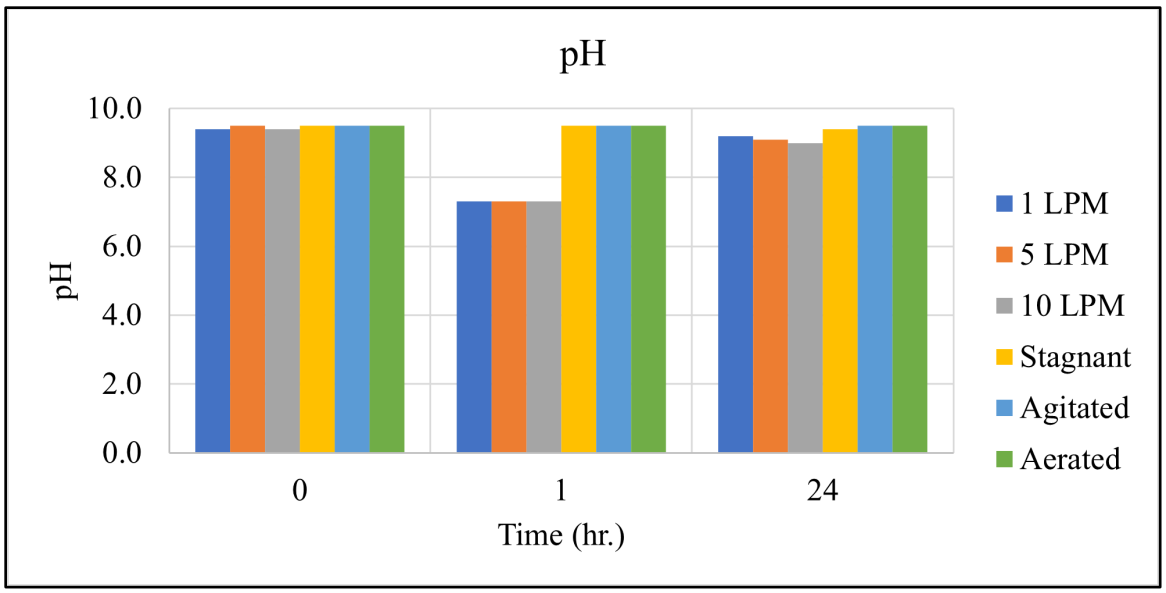


Figure 4.2 Benchtop urine carbonation pH profile.

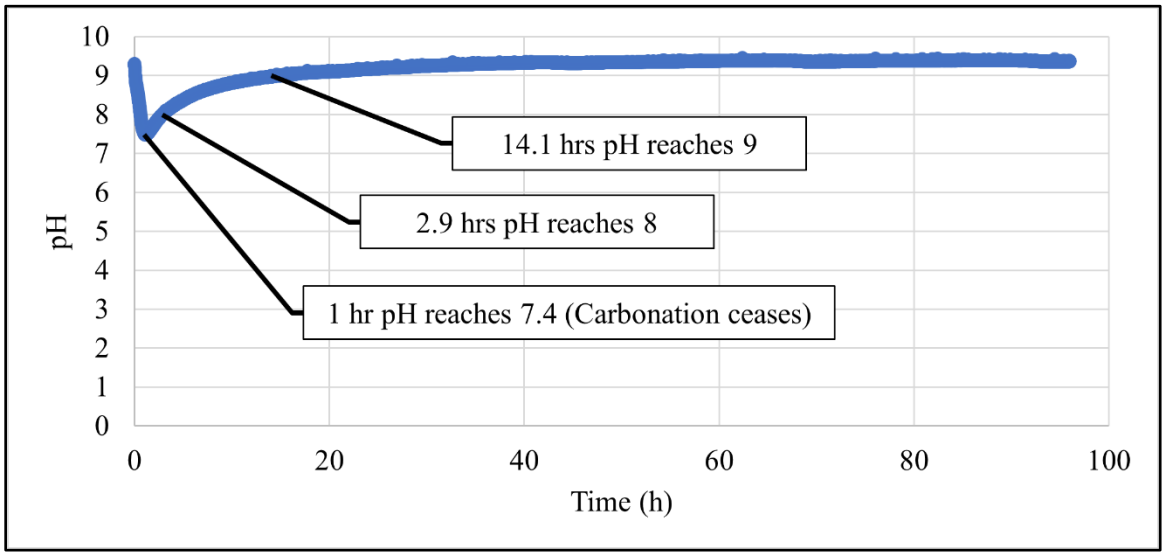


Figure 4.3 Benchtop urine carbonation detailed pH profile at 1 LPM.

Though there is some variation in COD of the samples throughout the experiment, the variation is minimal and unlikely to be significantly impacted by the carbon dioxide. There is the possibility of effects not captured by COD such as breaking down complex organics and would not be detected as a change in COD but would require deeper investigation. The COD values of the carbonated samples are significantly lower than the control or what would be expected of

hydrolyzed urine (~10,000 mg/L). As the urine is not sterile, the decrease in pH of the carbonated samples to more ideal levels possibly restimulated biological activity and degradation of COD compared to the control samples.

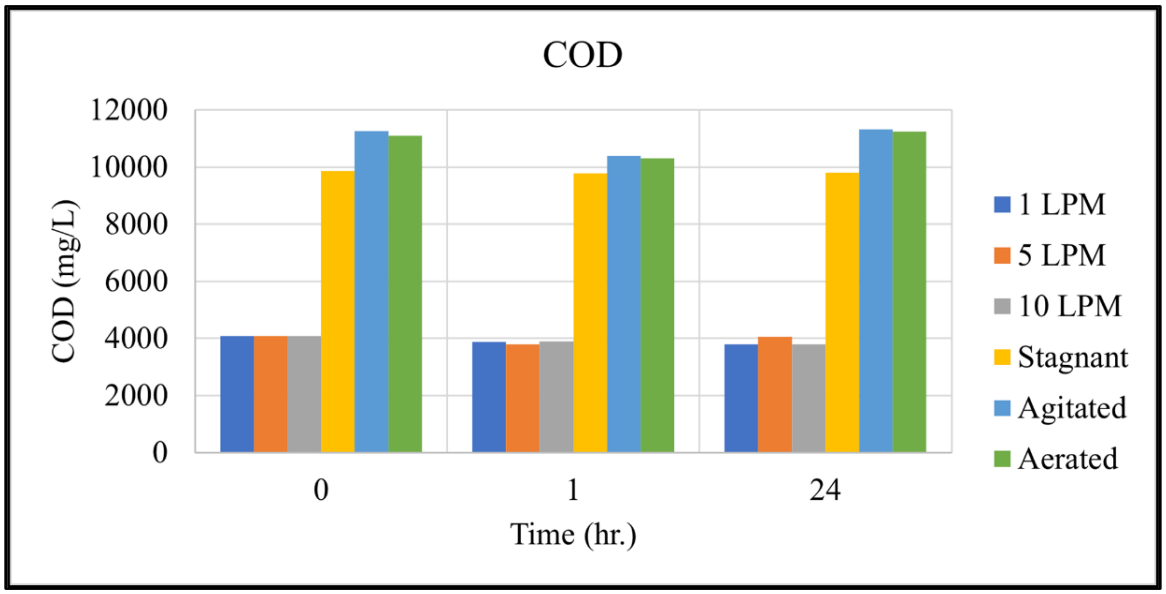


Figure 4.4 Benchtop urine carbonation COD profile.

The effect of carbonation on the ammonia nitrogen species is inconclusive as, similarly to COD, there is minimal variation between all samples. This is with the exception of the agitated control sample which has a slightly higher starting concentration that declined throughout the experiment (Fig. 4.5). This could be attributed to the high pH that maintained the dominant species as ammonia (instead of ammonium at lower pH) which likely also off gassed as the containers were all sealed after carbonation but was accidentally left unsealed for the agitated sample.

There did not appear to be any precipitation of any solids such as ammonium bicarbonate that would have resulted in a decrease in soluble ammonia nitrogen. Precipitation requires saturation of constituents and while at a high concentration, the combination of ammonium and

carbonate species may not be at saturation. There is a slight increase in alkalinity of the carbonated species (Fig. 4.6) indicating some of the dissolved carbon dioxide converted to carbonate. The opposite seems to have occurred for the control samples as there is a slight decrease, indicating some existing carbonate species may have converted and off gassed.

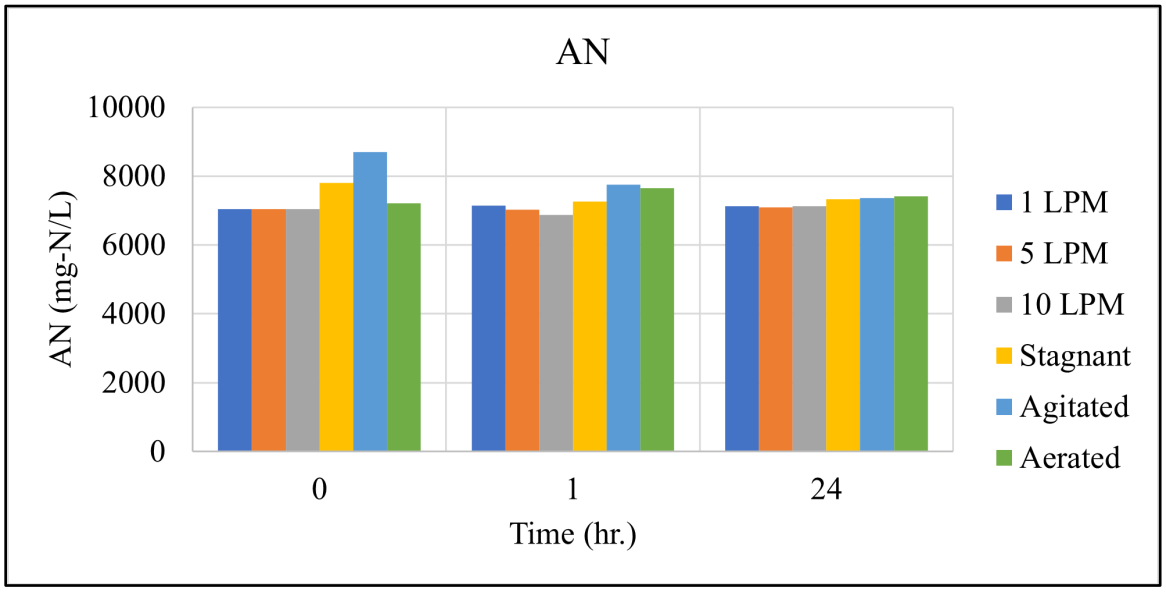


Figure 4.5 Benchtop urine carbonation ammonia nitrogen profile.

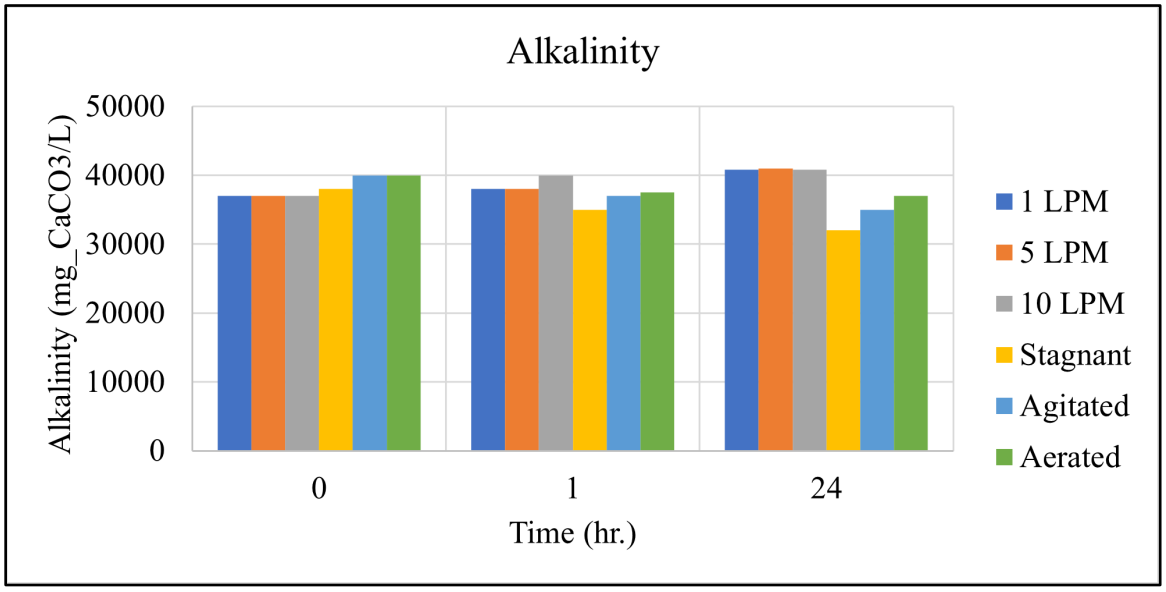


Figure 4.6 Benchtop urine carbonation alkalinity profile.

While additional benefits, such as AN removal via precipitation, were inconclusive, the use of carbon dioxide was shown to be effective at reducing the pH of urine. However, the pH reduction effect is limited to a few hours and would require the immediate treatment of the urine by the SAMBR system.

4.3 Carbonation Column

4.3.1 Design and Development

A full-scale system would need to continuously maintain the reduced pH by continuously carbonating the urine and then deliver it to the SAMBR system on demand so that it is immediately introduced into SAMBR. Under ambient conditions, carbonating continuously at 1 LPM would require approximately 3 kg of carbon dioxide per day. The Baseline Values and Assumptions Document (BVAD) lists the average carbon dioxide generated by a crew member as 1.04 kg per day, making the generation rate for a crew of four: 4.16 kg per day. The crew generation exceeds the demand to continuously carbonate, indicating this could be a sustainable source of carbon dioxide without the need to bring additional.

The design of the carbonation module needed to be compatible with ECLSS hardware design constraints, primarily in terms of dimensions. Simultaneously, the module needs to be optimized for maximum contact time and mass transfer. A cylindrical column of PVC pipe was fabricated with an inner column of small diameter PVC pipe to serve as a concentric baffle, termed as a “tube ‘n tube” design, displayed in Fig. 4.7.

The urine/flush influent was fed through the top of the module, directly into the inner column. As more influent entered the column, hydrostatic pressure drove the influent out and around bottom of the inner column where it would rise to the top of the outer column and exit through a port near the top to enter the SAMBR system. Gas cylinders of carbon dioxide

supplied the carbonation through a fine diffusion stone at the bottom of the inner tube. The fine bubbles of carbon dioxide rise, countercurrent to the influent, to the top of the inner column and out the top of the module. The excess carbon dioxide and stripped gasses were routed to the feed tank that provided the influent to the module, this was done to provide upstream carbonation to the influent even before entering the module and maximize the use of the carbon dioxide. The active volume of the column ranged between 1-2 L to allow for concentrated carbonation and quick delivery of the carbonated influent to the SAMBR system. A dye test confirmed the assumptions of this design, and the module was assembled and finalized (Fig. 4.8). The module was integrated into the SAMBR system during at the start of stage H-2, with the performance having been discussed in Ch. 3.

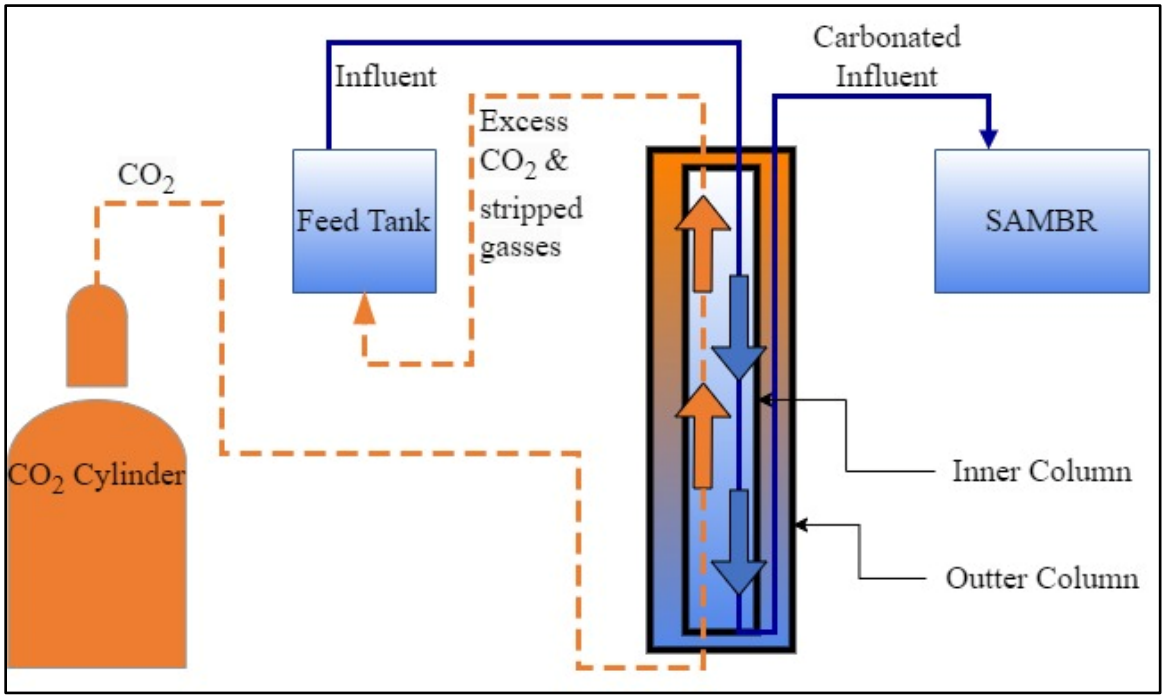


Figure 4.7 Process flow diagram of designed carbonation column for SAMBR.

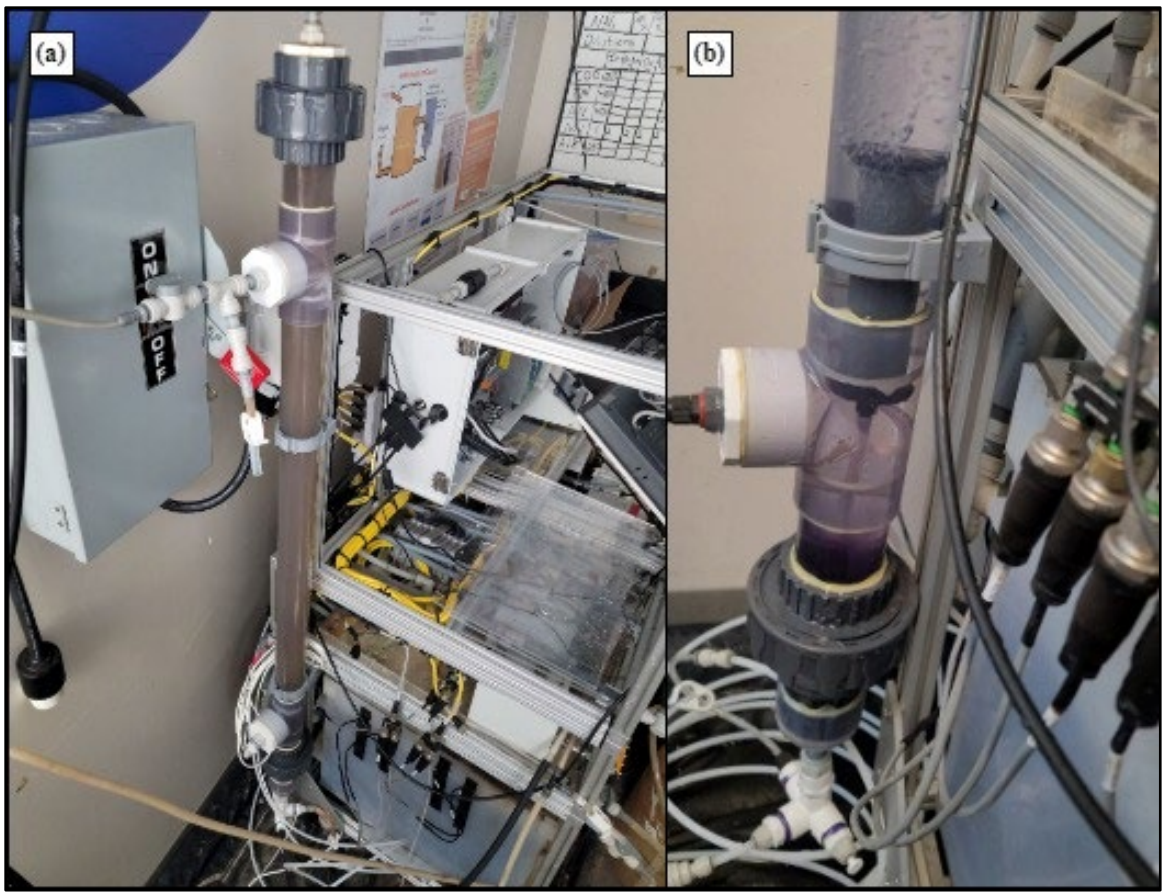


Figure 4.8 Carbonation system; (a) Assembled module; (b) Close up of fine diffusion stone for carbon dioxide delivery.

Chapter 5: Hygiene Graywater Conditioning

Unlike the hygiene graywater onboard the ISS, data and information of future hygiene graywater is speculative and subject to change. Estimates regarding the inputs, generation rates, and characterization of hygiene graywater under EPB/PGH scenarios have been made but are less certain as summarized previously in Table 1.1. Hygiene facilities are expected to be established. However, water rationing, and the capabilities of these facilities have yet to be determined with any degree of certainty(Muirhead et al., 2023; Muirhead et al., 2022a). The most fundamental and necessary may only be available in initial missions, while capabilities such as shave and laundry may not be available until a more mature environment is achieved. Water availability will determine the volume, available frequency, and waste streams characteristics.

As the SAMBR technology is intended for EPB/PGH scenarios, the incorporation of these future waste streams needed to be investigated. To fit within the scope of this dissertation, the handwash and shower streams were assessed for their potential effects of incorporating hygiene graywater with SAMBR. The selection of the handwash and shower streams was justified as being more critical compared to oral and shave. The handwash and shower constitute a large fraction of the hygiene graywater waste streams compared to the oral and shave, while laundry may not be incorporated until much later. Additionally, the handwash and shower streams are planned to use a similar modified commercial surfactant blend termed, ISS No-Rise, currently used onboard the ISS, simplifying the ability to generate these ersatz streams. Experimental characterization of the ISS No-Rinse is summarized in Table 5.1, where the CMC

was measured of using just ISS No-Rinse where the other parameters were measured of ISS No-Rinse after it had been used to wash a pair of hands once to represent the waste stream more accurately.

Table 5.1 ISS No-Rinse experimental characterization

ISS No-Rinse Solution	
CODs (mg/L)	667
ANs (mg-N/L)	3
NNs (mg-N/L)	2
pH	7.1
EC (uS/cm)	117
Turbidity (NTU)	17
Calculated Surfactant Concentration (mg/L)	214
Critical Micelle Concentration (mg/L)*	~104 mg/L

The No-Rinse is a surfactant blend, potential for foaming within SAMBR were known to be a possibility and may be necessary to address. Preliminary testing confirmed that foaming was substantial and would result in foam escaping the SAMBR system. Foaming is a common problem even in terrestrial wastewater treatment and has been studied extensively (Vesga-Baron et al., 2022). One study examined the use of UV irradiation and ultrasonication of bulk foaming at a wastewater treatment facility in Mexico (Hernández-Romano et al., 2022). While UV was not observed to have any significant effect, ultrasonication was shown to be effective at reducing foam (Hernández-Romano et al., 2022). NASA has also previously invested research into use membrane filtration as a method of surfactant treatment (Cartinella et al., 2006). Similar to urine conditioning, an investigation to develop a conditioning subsystem capable of mitigating the

foaming potential of the ISS No-Rinse. Literature has indicated that ultrasonication and membrane filtration have a potential and were investigated further to condition the ISS No-Rinse for SAMBR.

5.1 Materials and Methods

5.1.1 ISS No-Rinse Preparation

Concentrated ISS No-Rinse was obtained from NASA's Johnson Space Center and stored at 4°C in airtight containers when not in use. The concentrate was diluted with water to an approximate concentration of 0.5 g/L to achieve an ersatz handwash and shower formulation.

5.1.2 Sample Collection

Sample collection was conducted similar to the method described in Section. 3.1.1 with the following alteration: any foaming that was generated in the ersatz was allowed to dissipate before the sample was collected. This was done as surfactants concentrate in the foam and would create a disparity in concentration distribution.

5.1.3 Sample Preparation and Storage

Sample preparation was conducted similar to the method described in Section. 3.1.2 with the following alteration: any foaming that was generated in the ersatz was allowed to dissipate before the sample was prepared and stored. This was done for the same reason discussed in section 4.1.2.

5.1.4 Foaming Rate

As foaming from the No-Rinse was determined to be problematic for SAMBR, a method to measure it directly was necessary. Through standard methods exist, a custom method was developed that was informed by these existing methods. A 20mL sample in a 250 mL graduated cylinder was subjected to aeration at a rate of 1 LPM (controlled by a mechanical flowmeter)

with a fine diffusion stone. The volume of the foam and the time it took to reach it was recorded and was used to obtain a foaming rate in mL/s. The exact same equipment was used to test each sample after being thoroughly rinsed with deionized water. The setup for this method is shown in Fig. 5.1.

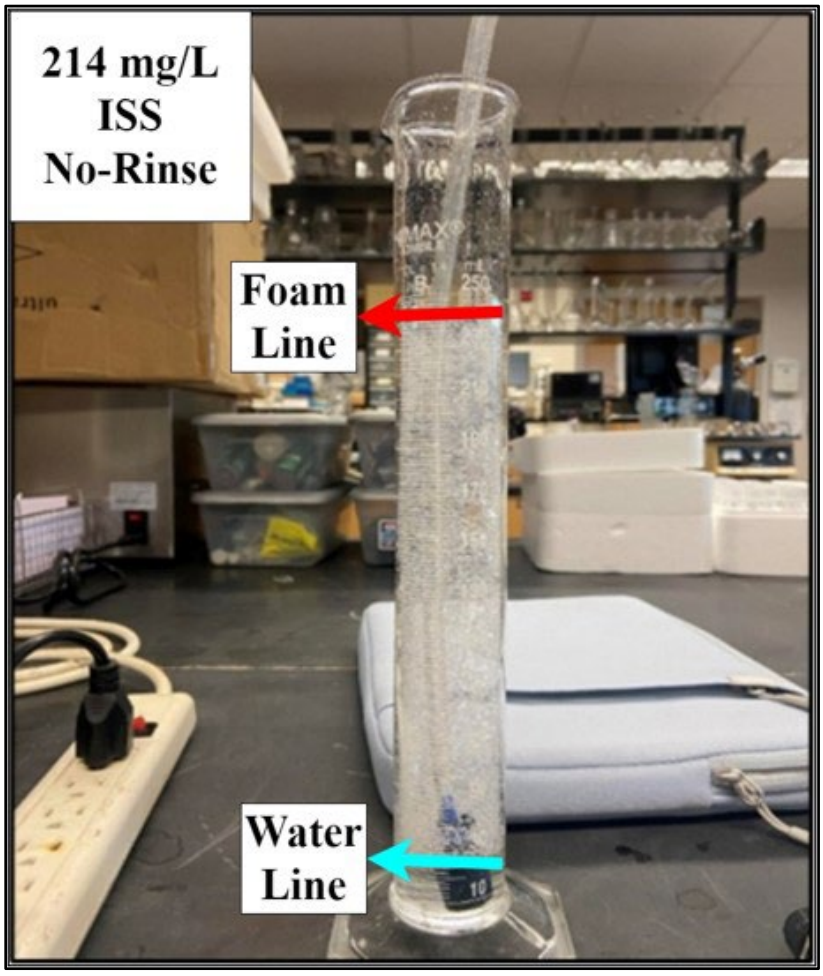


Figure 5.1 Experimental setup for surfactant foaming rate.

5.1.5 Chemical Oxygen Demand

COD analyses was conducted using the same method outlined in section 3.1.5.

5.1.6 Surfactant Calibration

Most methods for analyzing and characterizing surfactants can be time consuming, complex, and require expensive equipment such as chromatography, spectroscopy, spectrophotometry, and tensiometers. The foaming rate and COD were used to estimate the surfactant concentration in the samples. This was done by creating a calibration between the foaming rate, COD, and surfactant concentration. From the manufacturers Safety Data Sheet (SDS) the combined ersatz handwash and shower was known to have an approximate surfactant concentration of 214 mg/L. Samples of varying concentrations were created and tested and graphed to create linear trendlines to obtain a calibration for surfactants against both foaming rate and COD. Fig. 5.2 shows the calibration data between surfactant concentration and foaming rate. Eqn. 5.1 shows the resulting linear equation for estimating surfactant concentration from foaming rate. With an R^2 value of ~0.82, a relatively strong correlation between the surfactant concentration and foaming rate exists and can be used with a degree of confidence. Similarly, Fig. 5.3 similarly displays the calibration data and Eqn. 5.2 is the linear equation of estimated surfactant concentration and COD. A high R^2 value of ~0.98 suggests a very strong correlation between surfactant concentration and COD and a good indicator of estimating surfactant concentration.

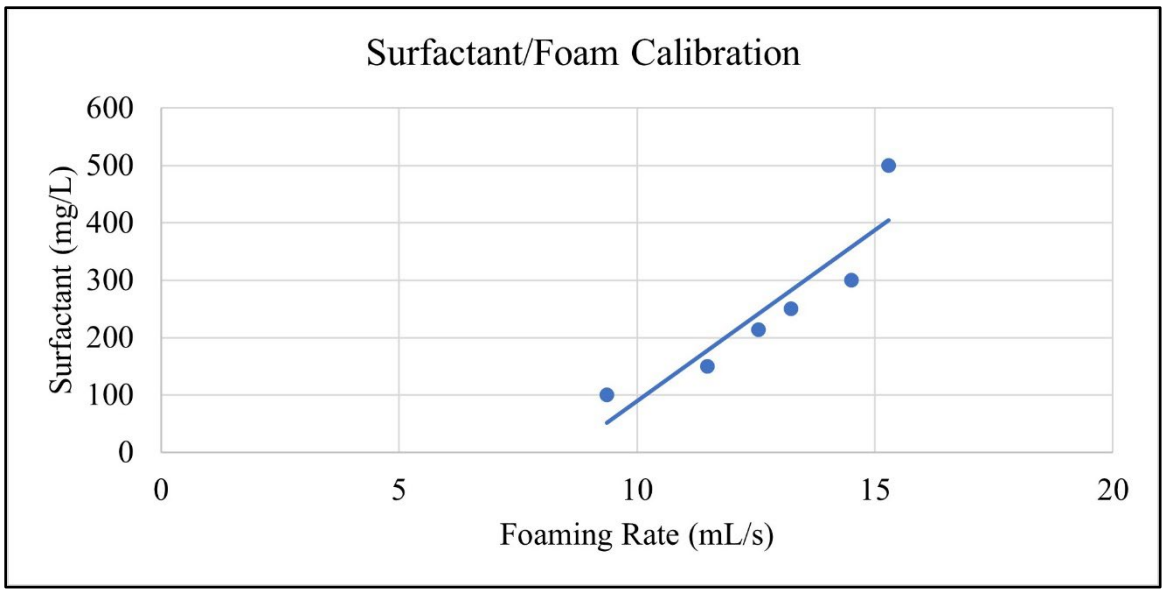


Figure 5.2 Linear calibration graph and trendline between surfactant concentration and foaming rate.

Equation 5.1 $C_s = 59.663 r_f - 507.65$

where:

C_s = estimated surfactant concentration (mg/L)

r_f = foaming rate (mL/s)

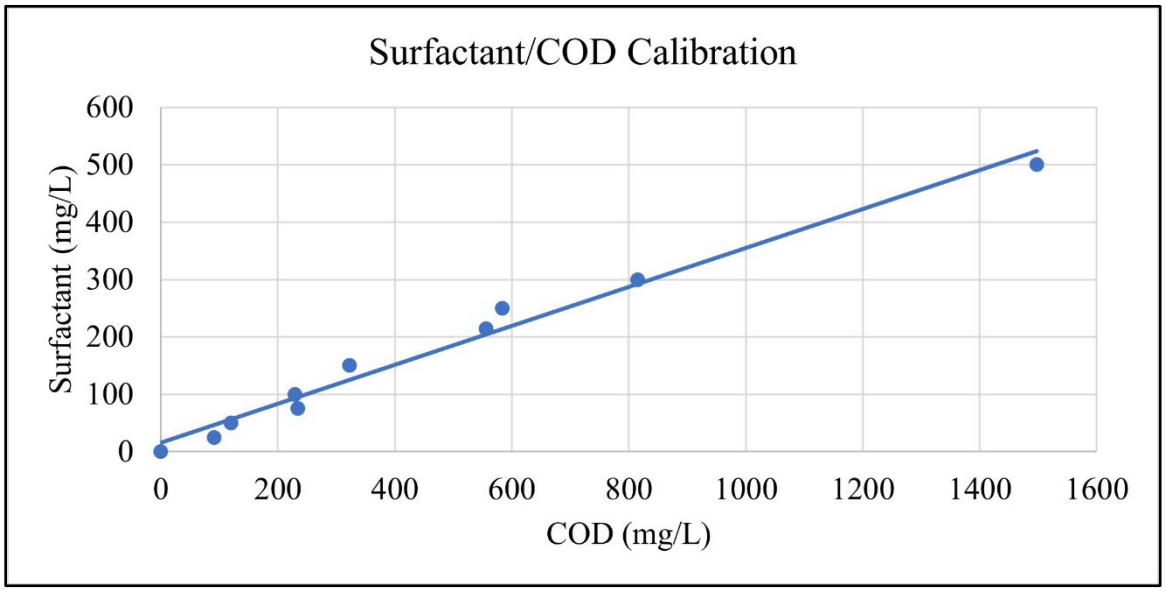


Figure 5.3 Linear calibration graph and trendline between surfactant concentration and COD.

Equation 5.2 $C_s = 0.3396C_{COD} + 15.327$

where:

C_{COD} = COD concentration (mg/L)

5.1.7 Transmembrane Pressure

TMP measurements were conducted similar to the method described in section. 3.1.14.

5.1.8 pH

pH measurement was conducted using the same method outlined in section 3.1.5.

5.1.9 Temperature

Temperature was measured with commercially available thermometers.

5.2 Benchtop Investigation

5.2.1 Ultrasonication

Fundamentally, sonication is the generation of sound energy typically to achieve disruption or agitation. Transducers convert the electrical input to high frequency mechanical vibrations by utilizing the piezoelectric effect (Allam et al., 2022; Sultanova et al., 2021). The high frequency waves are then often amplified to create ultrasonic waves directed at the target medium effect (Allam et al., 2022; Sultanova et al., 2021). The frequency is typically determined by the electrical signal and input to the transducer. In the case of a bath type system, transducers typically line the exterior sides and bottom of the bath and are covered by the external housing of the unit similar to that of Fig. 5.4.

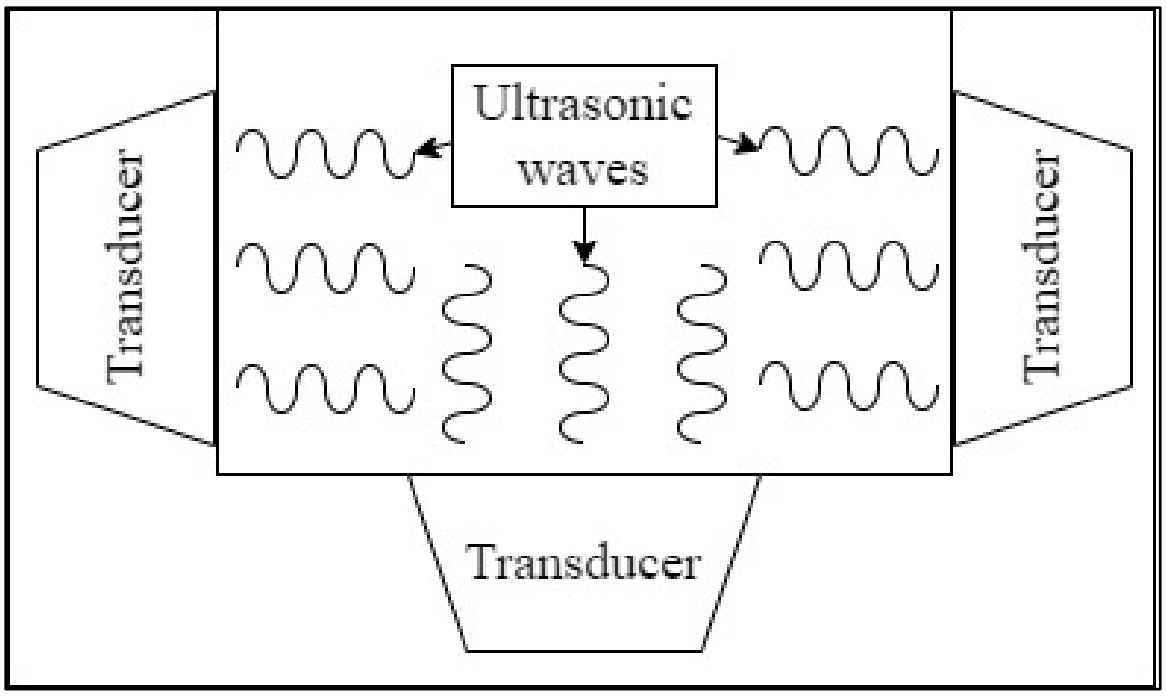


Figure 5.4 Diagram of a typical ultrasonic bath system.

Ultrasonic frequencies are typically considered anything over 20 kHz (Sajwan et al., 2018). In the case of the ultrasonic bath, the multi-directional waves create cavitation in the solution (Nie et al., 2021; Wang et al., 2018). These small, implosive bubbles momentarily create such drastic localized temperature and pressure that water dissociates into hydrogen and hydroxyl radicals as shown in Rxn. 5.1 (Nie et al., 2021; Wang et al., 2018). Both radicals have oxidizing potential but the hydroxyl radical's potential is greater and commonly used in Advanced Oxidation Processes (AOPs) to degrade organics, pathogens, and contaminants in wastewater treatment (Dong et al., 2022).

Water Dissociation: $H_2O \Leftrightarrow H\cdot + OH\cdot$ (5.1)

As the input(s) required is typically just power, this makes it an ideal candidate for sustainable space applications. (Powell et al., 2021) previously proposed ultrasonic drying of fecal matter as a method of extracting water in space applications. The use of sonication for the

treatment of surfactants has been extensively studied and shown to be effective when the surfactant concentration is below the Critical Micelle Concentration (CMC) (Dehghani et al., 2019; Jabariyan & Zanjanchi, 2012; Lin et al., 2016; Weavers et al., 2005). The CMC of a surfactant or blend of surfactants is the concentration at which surfactants form together with their hydrophilic heads directed outward and their hydrophilic tails tucked into the center. This collectively alters their properties and in the case of ultrasonication makes them significantly more resistant to inactivation. Fig. 5.5 visualizes the formation of micelles as surfactants reach CMC. Fig. 5.6 shows the theoretical chain effect of ultrasonication creating cavitation, consequently creating hydroxyl radicals that are capable of inactivating surfactant monomers.

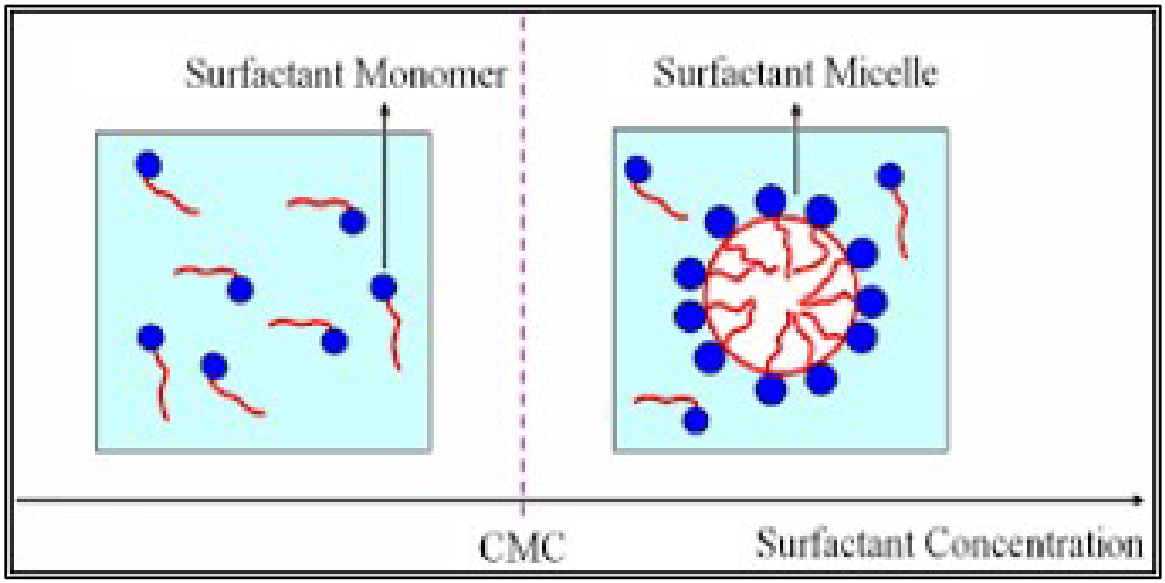


Figure 5.5 Surfactant and micelle comparison. (Lin, 2009)

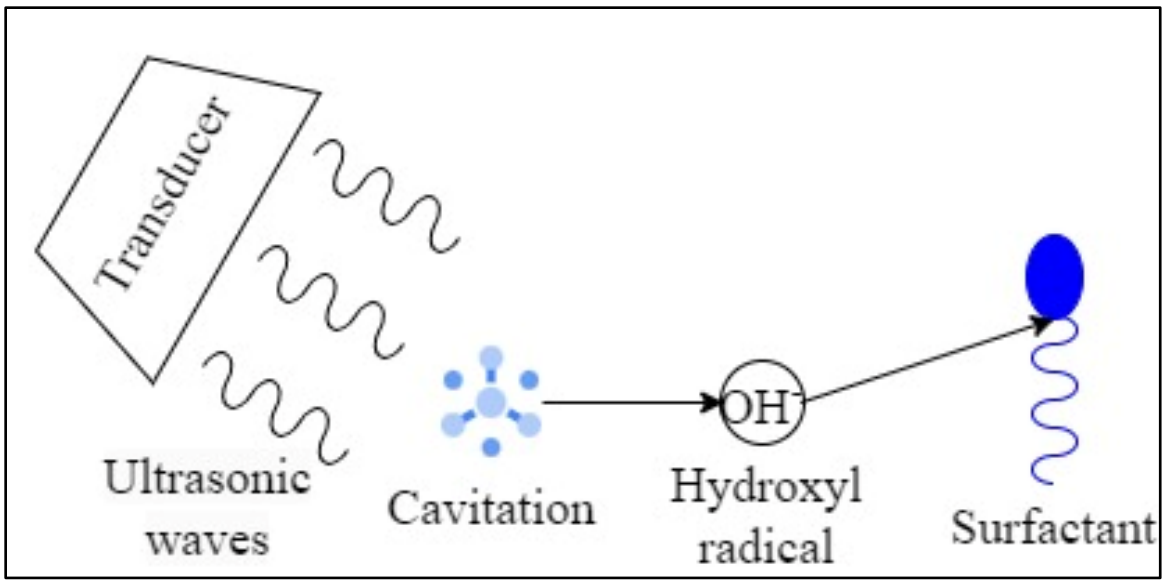


Figure 5.6 Ultrasonic inactivation of surfactant monomer.

As no conclusive CMC data of the ISS No-Rinse, including from the manufacturer, was available, experimental testing of ultrasonication was necessary to investigate its capability to inactivate the surfactants present at the typical concentration of ISS No-Rinse. (Lin et al., 2016) and (Dehghani et al., 2019) both subjected surfactants to sonication up to 120 minutes. To mimic this, ISS No-Rinse was subjected to sonication for 180 minutes with samples taken every 30 minutes. A commercial ultrasonic cleaning bath was used with the specifications and experimental conditions listed in Table 5.2.

Table 5.2 ISS No-Rinse ultrasonication specs and experiments.

Ultrasonication Bath Specs		Time Sample Taken (min.)
Active Volume (L)	10	0
Sonication Power (W)	480	30
Frequencies Available (kHz)	28/40	60
		90
		120
		150
		180

While the foaming rate and surfactant correlation has a lower R^2 value it likely is a more accurate representation as it is possible that any inactivation of the surfactants by sonication (i.e., cleaving the surfactant monomer) may still result in a similar COD measurement. The foaming rate/surfactant correlation would likely capture this change.

For COD, foaming rate, and correlating surfactant concentrations (Fig. 5.7 and 4.8) there appear to be insignificant changes at 28 kHz indicating that ultrasonication at this frequency is ineffective. However, the lack of significant change could also be attributed to insufficient sonication power relative to the volume treated or the surfactant concentration is at or above the CMC resulting in micelles which are largely unaffected by sonication. At 40 kHz (Fig. 5.9 and 4.10), a slight dip in the middle of the experiment is observed for the COD, foaming rate and correlating surfactant concentrations returns to starting levels by the end of the experiment. Possible variations in the distribution of cavitation bubbles could result in a variable surfactant distribution and be reflected as a dip in the COD/surfactant estimation. This can lead to a more effective breakdown or modification of surfactant molecules, reflected as a dip in the COD/estimated surfactant concentration. After sonication ceased, the surfactants quickly redistribute, resulting in an apparent recovery of the COD/estimated surfactant concentration. This dip could also be attributed to variability or error inherent in the analytical process.

The pH remained relatively steady around 7.2 for 28 kHz but rose from roughly 7.1 to 7.7 for 40 kHz. The hydroxyl radical is more likely to react with contaminants, but if the ISS No-Rinse is at or above the CMC, it may be unable to react with the micelles. The solution was prepared with deionized water (DI) which may result in little to no constituents available for the radical to react with. The hydrogen radical has the potential for a few different reactions such as forming hydroperoxyl radicals (Rxn. 5.2), hydrogen (Rxn. 5.3), or hydrogen peroxide (Rxn. 5.4).

The consumption of these hydrogen radicals and the lack of available reactive species for hydroxyl radicals could potentially result in an increase in pH. This could have occurred under both frequencies but only at a great enough rate at 40 kHz to make a measurable difference.

Hydroperoxy Radical Formation: $H\cdot + O_2 \Leftrightarrow HO_2\cdot$ (5.2)

Hydrogen Formation : $H\cdot + H\cdot \Leftrightarrow H_2$ (5.3)

Hydrogen Peroxide Formation: $H\cdot + HO_2\cdot \Leftrightarrow H_2O_2\cdot$ (5.4)

The temperature rose steadily at both frequencies and is expected as the cavitation produced by ultrasonication would increase the temperature. The temperature remains consistently higher at 28kHz than at 40 kHz which may be attributed to the frequency. A lower frequency possess a longer wavelength, which can lead to more effective pressure differentials. This can result in more vigorous and larger cavitation bubbles, leading to higher energy dissipation as heat.

The main metric to assess the change in surfactant and foaming potential was the foaming rate method developed for this experiment despite having a looser correlation compared to the COD as the foaming was a more direct measurement. For this experiment the COD method was potentially more subjective to interferences such as cleaved surfactant monomers still being captured by COD measurement. In either case there was minimal change observed in both methods, indicating that ultrasonication is ineffective at inactivating the surfactants in the ISS No-Rinse. The surfactants at the prepared ISS No-Rinse concentration are likely at or above the CMC and could be why no change was observed. The rise in temperature to 65 °C could potentially be beneficial for the biological consortia in SAMBR as (Baghchehsaraee et al., 2008) found that when activated sludge increased hydrogen production and growth yields at 65 °C compared to untreated activated sludge. The minor pH increase individually may not be

problematic, but coupled with the complications of the high urine pH it may be necessary to address if it were to be implemented. Further research into this would be necessary to find out to what extent the pH and possibly alkalinity would change with sonication. During testing, the ultrasonic bath emitted a rather loud and unpleasant noise that resulted in a noise dampening container placed over it when the system was operating. The nominal sound generation in the laboratory was measured to be 63 dB. When under operation the ultrasonication system was measured to be 90 dB and 72dB when uncovered and covered at 12 inches, respectively. This is in line with noise generated by other ultrasonic devices which measured from 60-97 dB (Khirnykh et al., 1999). Aboard the ISS, astronauts are required to use ear protection when exposed to sounds exceeding 72 dBA during working hours and 49 dBA or less is desired during sleeping periods (Limardo et al., 2017). If such a device were to be implemented, efforts to dampen the noise generated would be necessary.

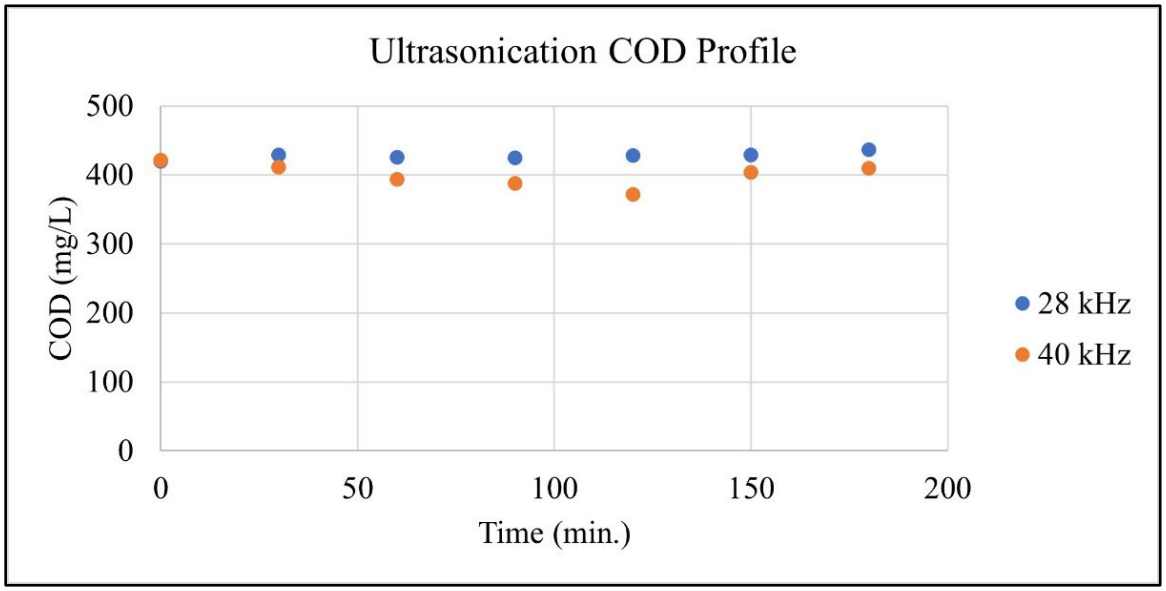


Figure 5.7 COD profile during ultrasonication of ISS No-Rinse.

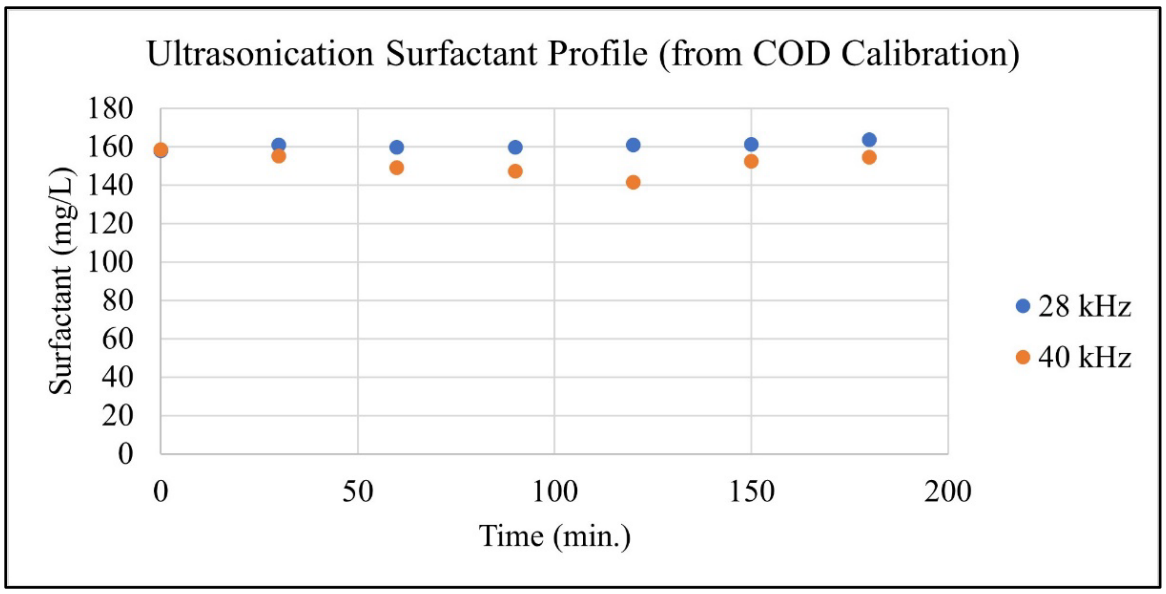


Figure 5.8 Surfactant profile estimated from COD during ultrasonication of ISS No-Rinse.

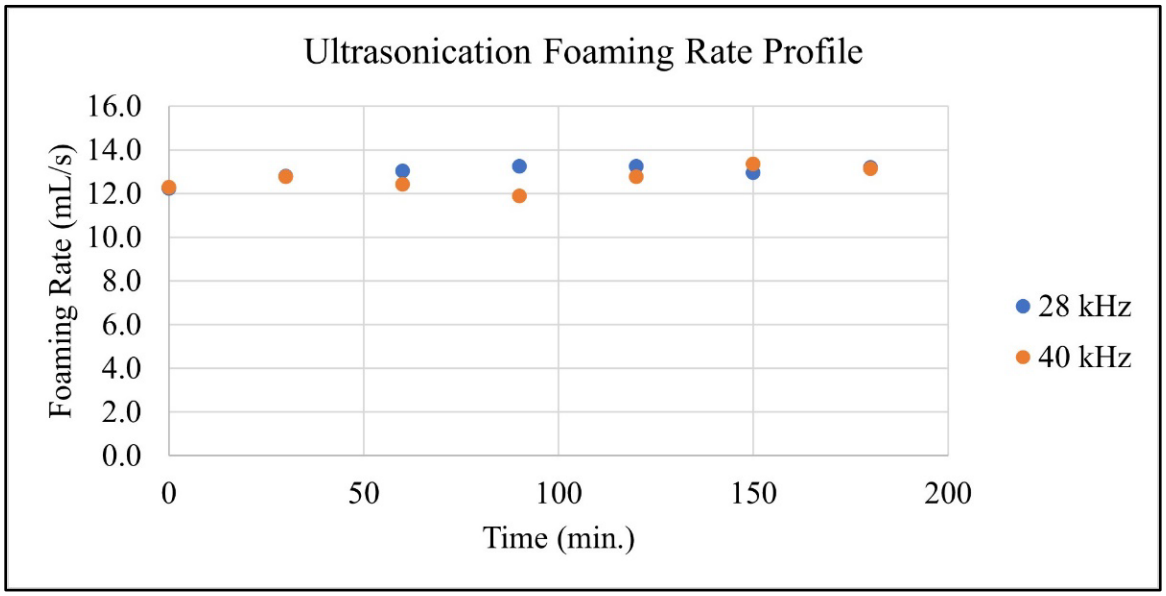


Figure 5.9 Foaming rate profile during ultrasonication of ISS No-Rinse.

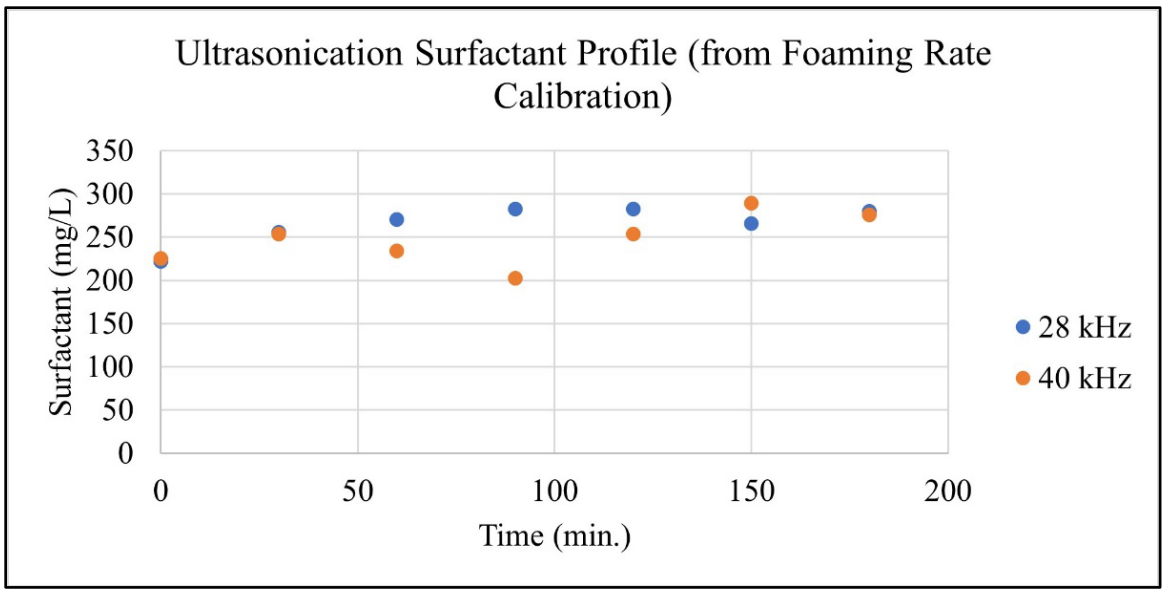


Figure 5.10 Surfactant profile estimated from foaming rate during ultrasonication of ISS No-Rinse.

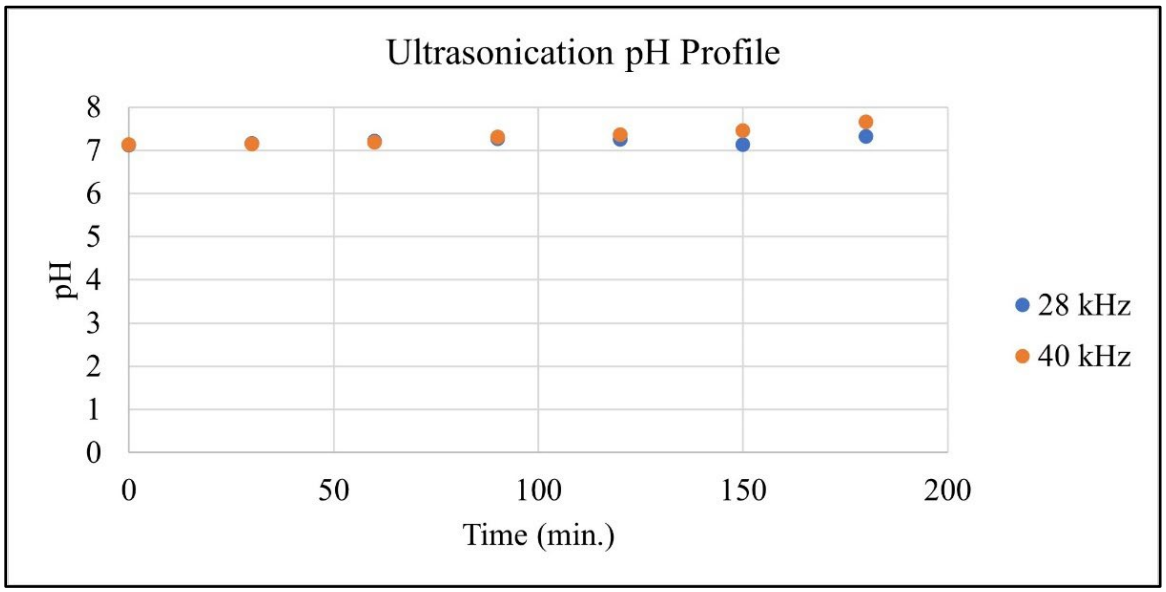


Figure 5.11 pH profile during ultrasonication of ISS No-Rinse.

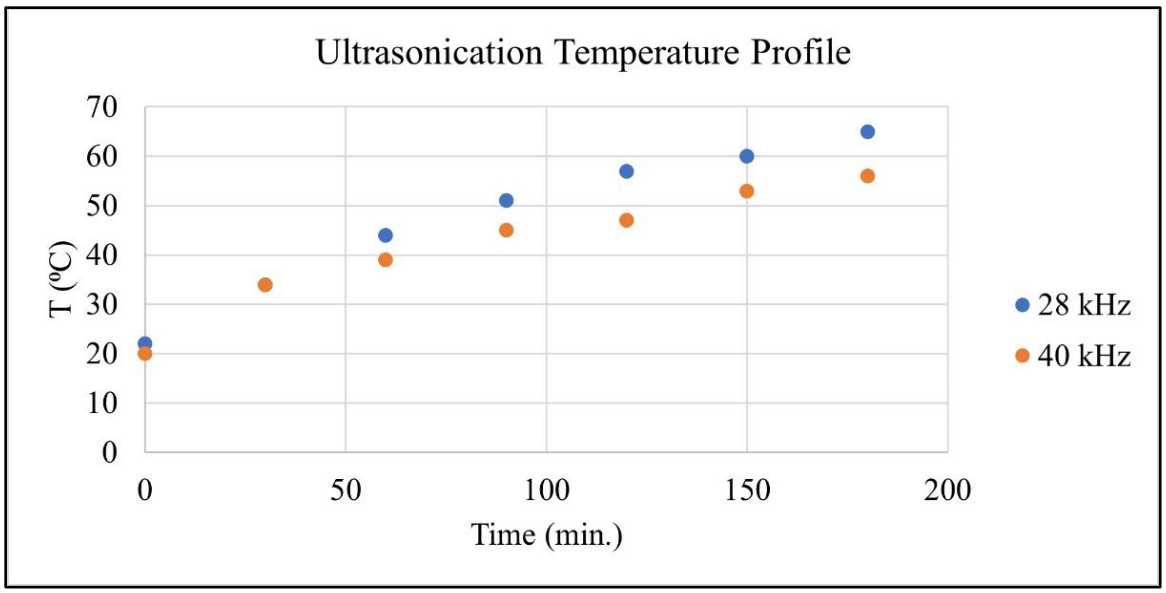


Figure 5.12 Temperature profile during ultrasonication of ISS No-Rinse.

5.2.2 Membrane Filtration

The five primary scales of filtration are macrofiltration, microfiltration, ultrafiltration, nanofiltration, and reverse osmosis. Pores provide a physical barrier, based on size exclusion, to reject the desired constituents and selectively allow the effluent to pass through. Fig. 5.13 depicts the range of pore sizes that constitutes these types of filtrations and their corresponding filtration capabilities. Smaller pore sizes increase the rejection of smaller constituents and yield a higher quality effluent. However, with the smaller pore size, a larger driving force (i.e., TMP) is needed for the solution to permeate through the membrane and consequently requires more energy.

Macro-, micro-, and ultrafiltration often operate at a pressure generally up to five bar and are primarily composed of a polymeric substance (Wang & Wang, 2019). Generally used for primary treatment to remove larger constituents, these membranes are used as a pretreatment to reduce the loading for downstream nanofiltration and reverse osmosis membranes. Nanofiltration membranes operate at a slightly higher pressures ranging from two to ten bar, reject virtually all

constituents barring ionic compounds, and utilize composite materials, cellulose, and acetate(Wang & Wang, 2019). Reverse osmosis, the smallest level of filtration, operate at high pressures, resulting in a high energy demand, but retain salts and ions utilizing similar materials as nanofiltration membranes(Wang & Wang, 2019). Due to their filtration scale, they are largely used in desalination processes but generally require substantial pretreatment or additional processing including pulse flow or backwashing, to reduce loading.

Many different surfactants exist and therefore vary greatly in characteristics and size from just a few nanometers to hundreds. The formation of micelles produce a particle significantly larger than the individual monomer that it is composed of. For example, (Wang et al., 2013) found that polyfluorinated-2-dodecenyl (3-sulfate) propyl dimethyl ammonium (PDSPDA) formed micelles 25.45 ± 3.74 nm in diameter, over 600% larger than individual monomer (~4 nm). The high variation in surfactant size coupled with lack of comprehensive surfactant data of ISS No-Rinse makes it difficult to identify the appropriate scale of membrane filtration.

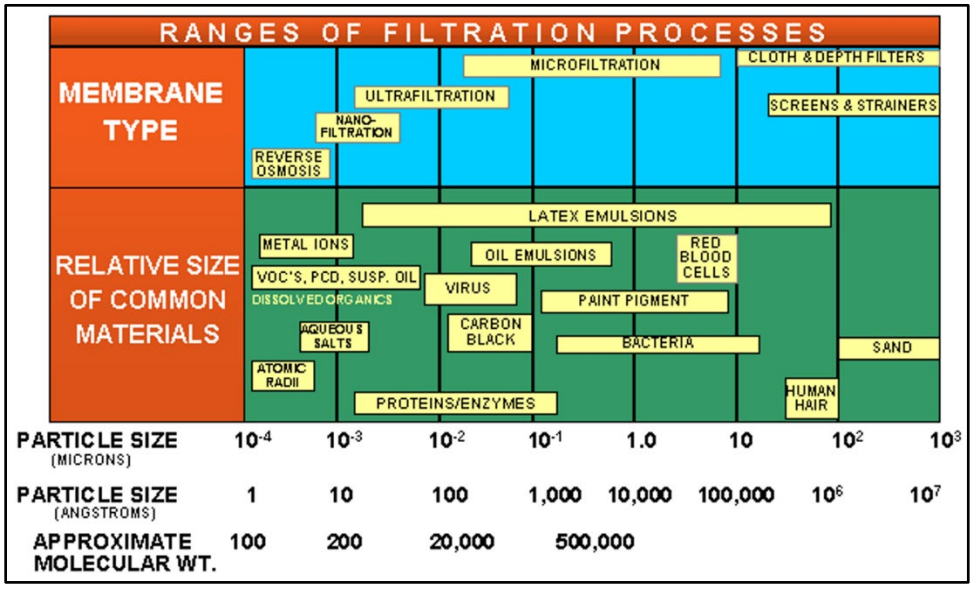


Figure 5.13 Membrane filtration type and rejection range.

As nano- and ultrafiltration span the largest portion of potential surfactant size, these two were selected to investigate as additional methods to treat the surfactants and mitigate foaming in the SAMBR system. Table 5.3 summarizes the specifications (obtained from the manufacturer) of the pre-fabricated membrane modules used.

It was decided to test the membranes in batch operation for ease of setup and testing. A batch solution of ISS No-Rinse was prepared and ran continuously through the membrane until the approximate final concentration factor was achieved (Fig. 5.14). The concentration factor describes the concentration in the feed solution relative to how much solution has passed through the membrane and is defined by Eqn. 5.3.

Table 5.3 Specifications of membranes used in benchtop studies.

Name	Pentair 33V Helix	Pentair HF RX300
Filtration Type	UF	NF
Membrane Type	Tubular	Hollow Fiber
Avg. Pore Size (μm)	0.03	0.002
MWCO (Da)	150,000	1,000
Max TMP (bar)	2.5	3
SA (m^2)	0.055	0.113

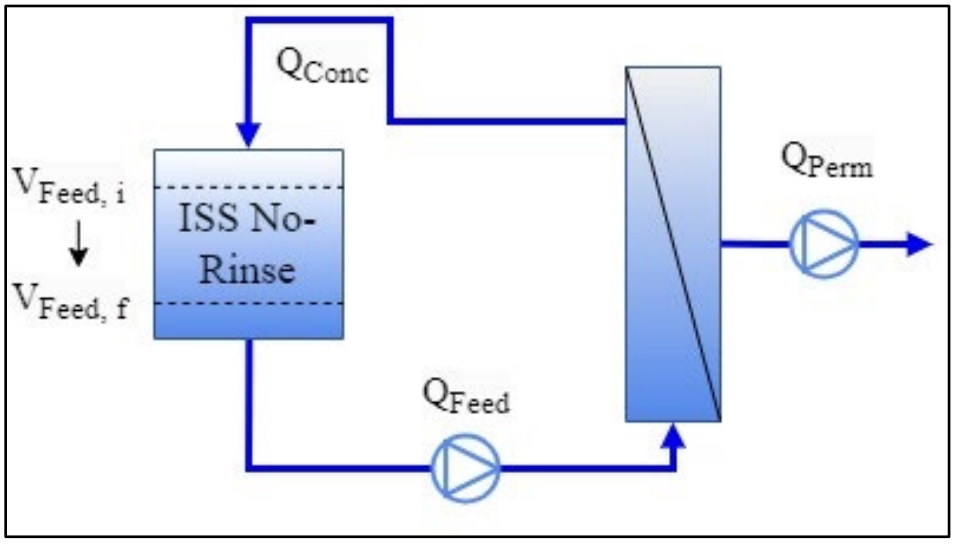


Figure 5.14 Diagram of batch filtration testing.

Equation 5.3 $CF = \frac{1}{1 - Recovery}$

where:

CF = concentration factor

Recovery = permeate volumetric flowrate/feed volumetric flowrate

The target concentration factor of 28 was chosen based on the 28 L/d of handwash and shower generated by a crew of four. Fig. 5.15 displays the estimated surfactant concentration in the permeate and concentrate over various CFs. The surfactant concentrations were estimated assuming a membrane filtration of sufficient Molecular Weight Cut-Off (MWCO) for the surfactants present in the ISS No-Rinse. MWCO is determined by the lowest molecular weight solute the membrane is capable of rejection 90% of said solute. The incremental CFs were graphed over each liter of the 28 L generated each day. For example, with 1 L of the 28 L having permeated through the membrane, the resulting CF was 1.04 and then 1.08 for 2 L of 28 L and so on. During each filtration experiment, operation was temporarily paused to obtain samples approximately at CFs of 1, 7, 14, 21, and 28.

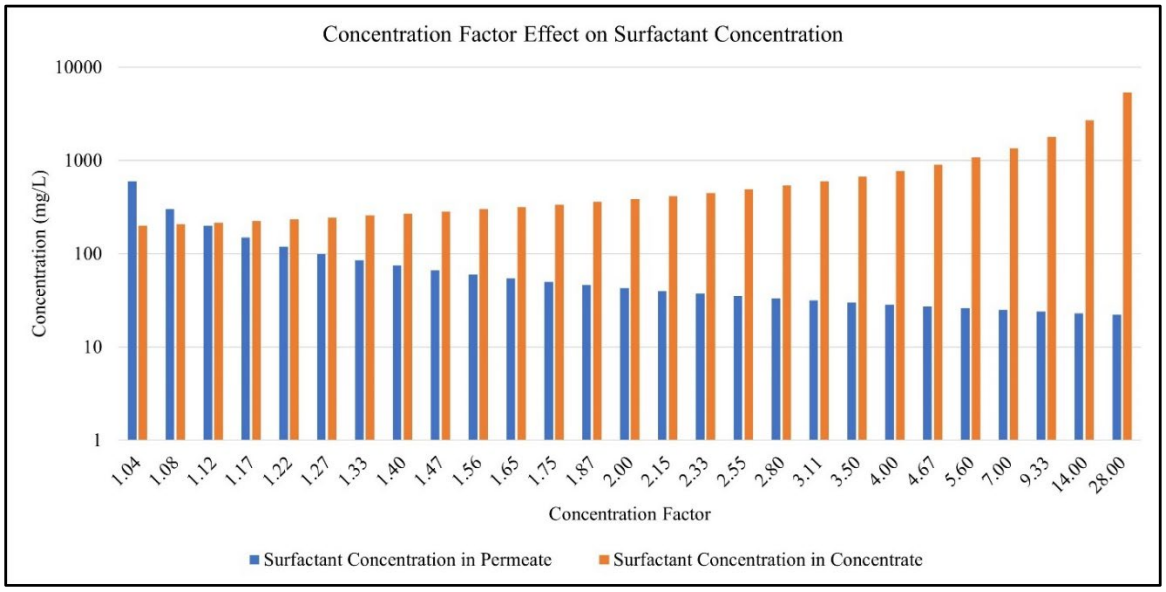


Figure 5.15 Estimated surfactant concentration in permeate and concentrate at various concentration factors.

5.2.2.1 Ultrafiltration

Fig. 5.16 Shows the testing setup for the UF membrane. A 4 L solution of the ISS No-Rinse was recirculated through the membrane with a permeation rate set to mimic the 28 L/d handwash and shower generated by a crew of four.



Figure 5.16 UF membrane setup for batch testing.

Transducers were used to measure the feed, concentrate, and permeate pressures of the membrane and prepared as described in section 4.1.7. Eqns. 5.4-5.6 are the resulting calibration equations.

Equation 5.4 $P_{feed} = 0.3805V_{feed} - 1.139$

where:

P_{feed} = feed pressure of membrane (bar)

V_{feed} = voltage measured by pressure transducer at the feed of the membrane (V)

Equation 5.5 $P_{concentrate} = 0.3585V_{concentrate} - 1.0281$

where:

$P_{concentrate}$ = concentrate pressure of membrane (bar)

$V_{concentrate}$ = voltage measured by pressure transducer at the concentrate of the membrane (V)

Equation 5.6 $P_{permeate} = 0.3805V_{permeate} - 1.1164$

where:

$P_{permeate}$ = permeate pressure of membrane (bar)

$V_{permeate}$ = voltage measured by pressure transducer at the permeate of the membrane (V)

Both correlations between foaming rate, COD, and surfactant are applicable in filtration as the removal of surfactants by filtration should be captured by both foaming and COD measurements. However, as the foaming rate/surfactant correlation has a lower R^2 value it is likely to have larger area and less reliable compared to the COD/surfactant correlation.

A reduction of COD and correlating surfactant concentrations is observed (Fig. 5.17 and 4.18). The permeate sample tested at each CF is consistently lower than its counterpart feed sample which increases in concentration as the membrane reject (or concentrate) is recycled back into the feed tank. A reduction as the permeate from the membrane is consistently lower than the starting feed values at a CF of 1. A similar observation can possibly be made from the foaming rate and correlating surfactant concentrations but there is considerably more error. At one point

the correlation results in a negative surfactant concentration and % membrane rejection of greater than 100%, indicating that it is not as reliable in this experiment (Fig. 5.19 and 4.20).

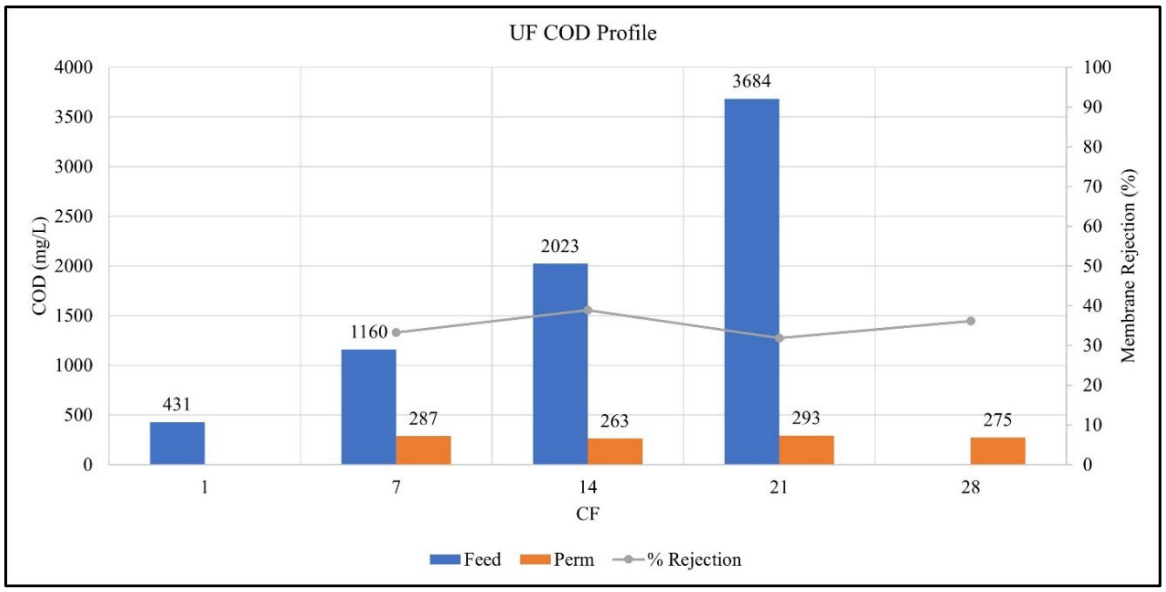


Figure 5.17 Ultrafiltration COD and rejection profile.

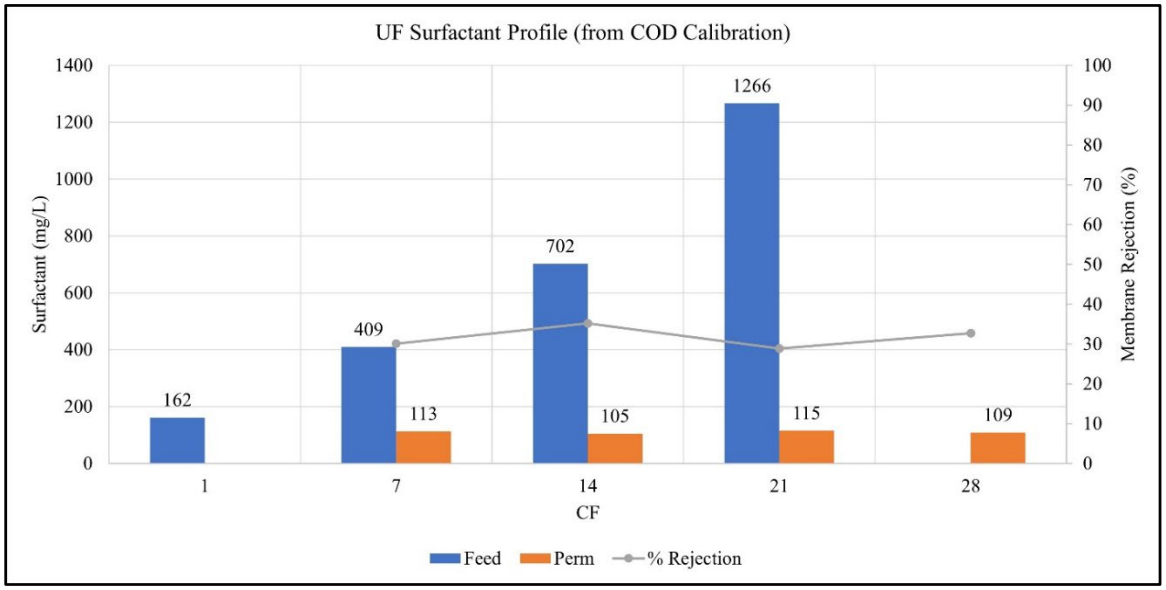


Figure 5.18 Ultrafiltration surfactant (from COD calibration) and rejection profile.

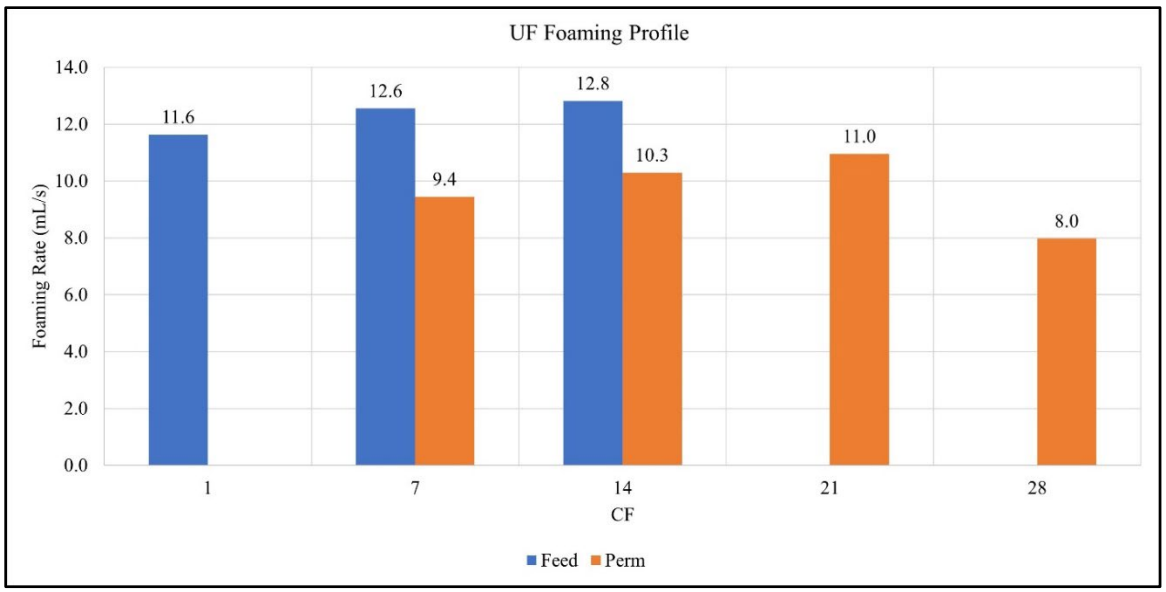


Figure 5.19 Ultrafiltration surfactant foaming profile.

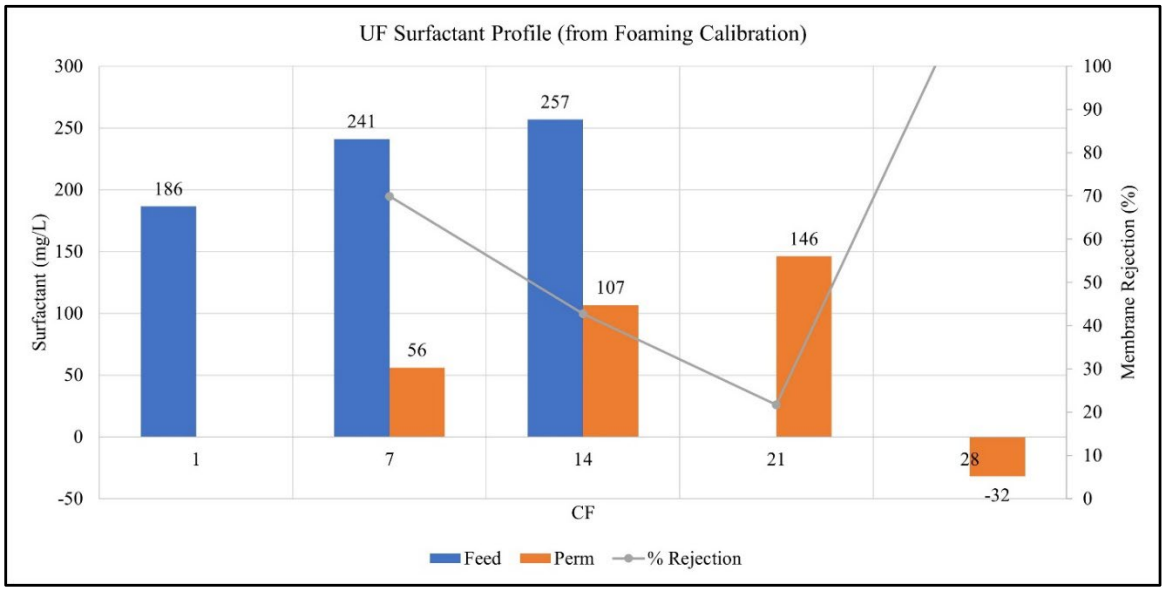


Figure 5.20 Ultrafiltration surfactant (from foaming calibration) and rejection profile.

The initial pH (the pH of the feed sample at CF 1) is slightly basic at 7.9 (Fig. 5.21). However, the pH of both the feed and permeate through the rest the sampling points consistently hover around a pH of 6. As surfactants are becoming more concentrated in the feed sample and consistently low in the permeate sample, this would suggest that the surfactants have little effect

on the pH or that complementary effects that lower the pH occurred in both solutions. This could include volatile compounds initially present in the solution from the manufacturer that evaporated or reacted during the agitation that occurred in operation and lowered the pH. If exposure to air prior to testing, was incidentally kept minimal, increased exposure to atmospheric carbon dioxide might lead to dissolution, alter the carbonate species, and lower pH.

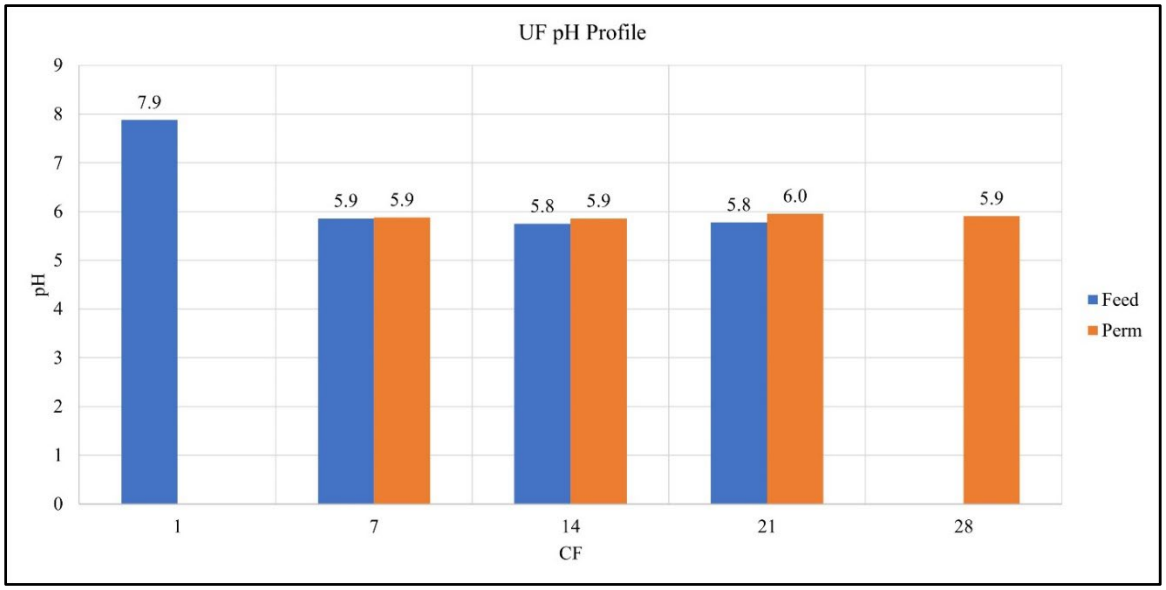


Figure 5.21 Ultrafiltration pH profile.

The TMP during operation averaged approximately 0.1 bar, demonstrating little strain on the membrane and minimal energy consumption. During some sampling events, the pressure logging system was forgotten to be paused and resulted in variable TMP drops observed in Fig. 5.22. The resulting flux and permeability of ~21 LMH and 209 LMH/bar are well within manufacturer specifications and general operational industry standards. However, a safety factor should still be incorporated to account for fouling over time and variations in the crew generated waste. Increasing the surface area (mainly by lengthening the membranes or adding more

membranes of said length and increasing the diameter of the module) to achieve a flux of 5-10 LMH should provide sufficient safety factor for continuous use on EPB/PGH scenarios.

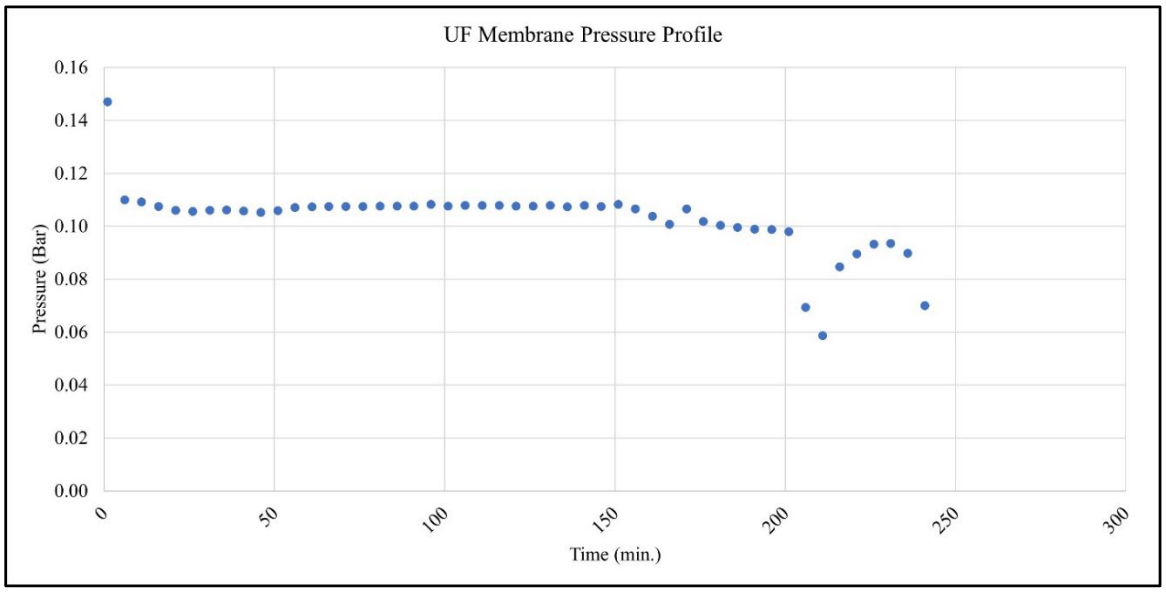


Figure 5.22 Ultrafiltration TMP profile.

The reduced concentration of surfactant metrics in the permeate indicates the UF membrane is capable of partial rejection of the surfactants, either as monomers or micelles, present in the ISS No-Rinse.

The highest MW ingredient, Disodium lauroampho diacetate (446 g/mol), listed in the ISS No-Rinse comprise an average 37 % weight, relative to the surfactants. Considering that the membrane rejection of COD and correlating surfactant concentrations is about 1/3, the rejected solution is likely comprised primarily of Disodium lauroampho diacetate. The produced permeate solution is consequently likely to be rich in the remaining lower molecular weight ingredients.

Despite the foaming reduction, sustained foaming was still observed, suggesting further investigation for UF is needed. Further testing, optimization, and possibly coupling with

additional treatment may be necessary to achieve sufficient reduction for practical applications in SAMBR.

5.2.2.2 Nanofiltration

Fig. 5.23a Shows the testing setup for the NF membrane. A 0.5 L solution of the ISS No-Rinse was recirculated through the membrane with a permeation rate set to mimic the 28 L/d handwash and shower generated by a crew of four.

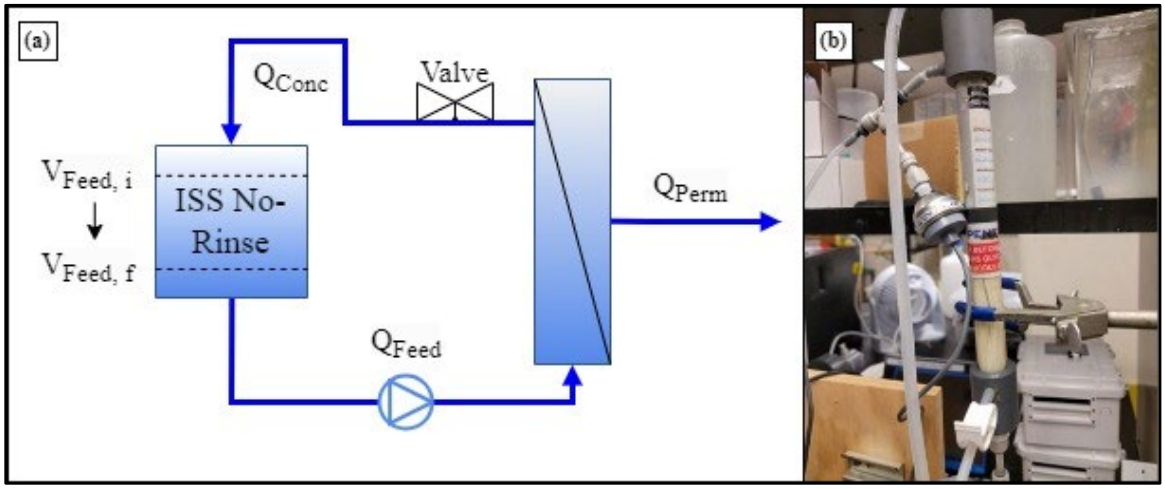


Figure 5.23 Nanofiltration; (a) Setup; (b) Diagram of batch testing.

Transducers were used to measure the feed and concentrate pressures of the membrane and prepared as described in section 4.1.7. Eqns. 5.7-5.8 are the resulting calibration equations. Fig. 5.23b shows the difference in setup compared to Fig. 5.14

$$\text{Equation 5.7 } P_{feed} = 0.3653V_{feed} - 0.8253$$

where:

P_{feed} = feed pressure of membrane (bar)

V_{feed} = voltage measured by pressure transducer at the feed of the membrane (V)

Equation 5.8 $P_{concentrate} = 0.4351V_{concentrate} - 1.2651$

where:

$P_{concentrate}$ = concentrate pressure of membrane (bar)

$V_{concentrate}$ = voltage measured by pressure transducer at the concentrate of the membrane (V)

Historical operation of this research has relied on the use of a second pump to drive filtration by drawing out the permeate. However, for the testing of the NF membrane, no secondary pump was used. Instead, filtration was driven by positive feed side pressure of the membrane, with a valve at the concentrate line to control said pressure. Permeate pressure was left open to ambient conditions and a gauge pressure of 0 bar was used for $P_{permeate}$.

Compared to UF, a larger reduction of COD and correlating surfactant concentration is observed (Fig. 5.24 and 4.25). The permeate sample tested at each CF is consistently low, hovering around 86 and 45 mg/L of COD and correlating surfactant respectively. The feed continuously increases in concentration as the membrane reject (or concentrate) is recycled back into the feed tank but seems to level out around 1244 and 437 mg/L of COD and correlating surfactant respectively.

This is further supported by the foaming rate and correlation surfactant concentration (Fig. 5.26 and 5.27). An increase in foaming rate/surfactant is observed in the feed with a low and steady profile for the permeate. It is worth noting that the foaming rate of the permeate samples did not capture the foaming's visual appearance. There was a much smaller degree of foaming observed that was not sustained but with the bubbles collapsing nearly as quickly as they were generated.

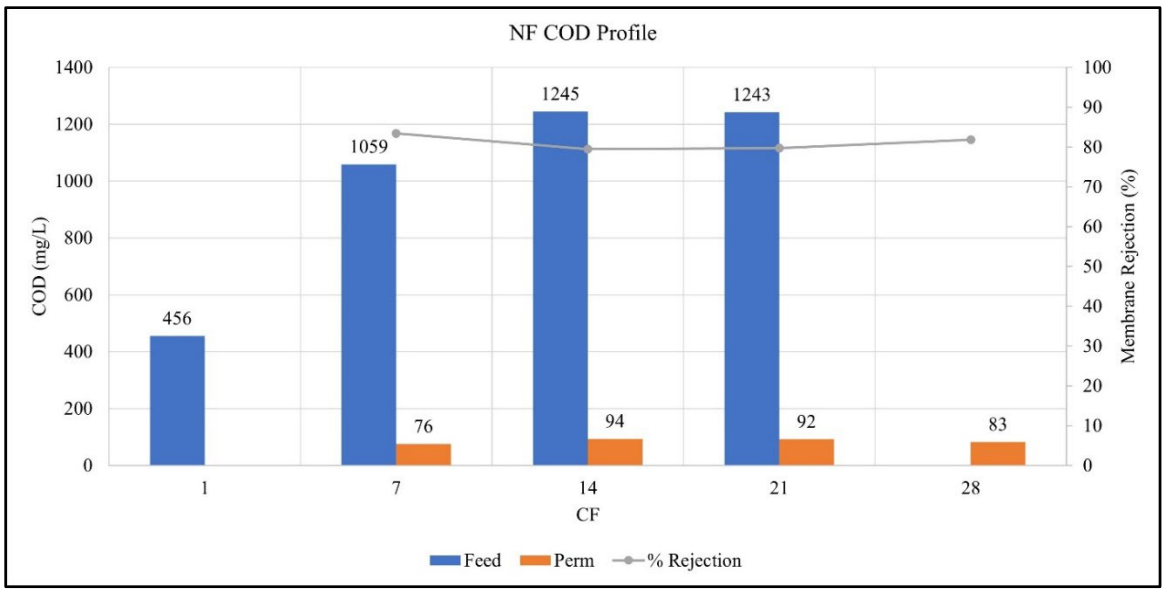


Figure 5.24 Nanofiltration COD and rejection profile.

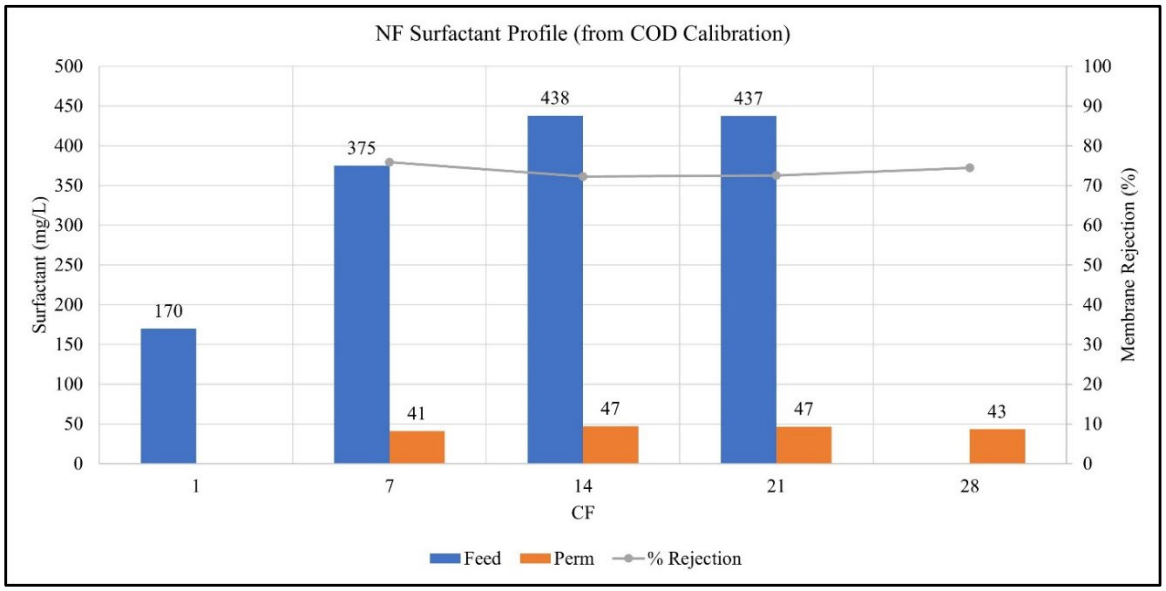


Figure 5.25 Nanofiltration surfactant (from COD calibration) and rejection profile.

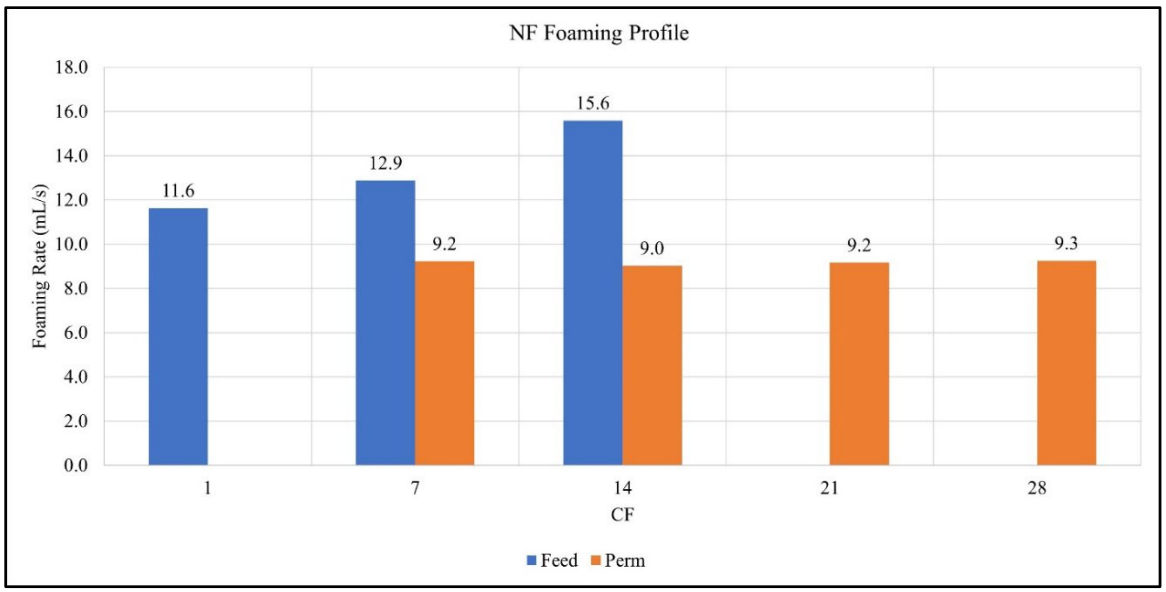


Figure 5.26 Nanofiltration surfactant foaming profile.

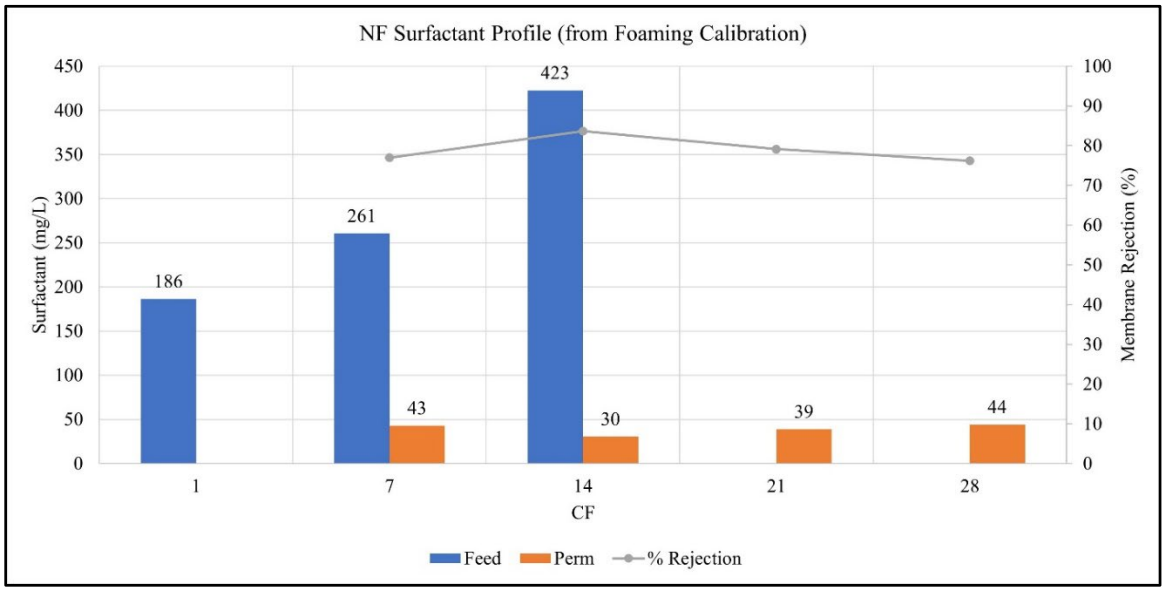


Figure 5.27 Nanofiltration surfactant (from foaming calibration) and rejection profile.

The pH profile was almost identical to that of the UF test, with the initial pH (the pH of the feed sample at CF 1) being slightly basic at 7.9 (Fig. 5.28). This is followed by a consistently lower pH (around 6) in both the feed and permeate. Both tests were operated in the same manner

and were attributed to volatile compounds evaporating and/or air exposure causing these changes in both tests.

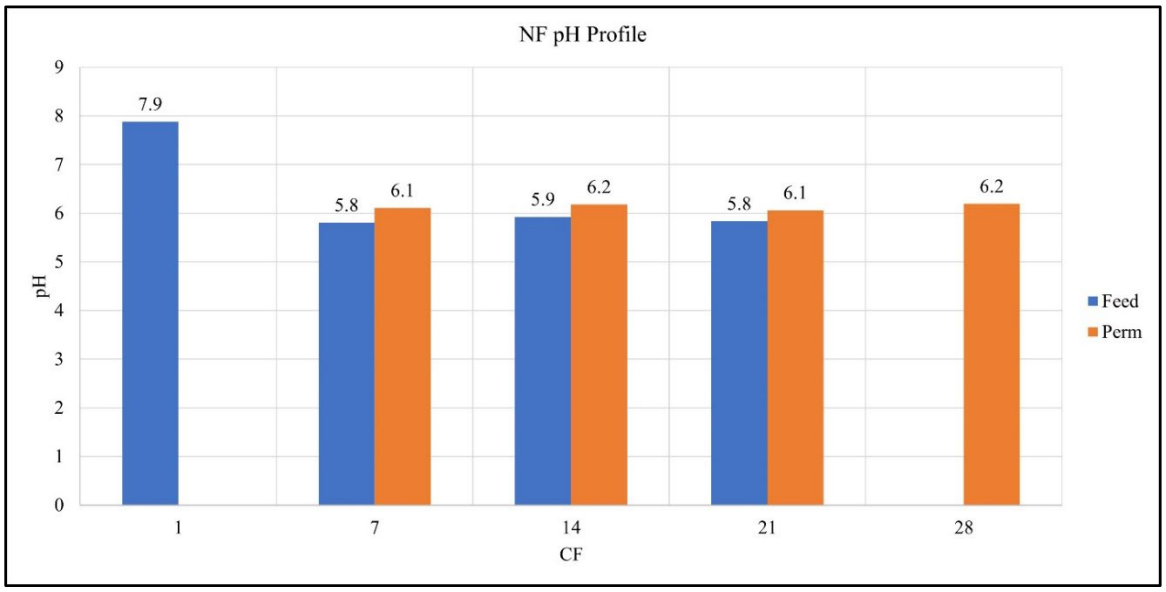


Figure 5.28 Nanofiltration pH profile.

The TMP average approximately 0.6 bar, an increase in strain compared to the UF membrane, but is to be expected given the smaller pore size (Fig. 5.29). The batch filtration was run at a flowrate equivalent to the 28 L/d of ISS No-Rinse generated by a crew of 4 and resulted in a flux and permeability of ~8 LMH and 13 LMH/bar respectively. The flux is within the manufacturer's specifications (<25 LMH) but the suggested operational permeability of 10 LMH/bar was exceeded. The high permeability suggests the membrane was strained. For full scale integration, a custom designed NF membrane would be necessary to increase the SA and decrease the flux with an appropriate safety factor.

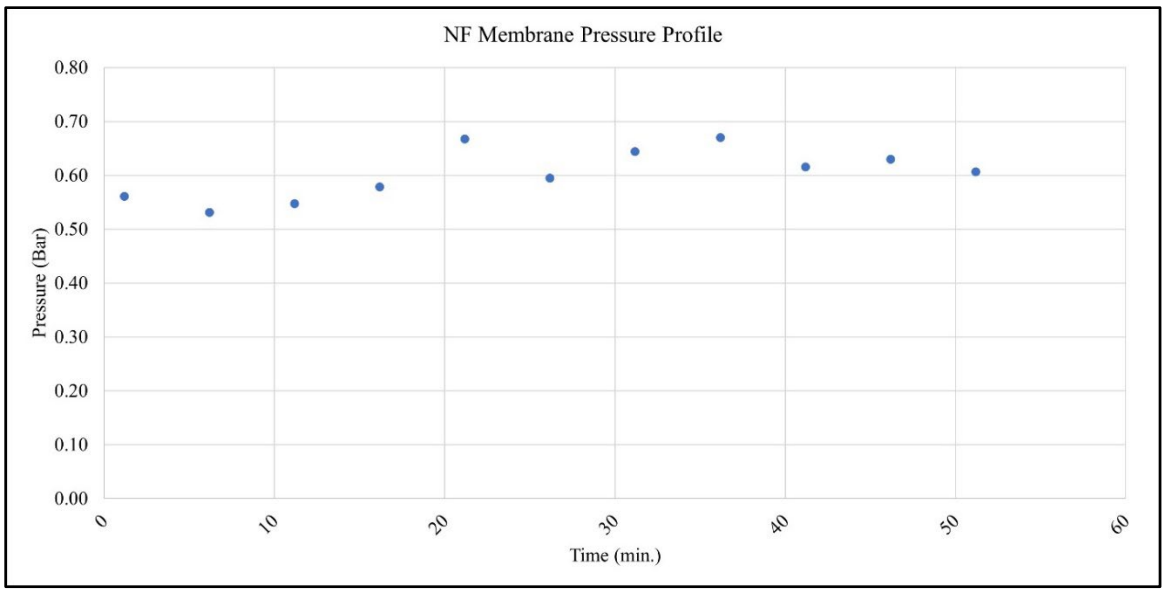


Figure 5.29 Nanofiltration TMP profile.

The NF membrane is capable of significant rejection of the surfactants present in the solution. This is supported by a reduced concentration of foaming rate, COD, and correlating surfactant concentrations in the permeate while the feed continuously increased in concentration. This partial rejection suggests many surfactants, either as monomers or micelles, of the ISS No-Rinse are large enough to be rejected by UF membrane.

The three highest molecular weight ingredients, Disodium lauroampho diacetate (446 g/mol), Sodium trideceth sulfate (STS) (435 g/mol), and Cocamidopropyl betaine (CAPB) (342 g/mol), listed in the ISS No-Rinse comprise an average 74 % weight, relative to the surfactants. Considering that the membrane rejection of surfactant (from COD) is also about 74%, the rejected solution is likely comprised primarily of these three. The produced permeate solution is consequently likely to be rich in the remaining lower molecular weight ingredients.

Visual observation of the foaming rate indicated the NF treated solution may be suitable for treatment in SAMBR without creating foaming overflow in the system. Further investigation

to scale and integrate the membrane with SAMBR could prove to be a sufficient stand-alone treatment for foaming reduction.

5.3 Modeling

As SAMBR was designed to incorporate future waste streams, adding handwash and shower are logical next steps for expanding SAMBR's range of treatment. Modeling SAMBR with the significant change in flowrate, HRT, and diluting effect will help inform its future research and development. While the diluting effect of the additional waste streams may be beneficial for reduced nitrogen inhibition, the reduction in HRT may result in reduced performance.

Fig. 5.30 shows a nearly identical setup when SAMBR was modeled treating just urine and flush. Table 5.4 summarizes the influent characteristics and was developed from literature and SAMBR's experimental data with the remaining settings and inputs left at default (Ray et al., 2018; Rose et al., 2015). The resulting model and balance was found to be quite similar to the performance when simulating just urine and flush. The addition of handwash and shower streams did not appear to significantly affect SAMBR's ability for COD removal as the same 95% was generated but moderately affected the TN removal, decreasing it to 42%.However, the same limitations of modeling the urine and flush seem to apply for incorporating hygiene as well. Similarly, significant portion of the treatment is placed on the membrane and inhibition parameters are likely to have been considered at concentrations lower than necessary for SAMBR. Refinement of these models to increase accuracy will be crucial for identifying potential future fail points and constraints.

Table 5.4 Summary of BioWin input data. (Ray et al., 2018; Rose et al., 2015)

Input Metric	U+F+H+S
Influent (L/d)	37.2
IR (L/d)	148.8
Solids wasting (L/d)	0.4
UF SA (m^2)	0.311
Aeration (LPM)	20
Membrane Feed (LPM)	2.4
COD (mg/L)	1692
TKN (mg-N/L)	3761
TP (mg-P/L)	258
pH	7.5
Alkalinity (mM CaCO3/L)	81

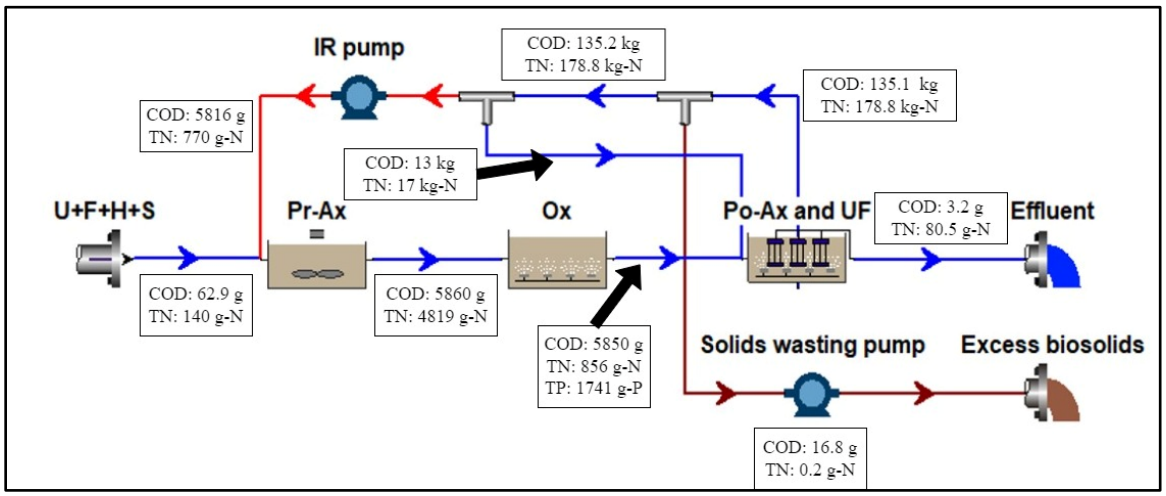


Figure 5.30 SAMBR mass balance treating U+F+H+S.

5.4 Discussion

The presence of surfactants in the ISS No-Rinse were determined to raise concerns regarding foaming within SAMBR. This presented a unique challenge to mitigate foaming caused by said surfactants. A review of literature revealed an array of potential technologies suitable for such an application. Ultimately, ultrasonication, ultrafiltration, and nanofiltration

were selected as the most promising and worthy of further investigation. Benchtop experiments were carried out to assess the feasibility of each one with the results summarized in Table 5.5.

The observed capability of ultrasonication to reduce the surfactant concentration and the foaming rate in the ISS No-Rinse solution was not observed. Analysis revealed that at both 28 and 40 kHz, there were insignificant changes in COD, foaming rate, and correlating surfactant concentrations, hence showing no effectiveness. The lack of observed change indicates the surfactant concentration was likely at or above the critical micelle concentration (CMC). A rise in temperature, up to 65 °C, was observed, likely due to cavitation, may benefit the biological consortia in SAMBR by potentially increasing hydrogen production and growth yields. The considerable noise generated, and the slight pH increase remain further points of investigation for integration with SAMBR.

Ultrafiltration demonstrated potential in reducing surfactant concentrations, as evidenced by reductions in COD and corresponding surfactant concentrations in the permeate samples. Efficient operation and scale is indicated by the low TMP and flux with minimal membrane strain. However, sustained foaming in the permeate indicates the need for further optimization or additional treatment steps for SAMBR integration. The analysis suggests UF membranes partially reject high molecular weight surfactants, making UF useful for reducing surfactant load before further treatment.

Nanofiltration proved most effective, significantly reducing COD, foaming rate, and surfactant concentrations in the permeate samples. Its ability to reduce surfactant concentrations to levels not sustaining foaming make it most suitable for integration with SAMBR. However, a custom design of the membrane would be necessary to incorporate significant safety factor into the SA to reduce the strain. In addition to scaling the NF, consideration would have to be given

to the steady stage operation as a purge stream would have to be incorporated to prevent accumulation. This purge stream could be routed for various fates including destruction such as biodegradation or recovery, purification, and reuse of the surfactants (Fig. 5.31).

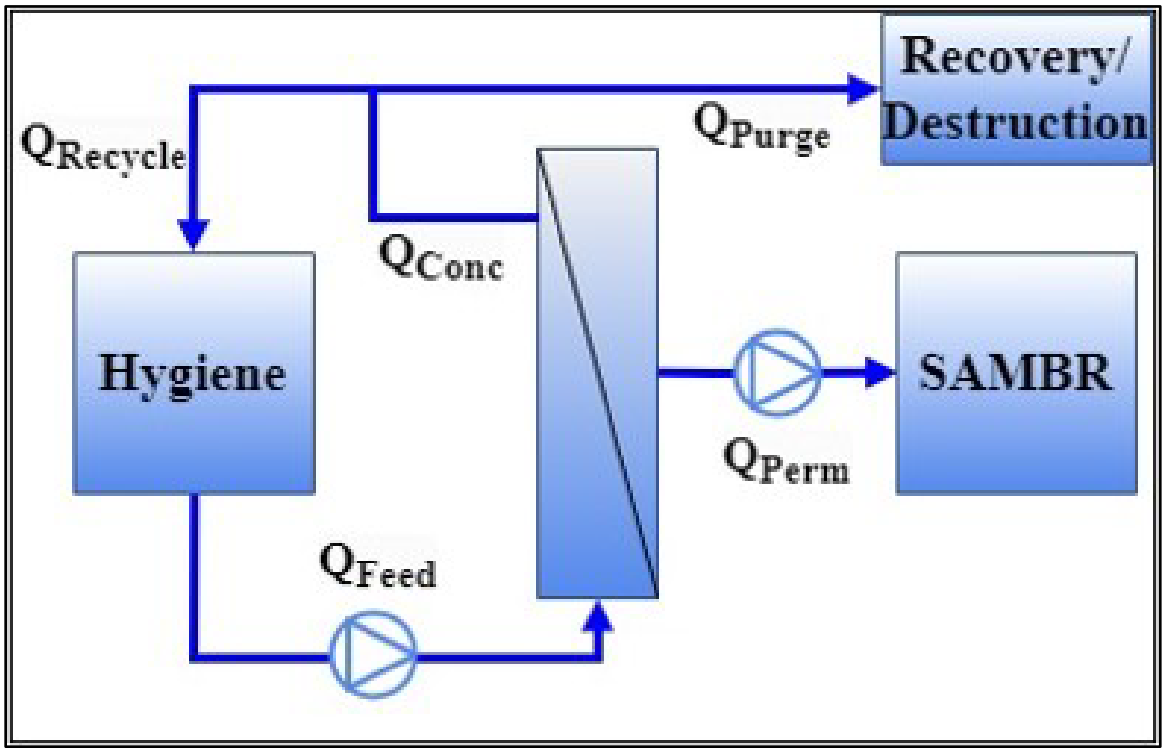


Figure 5.31 Envisioned integration of scaled foaming mitigation technology.

Table 5.5 Foaming mitigation technologies performance summary.

	Ultrasonication	Ultrafiltration	Nanofiltration
Surfactant removal (as COD)	Minimal	32%	74%
Surfactant removal (as foaming rate)	Minimal	63%	82%
TMP (bar)	N/A	0.1	0.6
Notes	Power is only input Literature indicates only effective when below surfactant CMC	Minimal strain on membrane would minimize the need for replacement membranes. Sustained foaming still observed despite surfactant removal.	Moderate strain on membrane. Custom design is needed to minimize fouling and replacements needed. No sustained foaming observed after treatment.

Chapter 6: Conclusions, Future Work, and Impact

6.1 Conclusions

As no flight-ready bioregenerative technology exists, SAMBR's development expanded the realm of bioregenerative waste treatment and resource recovery in space applications. In addition to expanding the knowledge of space waste treatment, SAMBR offers an alternative treatment approach to the traditional physical/chemical technologies. While efficient, reliable, and able to provide instantaneous treatment, many current technologies require constant consumable inputs and generate hazardous byproducts. Current bioregenerative technologies under development cannot treat urine directly, are not optimized for PGH, or are still being optimized for maximum nitrogen removal. If successful, the resulting technology can have an impact on sustainable urine treatment in space and on Earth with the help of NASA's Technology Transfer program, which outsources developed technologies for external use.

SAMBR was designed for a crew of four and designed within the physical constraints of an EXPRESS rack, supporting that it is compatible for space flight design. The system was developed to provide a similar high-quality treatment of currently utilized technologies while requiring less consumables and provide a sustainable alternative for long term habitation missions. The system was capable of tolerating a high-strength waste of up to ~3000 mg-N/L before conditioning was required, significantly higher than previously conducted studies. During this period SAMBR achieved significant COD (93%) and TN (76%) removal. The development and integration of a novel carbonation system was able to condition the urine dominated influent to more ideal conditions, primarily the pH, and prevent ammonia volatilization. Several

technologies were evaluated to identify a suitable subsystem for treating future waste streams. SAMBR was designed to incorporate graywater in addition to urine, but the surfactants present, primarily in handwash and shower, raised concerns about foaming in the bioreactor. The evaluation identified membrane filtration as the most promising candidate to mitigate the potential for foaming. The development of a model, using BioWin software, provided further insight for incorporating handwash and shower wastes streams. While accurate with respect to COD, further customization is needed to improve its ability to model SAMBR's treatment of nitrogen. However, the current model indicated that the incorporation did not significantly affect COD removal but moderately reduced TN removal. The culmination of this research has established SAMBR as a promising and emerging technology and with further investigation, will be a valuable tool to long-term space habitation.

6.2 Future Work

The scope of this work covers SAMBR's development from its initial concept development to a scaled prototype testing genuine feed media. NASA's Technology Readiness Level (TRL) tracks the advancement of technologies from concept to utilization as a proven and successful system. The design, fabrication, and verification testing advanced SAMBR from a conceptual TRL of 1 to 4. The switch to real urine provides realistic validation that cannot be obtained from simulants and advanced SAMBR to a TRL of 5. With the final stage of testing, SAMBR was tested in a relevant ground scenario where it was fed the urine and flush estimated for a crew of four, advancing it to its current TRL of 6.

While SAMBR achieved significant TRL advances during the course of this research, several milestones must be achieved for further advancement (Table 6.1). Continued testing and deeper analysis of SAMBR biological treatment and tolerance for varying inhibiting factors. In

addition to inhibition from nitrogen and pH, salinity could also be inhibitory. An additional approach to acclimating the consortia to develop higher tolerances, bioprospecting from natural sources under extreme conditions (i.e., marshes, brine channels, springs, etc.) to extract organisms already established as highly tolerant (D & Ak, 2022).

The next recommended phase would be continuing the development of the foaming mitigation subsystem. As filtration was identified to be the most suitable, a scaled membrane module would need to be designed with sufficient safety factor to handle the higher hygiene waste volumes. Considerations to the fate of the rejected constituents (primarily surfactants) will be necessary since continuous operation will result in accumulation. Further treatment expansion to include laundry water will further display SAMBR's ability to treat a range of wastes that vary in volume and strength. The eventual addition of laundry should be incorporated into the calculations when the foaming mitigation subsystem is designed. Long-term operation treating the various waste streams are necessary to establish not only SAMBR's treatment performance but reputation as a robust and reliable system, capable of consistent treatment equivalent to (and greater than) the anticipated length of future habitation missions. Analog mission testing in isolated conditions such as the CHAPEA habitat at Johnson Space Center or McMurdo station in Antarctica will assess SAMBR's ability to operate in a remote environment. Given a continuous successful evaluation and performance, SAMBR would reach advance to a TRL of 7 when tested in a space environment. A suborbital test flight would assess the effects of reduced gravity on SAMBR and provide insight for future areas of development for such operation.

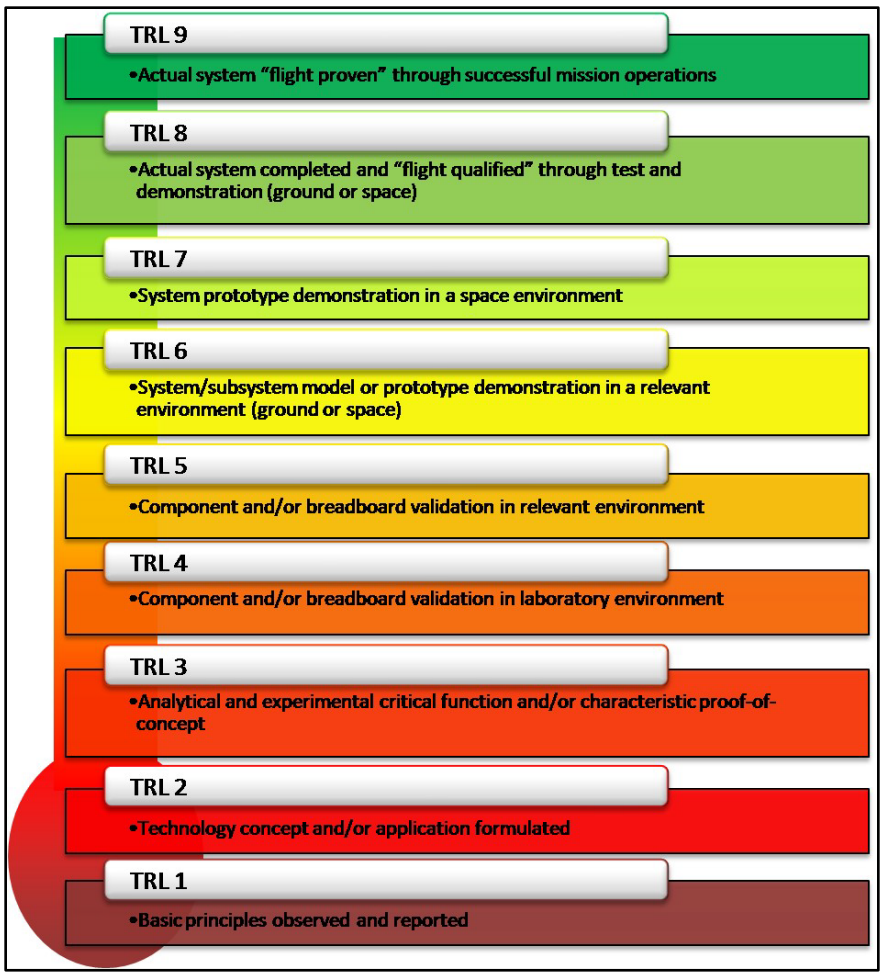


Figure 6.1 Technology Readiness Levels (TRLs).

Table 6.1 Suggested next steps for future SAMBR development.

Next suggested research phase	Description	TRL
1	Scaled testing and integration of foaming mitigation technologies with SAMBR	6
2	Operation of SAMBR with urine, flush, and hygiene waste streams	
3	Incorporation of laundry waste stream	
4	Long-term evaluation	
5	Analog mission testing	
6	Sub-orbital flight testing	7

6.3 Impact

The development of SAMBR provides a novel and sustainable alternative for wastewater treatment, primarily for space exploration as designed but also in terrestrial applications. While complimentary technologies exist, no spaceflight capable biological treatment system has been developed. Similar technologies in development require further refinement including scaled testing, nutrient recovery, and consumable requirement. The knowledge gained from SAMBR has provided expanded insight for developing biological treatment systems for space. My educational pathway culminating in the research presented here have expanded the boundary of human knowledge (Fig. 6.2).

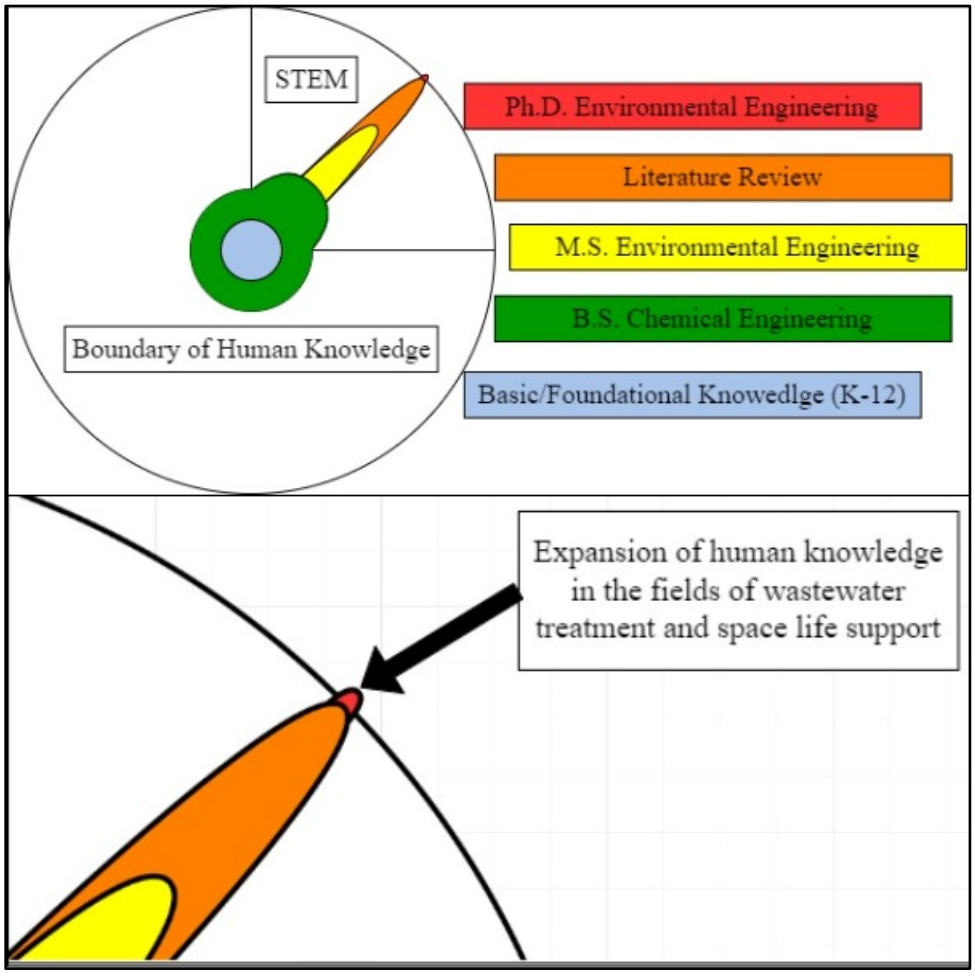


Figure 6.2 Impact of Ph.D. on boundary of human knowledge. Adapted from (Might, 2024)

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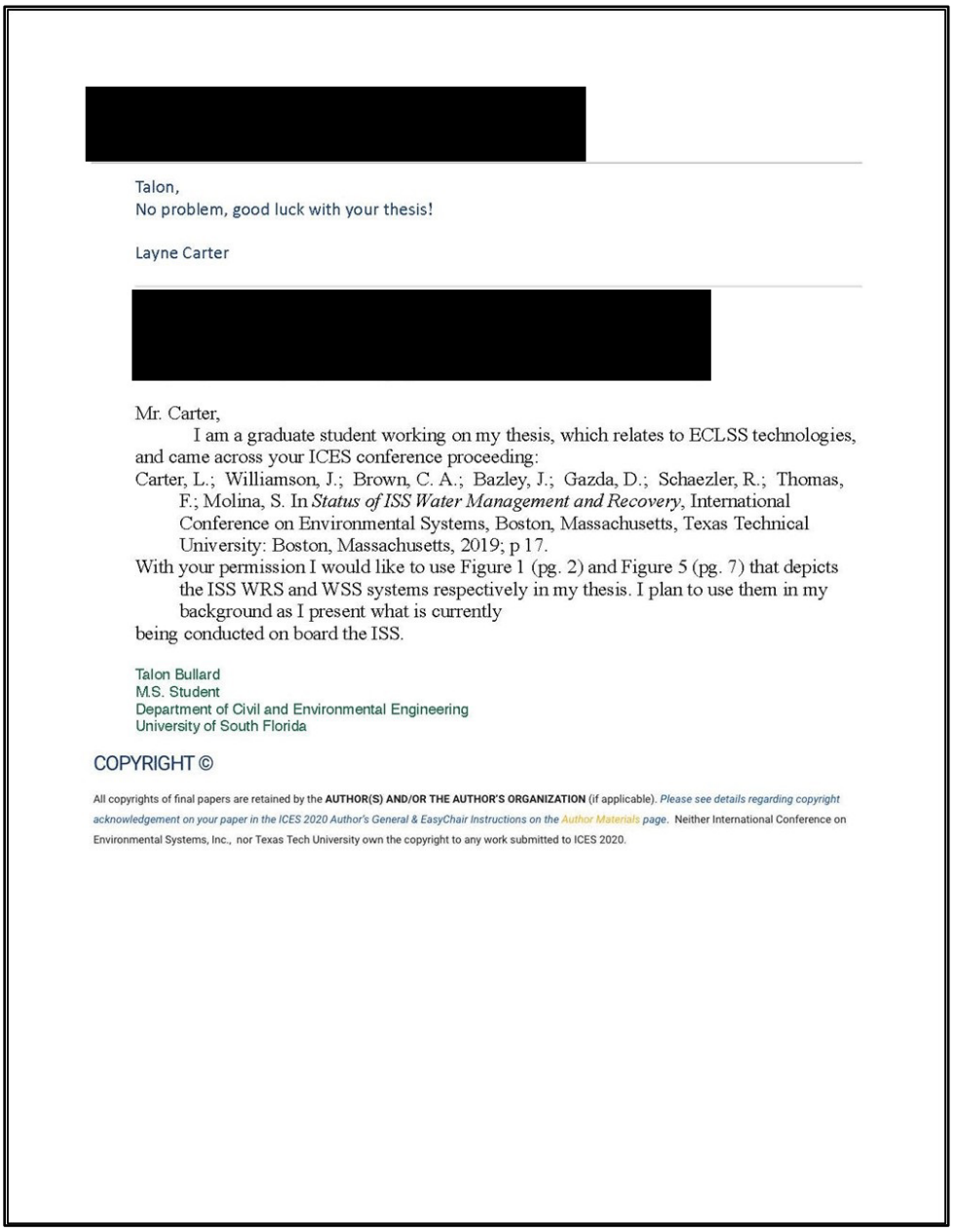
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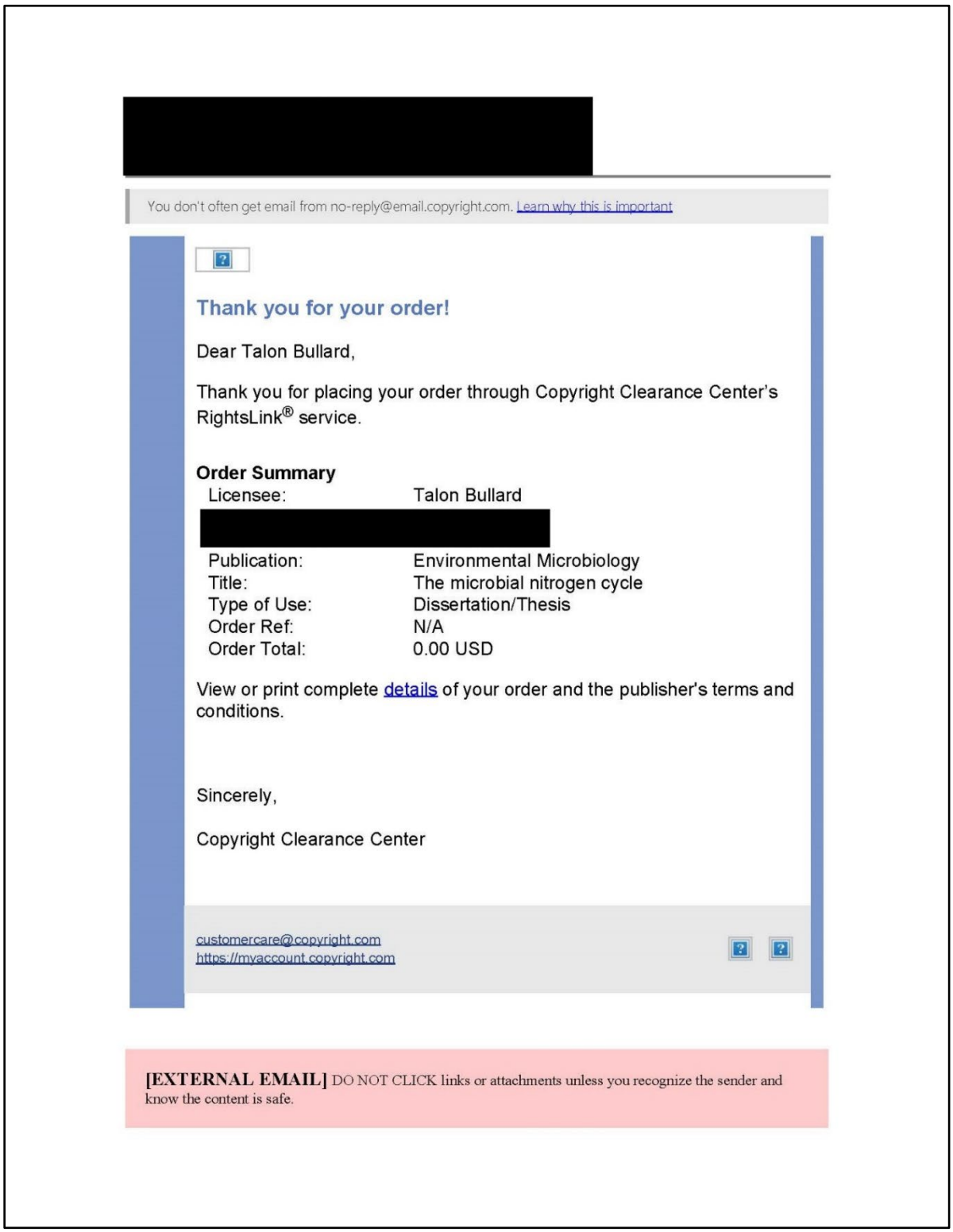
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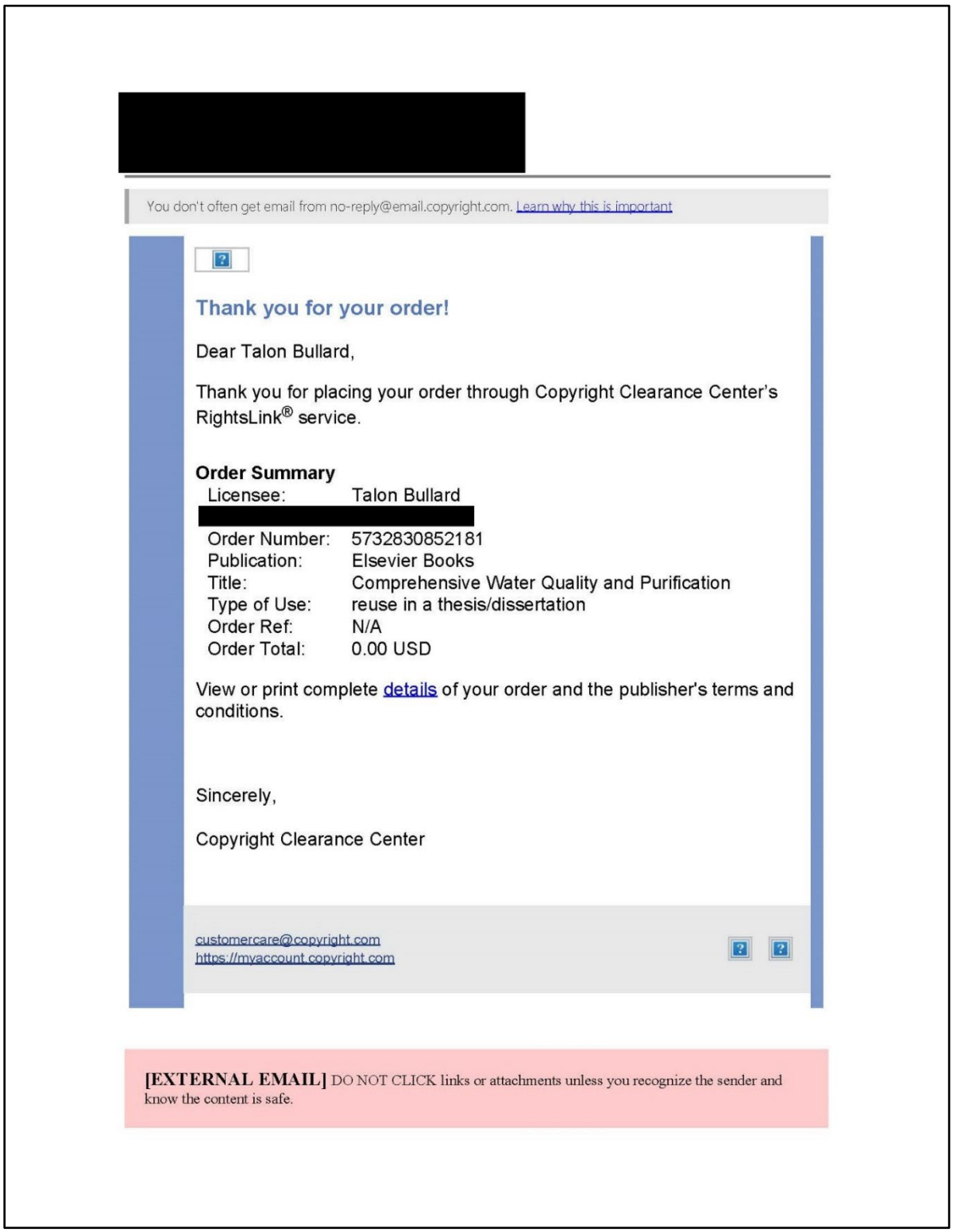
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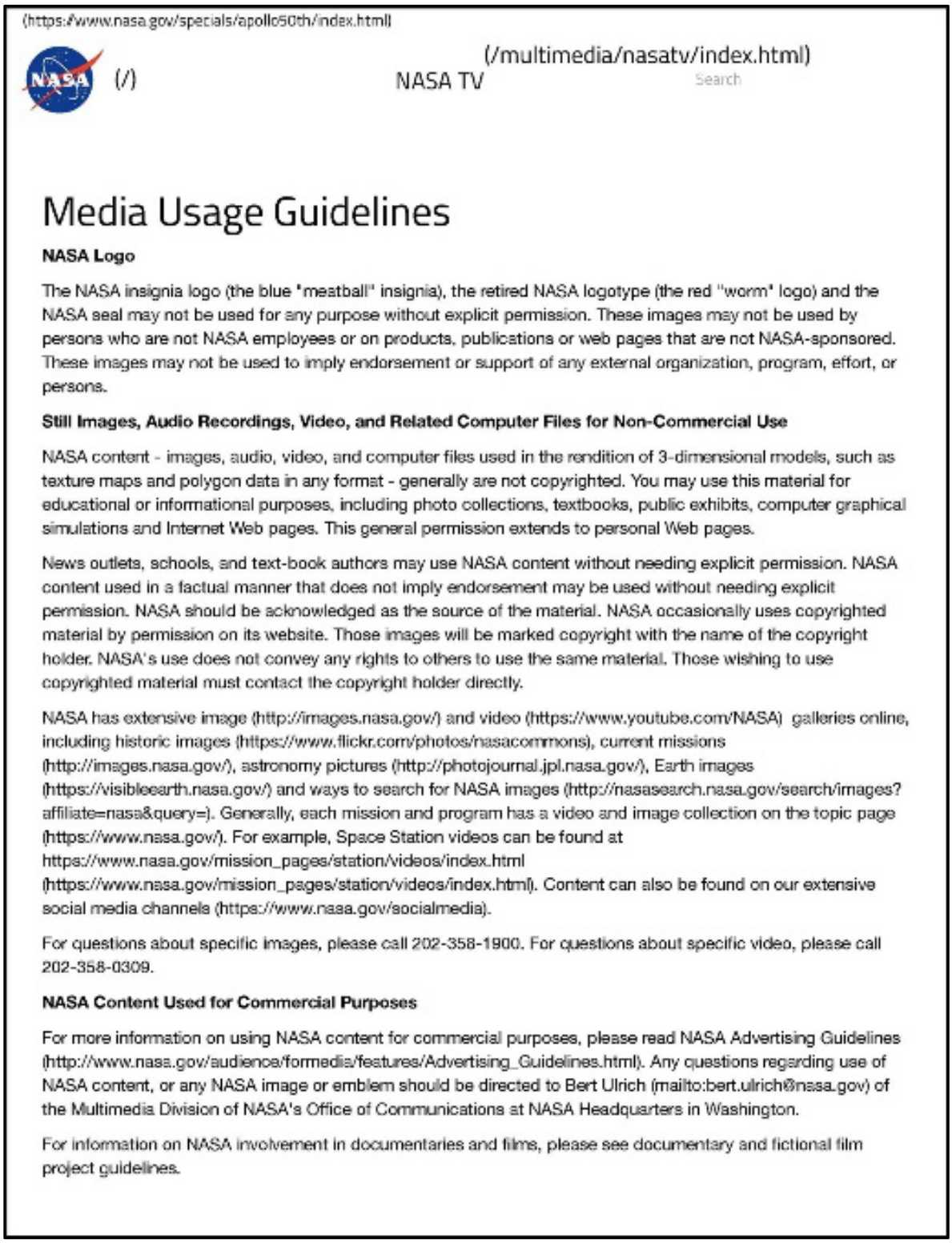
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Appendix B: Supplemental Data

B.1 Oxidation Reduction Potential

ORP measurements gave an indication of how oxidative or reductive the zones were in SAMBR. For the oxidation of ammonia to nitrate (nitrification), a positive measurement is expected while the reduction of nitrate to nitrogen (denitrification) should result in a negative measurement. The expected measurements for the oxic conditions were generally what was observed in the initial stages (Fig. B.1) with consistently positive ORP measurements. The anoxic zones were weakly positive due to the events described in section 5.4.4. The ORP of the permeate was consistently the highest, due to the retention of reducing constituents by the membrane.

In the later stages the anoxic zones became more established as reducing environments but were later joined by the oxic zone and permeate as the feed strength increased and the aeration event in stage G. The aeration restoration and pH management strategies had a positive effect on the oxic zone and permeate, returning them to positive ranges.

B.2 Electrical Conductivity

EC measurements were incorporated into the late stages, G, H-1 and H-2, following the events of stage G. The EC values measured gave an indication of the salinity, total dissolved solids (TDS), and ultimately the system's ionic strength. Stage G displayed a mostly consistent EC around 22,000 uS/cm but increased with the transition to full-strength urine in stages H-1 and H-2 (Fig. B.2). The observed increase was accompanied with increased variability as the EC ranged from 51,000 to 180,000 uS/cm.

B.3 Turbidity

The turbidity, an indicative measure of clarity, of the permeate was very low and averaged out to below NTU in most instances. The membrane proved to be effective as an absolute barrier to particulate retention (Fig. B.3). The switch and subsequent increased strength of urine introduced a slight increase in value and variability but was still maintained to single digit values. The sharp increase in turbidity observed in stage G could be attributed to the formation of precipitants such as struvite and species of phosphates. Though the UF membrane rejects particulate constituents, the ions that comprise such precipitants pass through and have the capacity to precipitate post-filtration. As the pH recovered in stages H-1 and H-2, turbidity returned to previously nominal levels just as quickly. Though not reported, the turbidity of the influent was attempted to be measured on several occasions, however at each occurrence it measured beyond the range of the meter (>10000 NTU). In comparison to the influent turbidity, the production of measurable turbidity levels at the effluent is a significant improvement.

B.4 Transmembrane Pressure

TMP ranged from 0.27 to 0.48 bar, averaging 0.33 bar. Far below the manufacturer listed maximum of 2.4 bar, the membrane separated the treated effluent from the reactor contents without overloading the membrane as can be observed in Figure B.4. Given that the membrane was designed to be oversized to allow for shock loading and gradual fouling, the TMP was expected to be low. The data logging system was initially set to record data at intervals on the scale of seconds. However, the interval captured a lot of dynamic noise from the relaxation and backwashing events. The sampling interval widened, capturing and averaging wider sets of measurements, resulting in variation for stages H-1 and H-2.

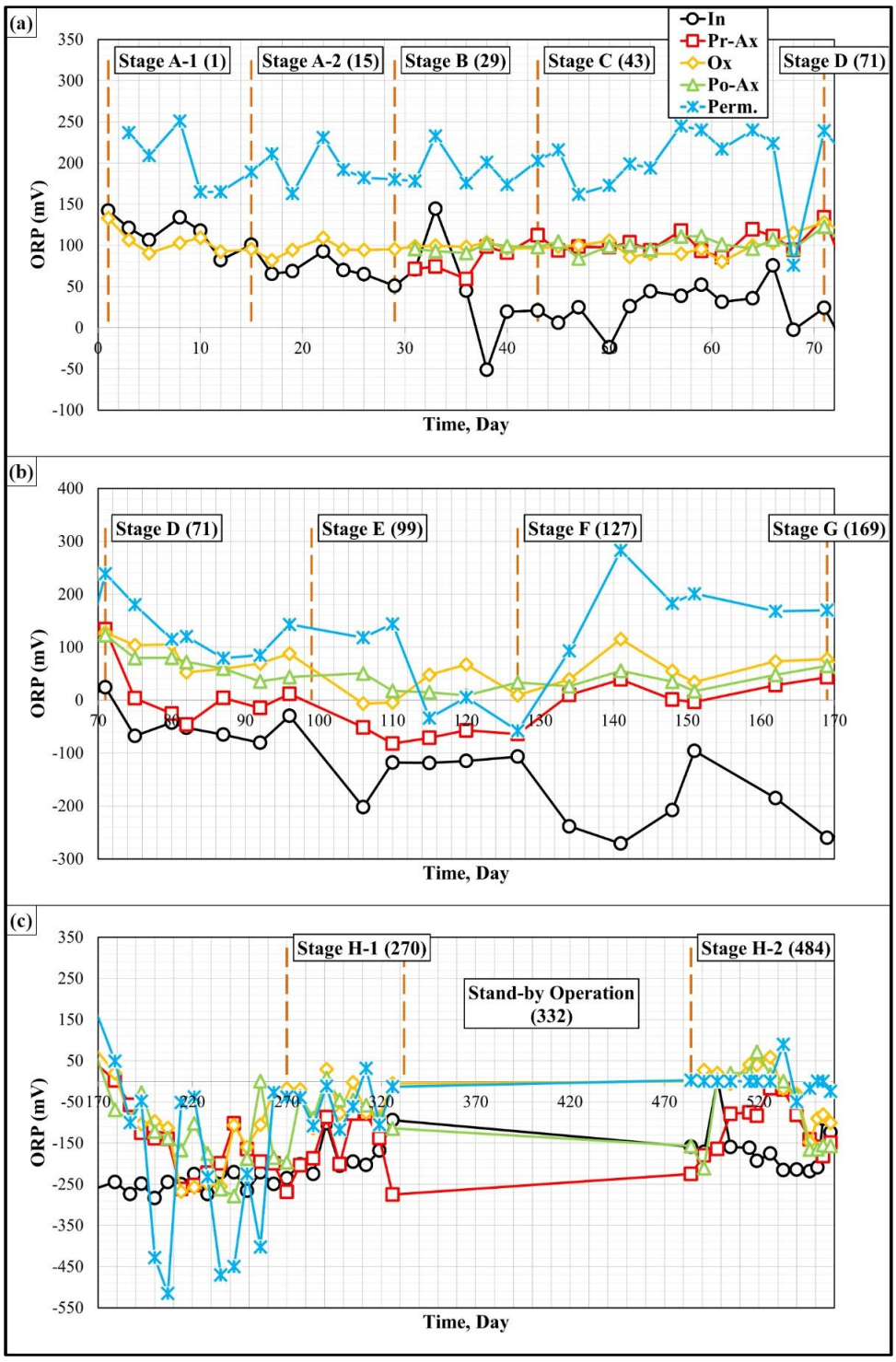


Figure B.1 ORP profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

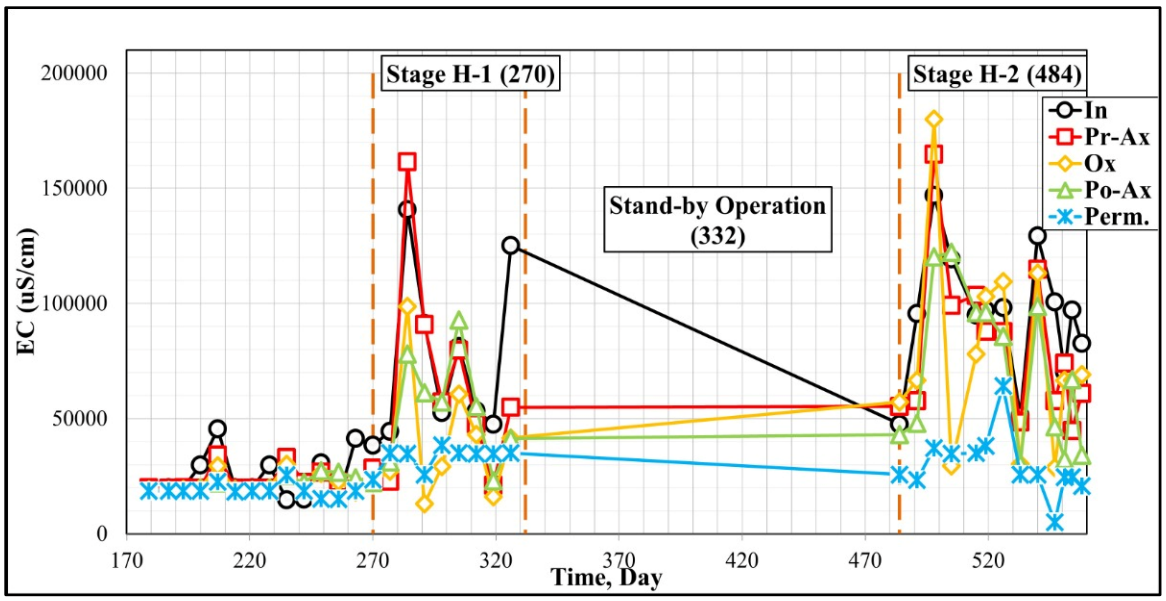


Figure B.2 EC profile of SAMBR during stages G to H-2 of testing.

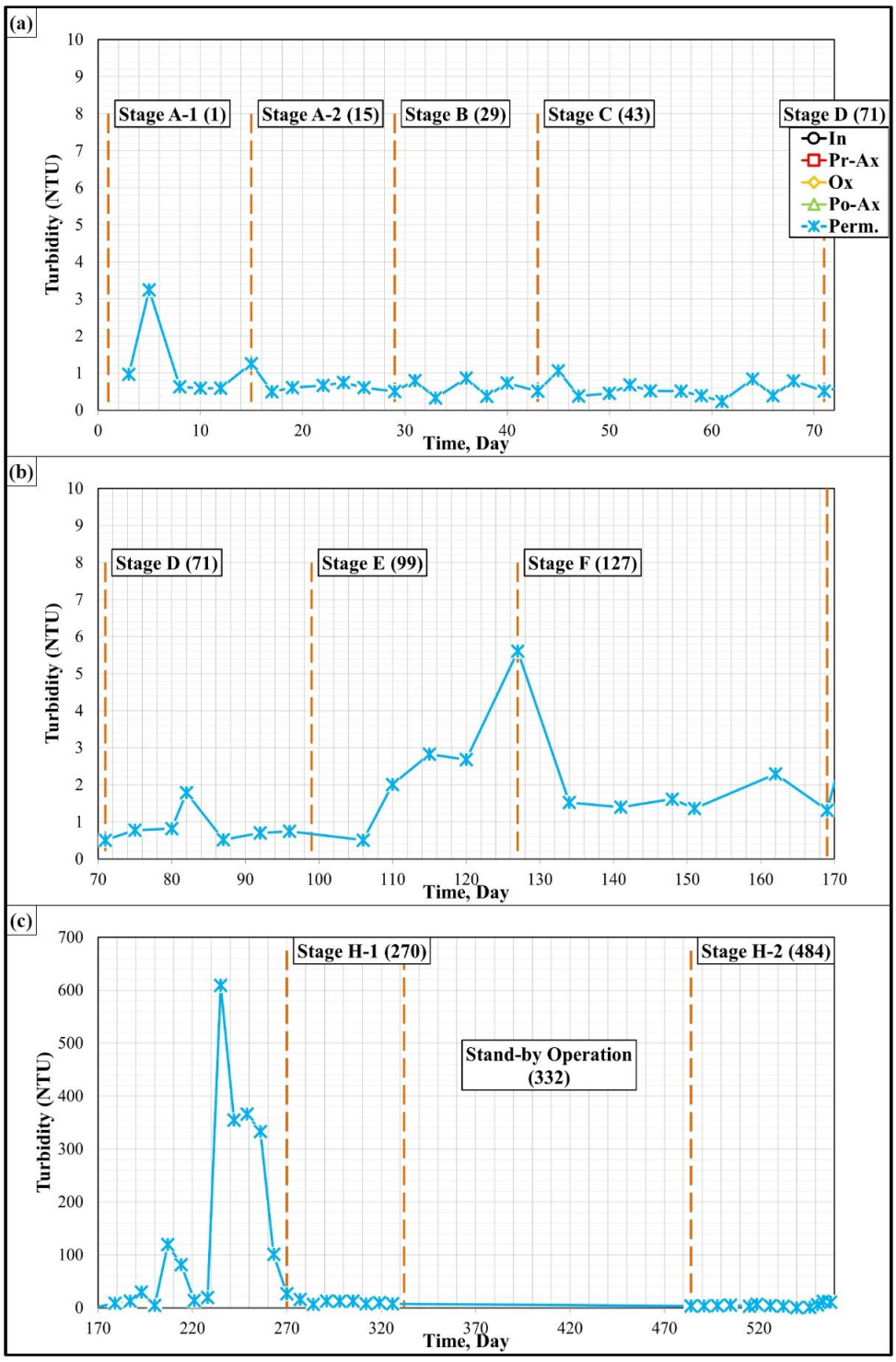


Figure B.3 Turbidity profile of SAMBR; (a) stages A-1 to C; (b) stages D to F; (c) stages G to H-2 of testing.

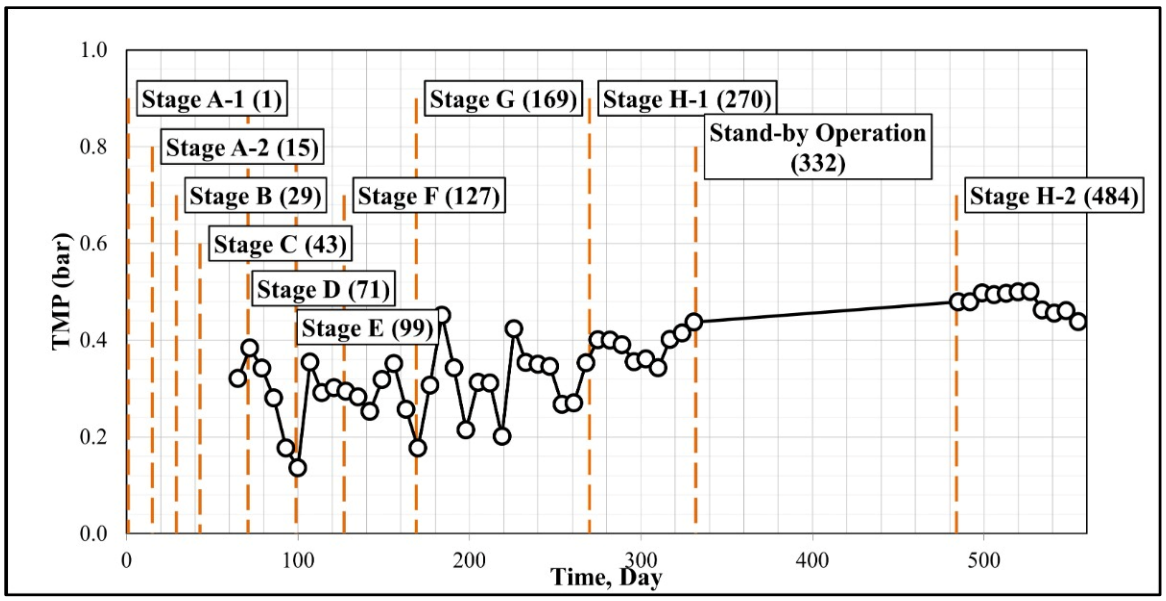


Figure B.4 TMP profile of SAMBR.